

Key Attack on the ACDGV Matrix Encryption Scheme

(full version)^{*}

Anmoal Porwal¹ (✉) , Antonia Wachter-Zeh¹ , and Pierre Loidreau^{2,3,4} 

¹ Technical University of Munich, Munich, Germany
`{anmoal.porwal, antonia.wachter-zeh}@tum.de`

² DGA and University of Rennes, Rennes, France

³ CNRS, Rennes, France

⁴ IRMAR – UMR 6625, Rennes, France
`pierre.loidreau@univ-rennes.fr`

Abstract. We present an exponential-time key recovery attack on the public-key encryption scheme using matrix codes proposed by Aragon *et al.* [1] at Asiacrypt 2024. The secret key is a Gabidulin code expanded using an $\mathbb{F}_q$ -basis of $\mathbb{F}_{q^m}$ to obtain a matrix code, which is then hidden by appending random rows and columns and by left- and right-multiplication with invertible matrices. Our attack does not rely on the Gabidulin structure and hence applies to most $\mathbb{F}_{q^m}$ -linear codes hidden by their transform. Its complexity is better than the previously best-known distinguisher and significantly better than the naive key recovery algorithm. Our attack breaks some of their proposed parameters. For example, a parameter set targeting 192-bit security is reduced to about 161 bits, and a 256-bit set to about 223 bits.

Keywords: Rank metric · Matrix codes · McEliece scheme · Public-key encryption · Structural attack

1 Introduction

Code-based cryptography has become an important post-quantum alternative to lattice-based schemes. In recent work [1], Aragon, Couvreur, Dyseryn, Gaborit, and Vinçotte (ACDGV) introduced a new code-based public-key encryption scheme within the McEliece framework that uses matrix codes with a polynomial-time decoder and whose message security relies on the so-called Min-Rank problem. The public key is obtained by expanding a secret $\mathbb{F}_{q^m}$ -linear Gabidulin code using an $\mathbb{F}_q$ -basis of $\mathbb{F}_{q^m}$, and then hiding the resulting matrix code by appending random rows and columns and applying left- and right-multiplication with invertible matrices.

Despite using matrix codes, the scheme achieves significantly smaller key sizes than Classic McEliece [2]. For example, at the 128-bit security level, Classic McEliece requires a public key of about 261 kB and a ciphertext of 96 B,

^{*} Errors in the published version are listed in Appendix I.

whereas the ACDGV scheme requires only 41 kB and 138 B, respectively. This is partly because message security relies on the MinRank problem, which for similar parameters is more expensive to solve than decoding in the Hamming metric, and partly because the key distinguisher introduced in [1] has high complexity, allowing the code to be hidden with relatively few random rows and columns.

1.1 Contribution

We present an exponential-time attack (Algorithm 4 in Section 5.3) that recovers the secret key of ACDGV [1] with cost lower than the best algorithm (by [1] themselves) to distinguish the public key from random, ignoring polynomial factors. This breaks some of their proposed parameters; see Table 1 for examples and Table 3 in Section 6 for the full list. We also formalize the search and distinguishing problems corresponding to the key-recovery problem (Section 2) and rigorously analyze a naive algorithm and ACDGV’s distinguisher (Section 5) for comparison. Since their transform to hide the secret code is quite natural, we view this as an interesting coding-theoretic problem in its own right. Notably, our attack does not rely on the Gabidulin structure and applies to most $\mathbb{F}_{q^m}$ -linear codes (Remark 2).

As a secondary contribution, Algorithm 1 in Section 4 efficiently solves the following subproblem⁵: given a matrix code that is the expansion of an unknown $\mathbb{F}_{q^m}$ -linear code, find an $\mathbb{F}_q$ -basis of $\mathbb{F}_{q^m}$ used to expand the code. This is an extension of the results of Couvreur, Debris-Alazard and Gaborit [5] who show how to decide whether two matrix codes, obtained by expanding two unknown $\mathbb{F}_{q^m}$ -linear codes, are equivalent under left- and right-multiplication by invertible matrices. In particular, we use results (Lemmas 1 and 2) that are similar to the ones proved in [5, Lemma 26]. While the extension of their results to recovering the $\mathbb{F}_{q^m}$ -linear code was generally believed possible, the proof has so far been missing in the literature.

We provide Python code (Appendix H) to demonstrate a proof of concept of our attack, verify the assumptions it relies on, and reproduce the concrete costs in Table 3 in Section 6. It is available at <https://github.com/mnap/acdgv-key-attack> and will also be submitted to the IACR Artifact Archive.

1.2 Related Work

The first rank-metric cryptosystem, GPT [7], was a McEliece-based scheme using $\mathbb{F}_{q^m}$ -linear vector codes with Gabidulin codes as secret. This was broken by structural attacks of Gibson [8] and Overbeck [18], the latter of which led to the cryptanalysis of several later GPT variants, such as the column-scrambler version [17]. Overbeck’s attack works by applying the Frobenius automorphism to the codeword entries of the public $\mathbb{F}_{q^m}$ -linear code. However, this technique seems inapplicable to the ACDGV scheme [1], since the hiding transform completely distorts the $\mathbb{F}_{q^m}$ structure; indeed, this was one of the motivations behind the scheme’s design.

⁵ In the case where the stabilizer algebra (Definition 6) has dimension m .

Table 1. ACDGV [1] proposed parameter sets with the three smallest public key sizes for each NIST security level. The work factor of our attack is shown, with values below security level in bold.

NIST Security Level	public key size (kB)	ciphertext size (B)	Our Work factor (bits)
143 bits (AES-128)	39	242	196
	41	138	176
	70	111	112
207 bits (AES-192)	133	134	161
	173	112	214
	230	106	214
272 bits (AES-256)	87	334	223
	108	218	295
	192	177	240

1.3 Organization

Section 2 defines the search and distinguishing problems, which correspond to the key-recovery problem for the ACDGV scheme when the code is specialized to a Gabidulin code. Section 3 provides an intuitive explanation of our attack. Section 4 introduces the **BasisRecover** subroutine (Algorithm 1) that is used later. Section 5 presents a naive key-recovery algorithm, the distinguisher of ACDGV, and our new attack. Section 6 compares these algorithms and shows which ACDGV parameter sets are affected.

1.4 Notation

Coding Theory and Linear Algebra We assume column vectors. An $[n, k]_{\mathbb{F}}$ code $\mathcal{C}$ is a k -dimensional subspace of blocklength n over the field $\mathbb{F}$, which will usually be the finite fields $\mathbb{F}_q$ or $\mathbb{F}_{q^m}$. A vector code is a subspace of $\mathbb{F}^n$ and a matrix code is a subspace of $\mathbb{F}^{m \times n}$. The row and column span over $\mathbb{F}$ of a matrix $\mathbf{A} \in \mathbb{F}^{m \times n}$ are denoted $\text{rowspan}(\mathbf{A})$ and $\text{colspan}(\mathbf{A})$ respectively. For a subspace $\mathcal{U} \subseteq \mathbb{F}^n$, its orthogonal complement $\mathcal{U}^\perp$ is $\{\mathbf{v} \in \mathbb{F}^n \mid \sum_{i=1}^n v_i u_i = 0 \text{ for all } \mathbf{u} \in \mathcal{U}\}$. Semilinear $\mathbb{F}_q$ -equivalence and the stabilizer algebra (Stab) of a matrix code are defined in Definitions 5 and 6 (Section 2), respectively. The notation $\text{Stab}_{\text{samp}}$ is defined in Definition 7 (Section 4).

Field Theory For $x \in \mathbb{F}_{q^m}$, we denote the j -th Frobenius power x^{q^j} by $x^{[j]}$. Let $\boldsymbol{\gamma} = (\gamma_1, \dots, \gamma_m)^\top \in \mathbb{F}_{q^m}^m$ be an $\mathbb{F}_q$ -basis of $\mathbb{F}_{q^m}$. The expansion map and

its inverse are

$$\begin{aligned} \text{ext}_\gamma : \mathbb{F}_{q^m} &\rightarrow \mathbb{F}_q^m & \text{s.t.} & \sum_{i=1}^m x_i \gamma_i \mapsto (x_1, \dots, x_m)^\top, \\ \text{ext}_\gamma^{-1} : \mathbb{F}_q^m &\rightarrow \mathbb{F}_{q^m} & \text{s.t.} & (x_1, \dots, x_m)^\top \mapsto \sum_{i=1}^m x_i \gamma_i. \end{aligned}$$

Operation Extension Let $\mathcal{C}_{\text{mat}} \subseteq \mathbb{F}_q^{m \times n}$ and $\mathcal{C}_{\text{vec}} \subseteq \mathbb{F}_{q^m}^n$. For $\mathbf{S}' \in \mathbb{F}_q^{m \times m}$ and $\mathbf{S} \in \mathbb{F}_q^{n \times n}$ and any integer j , define

$$\begin{aligned} \mathcal{C}_{\text{mat}} \mathbf{S} &:= \{\mathbf{C} \mathbf{S} \mid \mathbf{C} \in \mathcal{C}_{\text{mat}}\} \subseteq \mathbb{F}_q^{m \times n}, \\ \mathbf{S}' \mathcal{C}_{\text{mat}} &:= \{\mathbf{S}' \mathbf{C} \mid \mathbf{C} \in \mathcal{C}_{\text{mat}}\} \subseteq \mathbb{F}_q^{m \times n}, \\ \mathbf{S} \mathcal{C}_{\text{vec}} &:= \{\mathbf{S} \mathbf{c} \mid \mathbf{c} \in \mathcal{C}_{\text{vec}}\} \subseteq \mathbb{F}_{q^m}^n \quad (\text{recall we assume column vectors}), \\ \mathcal{C}_{\text{vec}}^{[j]} &:= \{\mathbf{c}^{[j]} \mid \mathbf{c} \in \mathcal{C}_{\text{vec}}\} \subseteq \mathbb{F}_{q^m}^n \quad \text{where} \quad \mathbf{c}^{[j]} := (c_1^{[j]}, c_2^{[j]}, \dots, c_n^{[j]})^\top. \end{aligned}$$

We also extend the maps ext_γ and ext_γ^{-1} as

$$\begin{aligned} \text{ext}_\gamma(\mathcal{C}_{\text{vec}}) &:= \{(\text{ext}_\gamma(c_1), \dots, \text{ext}_\gamma(c_n)) \mid \mathbf{c} \in \mathcal{C}_{\text{vec}}\} \subseteq \mathbb{F}_q^{m \times n}, \\ \text{ext}_\gamma^{-1}(\mathcal{C}_{\text{mat}}) &:= \{(\text{ext}_\gamma^{-1}(\mathbf{C}_{:,1}), \dots, \text{ext}_\gamma^{-1}(\mathbf{C}_{:,n}))^\top \mid \mathbf{C} \in \mathcal{C}_{\text{mat}}\} \subseteq \mathbb{F}_{q^m}^n, \end{aligned}$$

where $\mathbf{C}_{:,i} \in \mathbb{F}_q^m$ denotes the i th column of $\mathbf{C} \in \mathbb{F}_q^{m \times n}$.

Probability and Complexity If $\mathcal{X}$ is a distribution (resp. a set), then $x \stackrel{\$}{\leftarrow} \mathcal{X}$ means to sample x from $\mathcal{X}$ (resp. uniformly at random from $\mathcal{X}$). We may omit the word “uniformly” on occasion, for instance we may speak of a random full-rank matrix in $\mathbb{F}_q^{m \times n}$, but the uniform distribution is always implied in such cases. If an event happens “with at least constant probability”, we mean with probability at least p where $p > 0$ is a fixed constant. If an event happens “except with negligible probability in m ”, its probability is $1 - f(m)$ where, for all positive integers d , for sufficiently large m , we have $f(m) < 1/m^d$. If the complexity of an algorithm is at most T “up to a constant factor”, we mean it is at most cT for some fixed $c > 0$. The matrix multiplication exponent, also called the linear algebra constant, is $\omega \in [2, 3]$.

2 The Search Problem

2.1 Main Problems

Definition 1 (Hiding Transform [1, Def. 14]). *The input is a matrix code $\mathcal{C}_{\text{mat}} \subseteq \mathbb{F}_q^{m \times n}$ of dimension K . Draw a random basis $\mathbf{A}_1, \dots, \mathbf{A}_K$ of $\mathcal{C}_{\text{mat}}$. Draw two random invertible matrices $\mathbf{P} \in \mathbb{F}_q^{(m+\ell_1) \times (m+\ell_1)}$ and $\mathbf{Q} \in \mathbb{F}_q^{(n+\ell_2) \times (n+\ell_2)}$.*

For each $i = 1, \dots, K$, draw random matrices $\mathbf{R}_i \in \mathbb{F}_q^{m \times \ell_2}$, $\mathbf{R}'_i \in \mathbb{F}_q^{\ell_1 \times n}$ and $\mathbf{R}''_i \in \mathbb{F}_q^{\ell_1 \times \ell_2}$. Let

$$(\mathbf{B}_1, \dots, \mathbf{B}_K) := \left(\mathbf{P} \begin{bmatrix} \mathbf{A}_1 & \mathbf{R}_1 \\ \mathbf{R}'_1 & \mathbf{R}''_1 \end{bmatrix} \mathbf{Q}, \dots, \mathbf{P} \begin{bmatrix} \mathbf{A}_K & \mathbf{R}_K \\ \mathbf{R}'_K & \mathbf{R}''_K \end{bmatrix} \mathbf{Q} \right).$$

The output is the matrix code $\mathcal{C}_{pub} := \langle \mathbf{B}_1, \dots, \mathbf{B}_K \rangle_{\mathbb{F}_q} \subseteq \mathbb{F}_q^{(m+\ell_1) \times (n+\ell_2)}$.

Definition 2 (Enhanced Matrix Code Distribution (EMC($\mathcal{D}$))-distribution). Let $\mathcal{D}$ be a distribution over the set of $[n, k]_{\mathbb{F}_{q^m}}$ codes. Draw a random $\mathbb{F}_q$ -basis γ of $\mathbb{F}_{q^m}$. Draw $\mathcal{C}_{vec} \xleftarrow{\$} \mathcal{D}$ and let $\mathcal{C}_{mat} := \text{ext}_{\gamma}(\mathcal{C}_{vec}) \subseteq \mathbb{F}_q^{m \times n}$. The EMC($\mathcal{D}$)-distribution outputs the matrix code $\mathcal{C}_{pub} \subseteq \mathbb{F}_q^{(m+\ell_1) \times (n+\ell_2)}$ of dimension km obtained by applying the hiding transform (Definition 1) to $\mathcal{C}_{mat}$.

In the ACDGV scheme, $\mathcal{D}$ is specialized to be the uniform distribution over all $[n, k]_{\mathbb{F}_{q^m}}$ Gabidulin codes. Key recovery in the general case amounts to the following problem and is thus the main focus of our work.

Definition 3 (Hidden Vector Code Search (HVCS) Problem for $\mathcal{D}$). Sample a matrix code $\mathcal{C}_{pub} \subseteq \mathbb{F}_q^{(m+\ell_1) \times (n+\ell_2)}$ of dimension km from the EMC($\mathcal{D}$)-distribution. Find

$$(\mathbf{V}, \mathbf{W}, \gamma') \in \mathbb{F}_q^{(m+\ell_1) \times m} \times \mathbb{F}_q^{(n+\ell_2) \times n} \times \mathbb{F}_{q^m}^m$$

such that γ' is an $\mathbb{F}_q$ -basis of $\mathbb{F}_{q^m}$ and

$$\mathcal{C}'_{vec} := \text{ext}_{\gamma'}^{-1}(\mathbf{V}^T \mathcal{C}_{pub} \mathbf{W}) \subseteq \mathbb{F}_{q^m}^n$$

is an $[n, k]$ code over $\mathbb{F}_{q^m}$.

The matrices $\mathbf{V}$ and $\mathbf{W}$ get rid of the random-rows and random-columns part from $\mathcal{C}_{pub}$ respectively. We require finding $(\mathbf{V}, \mathbf{W}, \gamma')$ rather than only $\mathcal{C}'_{vec}$, since in the ACDGV scheme the ciphertext is a corrupted codeword of $\mathcal{C}_{pub}$. Recovering $(\mathbf{V}, \mathbf{W}, \gamma')$ enables conversion of this ciphertext into a corrupted codeword in $\mathcal{C}'_{vec}$ which can then be decoded to obtain the message, hence serving as an equivalent secret key. Appendix A describes the ACDGV scheme and shows how key recovery reduces to the HVCS problem.

The following decisional version is potentially easier.

Definition 4 (Hidden Vector Code Distinguishing (HVCD) Problem for $\mathcal{D}$). Draw $b \xleftarrow{\$} \{0, 1\}$. If $b = 1$, sample $\mathcal{C}_{pub}$ from the EMC($\mathcal{D}$)-distribution, else sample $\mathcal{C}_{pub}$ from the uniform distribution over the set of km -dimensional subspaces of $\mathbb{F}_q^{(m+\ell_1) \times (n+\ell_2)}$. Given $\mathcal{C}_{pub}$, decide whether $b = 0$ or $b = 1$.

2.2 Equivalent Solutions

Let $\mathbf{S} \in \mathbb{F}_q^{n \times n}$ be invertible. If $\mathcal{C}_{mat} = \text{ext}_{\gamma}(\mathcal{C}_{vec})$, then also $\mathcal{C}_{mat} = \text{ext}_{\gamma[j]}(\mathcal{C}_{vec}^{[j]})$ for all $j \in \{0, \dots, m-1\}$. Further, since the hiding transform uniformly right-scrambles its input $\mathcal{C}_{mat}$, its output distribution is the same for both the inputs

$\mathcal{C}_{\text{mat}}$ and $\mathcal{C}_{\text{mat}}\mathbf{S}$. Thus, if an HVCS instance $\mathcal{C}_{\text{pub}}$ has solution $\mathcal{C}_{\text{vec}}$, then $\mathbf{S}\mathcal{C}_{\text{vec}}^{[j]}$ is another solution.

Proposition 1 (Multiple Solutions). *Let $\mathbf{V} \in \mathbb{F}_q^{(m+\ell_1) \times m}$, $\mathbf{W} \in \mathbb{F}_q^{(n+\ell_2) \times n}$ and $\gamma \in \mathbb{F}_{q^m}^m$ be a solution to an HVCS instance $\mathcal{C}_{\text{pub}} \subseteq \mathbb{F}_q^{(m+\ell_1) \times (n+\ell_2)}$. Let $\mathcal{C}_{\text{vec}} := \text{ext}_{\gamma}^{-1}(\mathbf{V}^T \mathcal{C}_{\text{pub}} \mathbf{W}) \subseteq \mathbb{F}_{q^m}^n$ be the corresponding vector code. Then for all $j \in \{0, \dots, m-1\}$ and invertible $\mathbf{S} \in \mathbb{F}_q^{n \times n}$, the instance also has solution $(\mathbf{V}, \mathbf{W}\mathbf{S}^T, \gamma^{[j]})$ with corresponding vector code $\mathbf{S}\mathcal{C}_{\text{vec}}^{[j]} = \text{ext}_{\gamma^{[j]}}^{-1}(\mathbf{V}^T \mathcal{C}_{\text{pub}} \mathbf{W}\mathbf{S}^T)$.*

See Appendix D for the proof. We consider all these solutions “equivalent”.

Definition 5 (Semilinear $\mathbb{F}_q$ -equivalence). *On the set of all $[n, k]_{\mathbb{F}_{q^m}}$ codes, $\mathcal{C}_{\text{vec}}$ and $\mathcal{C}'_{\text{vec}}$ are semilinear $\mathbb{F}_q$ -equivalent if and only if $\mathcal{C}'_{\text{vec}} = \mathbf{S}\mathcal{C}_{\text{vec}}^{[j]}$ for some $j \in \{0, \dots, m-1\}$ and invertible $\mathbf{S} \in \mathbb{F}_q^{n \times n}$.*

Remark 1. If an HVCS instance has N solutions up to semilinear $\mathbb{F}_q$ -equivalence, the stated complexity T of each attack in Section 5 becomes⁶ T/N . However, we conjecture $N = 1$ with high probability (w.h.p.). Moreover, if an attack has complexity T/N then ACDGV key recovery still costs $\frac{T}{N}N = T$ (Proposition 4 in Appendix A), since decryption requires a code semilinear $\mathbb{F}_q$ -equivalent to a Gabidulin code and other equivalence classes might not yield efficiently decodable codes. Hence N becomes irrelevant in this setting.

2.3 Stabilizer Algebra

The attack we present in Section 5.3 for the HVCS problem makes an assumption about the distribution $\mathcal{D}$ over the set of $[n, k]_{\mathbb{F}_{q^m}}$ codes.

Definition 6 (Stabilizer Algebra ([5, Def. 9])). *Let $\mathcal{C}_{\text{mat}} \subseteq \mathbb{F}_q^{m \times n}$ be a matrix code. The (left) stabilizer algebra of $\mathcal{C}_{\text{mat}}$ is*

$$\text{Stab}(\mathcal{C}_{\text{mat}}) := \{\mathbf{T} \in \mathbb{F}_q^{m \times m} \mid \mathbf{T}\mathcal{C}_{\text{mat}} \subseteq \mathcal{C}_{\text{mat}}\}.$$

Remark 2. Our attack assumes that, for any $\mathcal{C}$ sampled from $\mathcal{D}$, we have

$$\dim \text{Stab}(\text{ext}_{\gamma}(\mathcal{C})) = m$$

where γ is an $\mathbb{F}_q$ -basis of $\mathbb{F}_{q^m}$. This however is a mild restriction. In general, for any $\mathbb{F}_{q^m}$ -linear code $\mathcal{C}_{\text{vec}}$, we always have $\dim \text{Stab}(\text{ext}_{\gamma}(\mathcal{C}_{\text{vec}})) \geq m$ (see first part of Theorem 1’s proof). But experimentally random $\mathbb{F}_{q^m}$ -linear codes and random Gabidulin codes satisfy $\dim \text{Stab}(\text{ext}_{\gamma}(\mathcal{C}_{\text{vec}})) = m$ w.h.p., assuming non-trivial parameters and regardless of whether $n > m$ or $n \leq m$. In fact, one can prove this for all Gabidulin codes whenever $m \leq n$ (Proposition 3 in Appendix A). All the proposed ACDGV parameters use $m = n$.

⁶ Or T/N' if only $N' \leq N$ solutions satisfy the attack conditions (see Thm 2 and 4).

3 Intuition for the Attack

3.1 High-Level

Solving the HVCS problem (Definition 3) can be broken down into two parts. Let $\mathcal{C}_{\text{vec}}$ be an unknown $[n, k]_{\mathbb{F}_{q^m}}$ code and $\gamma \in \mathbb{F}_{q^m}^m$ an unknown $\mathbb{F}_q$ -basis of $\mathbb{F}_{q^m}$. Let $\mathcal{C}_{\text{mat}} := \text{ext}_\gamma(\mathcal{C}_{\text{vec}}) \subseteq \mathbb{F}_q^{m \times n}$, and let $\mathcal{C}_{\text{pub}} \subseteq \mathbb{F}_q^{(m+\ell_1) \times (n+\ell_2)}$ be the result after applying the hiding transform.

The first part of the attack recovers from $\mathcal{C}_{\text{pub}}$ a matrix code equivalent to $\mathcal{C}_{\text{mat}}$, namely $\mathbf{S}'\mathcal{C}_{\text{mat}}\mathbf{S}$ for some invertible $\mathbf{S}' \in \mathbb{F}_q^{m \times m}$ and $\mathbf{S} \in \mathbb{F}_q^{n \times n}$. The second part then recovers from this a vector code equivalent to $\mathcal{C}_{\text{vec}}$, i.e., $\mathbf{S}\mathcal{C}_{\text{vec}}^{[j]}$ for some $j \in \{0, \dots, m-1\}$.

Section 4 introduces **BasisRecover** (Algorithm 1), a polynomial-time algorithm to solve the second part. Section 5 then describes two complete HVCS solvers: a naive algorithm and our new attack. Both rely on **BasisRecover** for the second part, but differ in how they handle the first. Since the second part is efficient, the overall complexity depends entirely on the first step, i.e., on recovering an equivalent $\mathcal{C}_{\text{mat}}$, which is exponential for both algorithms (but with different costs).

3.2 The MinRank-like Problem

We now describe how our algorithm recovers an equivalent $\mathcal{C}_{\text{mat}}$ from $\mathcal{C}_{\text{pub}}$. Given a matrix code $\mathcal{C} \in \mathbb{F}_q^{m \times n}$, its (left) stabilizer algebra is

$$\text{Stab}(\mathcal{C}) := \{\mathbf{T} \in \mathbb{F}_q^{m \times m} \mid \mathbf{T}\mathcal{C} \subseteq \mathcal{C}\}.$$

For a random matrix code, experiments suggest this equals the space of all scalar matrices w.h.p., i.e., $\{\lambda\mathbf{I} \mid \lambda \in \mathbb{F}_q\}$, so that $\dim \text{Stab}(\mathcal{C}) = 1$. But if $\mathcal{C} = \text{ext}_\gamma(\mathcal{C}_{\text{vec}})$, then $\text{Stab}(\mathcal{C})$ provably has dimension at least m (in fact it contains a subspace isomorphic to $\mathbb{F}_{q^m}$). Thus one can distinguish whether a code arises from an $\mathbb{F}_{q^m}$ -linear code. For $\mathcal{C} = \mathcal{C}_{\text{pub}}$, experiments again give dimension 1. This is to be expected since adding random rows and columns destroys the hidden $\mathbb{F}_{q^m}$ -structure.

Now let $\mathbf{B}_1, \dots, \mathbf{B}_{km} \in \mathbb{F}_q^{(m+\ell_1) \times (n+\ell_2)}$ be a basis of $\mathcal{C}_{\text{pub}}$ and, as in Definition 2 (Hiding Transform), write

$$\mathbf{B}_s = \mathbf{P} \begin{bmatrix} \mathbf{A}_s & \mathbf{R}_s \\ \mathbf{R}'_s & \mathbf{R}''_s \end{bmatrix} \mathbf{Q} \quad \text{for } s = 1, \dots, km,$$

where $\mathbf{A}_1, \dots, \mathbf{A}_{km} \in \mathbb{F}_q^{m \times n}$ is a basis of $\mathcal{C}_{\text{mat}}$. We have seen $\mathbf{F}\mathcal{C}_{\text{pub}} \subseteq \mathcal{C}_{\text{pub}}$ implies w.h.p. that $\mathbf{F}$ is a scalar matrix. Consider instead the choice

$$\mathbf{F} = \mathbf{P} \begin{bmatrix} \mathbf{T} & \mathbf{0} \\ * & * \end{bmatrix} \mathbf{P}^{-1} \quad \text{where } \mathbf{T} \in \text{Stab}(\mathcal{C}_{\text{mat}}),$$

and where the asterisks denote unspecified block matrices. Then, for all $s = 1, \dots, km$,

$$\mathbf{FB}_s = \mathbf{FP} \begin{bmatrix} \mathbf{A}_s & \mathbf{R}_s \\ \mathbf{R}'_s & \mathbf{R}''_s \end{bmatrix} \mathbf{Q} = \mathbf{P} \begin{bmatrix} \mathbf{T} & \mathbf{0} \\ * & * \end{bmatrix} \begin{bmatrix} \mathbf{A}_s & \mathbf{R}_s \\ \mathbf{R}'_s & \mathbf{R}''_s \end{bmatrix} \mathbf{Q} = \mathbf{P} \begin{bmatrix} \mathbf{TA}_s & * \\ * & * \end{bmatrix} \mathbf{Q}.$$

Notice $\mathbf{TA}_s \in \mathcal{C}_{\text{mat}}$ and so there exists a matrix $\mathbf{C}_s \in \mathcal{C}_{\text{pub}}$ which agrees with $\mathbf{FB}_s$ up to the asterisk blocks. That is,

$$\mathbf{FB}_s - \mathbf{C}_s = \mathbf{P} \begin{bmatrix} \mathbf{0} & * \\ * & * \end{bmatrix} \mathbf{Q}.$$

We will set ourselves the goal of finding $\mathbf{F} \in \mathbb{F}_q^{(m+\ell_1) \times (m+\ell_1)}$ and scalars $\mu_{s,i} \in \mathbb{F}_q$ such that

$$\mathbf{FB}_s - \sum_{i=1}^{km} \mu_{s,i} \mathbf{B}_i = \mathbf{P} \begin{bmatrix} \mathbf{0} & * \\ * & * \end{bmatrix} \mathbf{Q} \in \mathbb{F}_q^{(m+\ell_1) \times (n+\ell_2)} \quad \text{for } s = 1, \dots, km, \quad (1)$$

for some invertible $\mathbf{P}$ and $\mathbf{Q}$. We will call this a “MinRank-like problem” for reasons that will be clear soon. At present, two questions remain:

- Q1. How do we solve (1)?
- Q2. How does solving (1) help recover $\mathcal{C}_{\text{mat}}$?

3.3 Applying Kernel Search (Q1)

We start with the first question. The ACDGV scheme chooses small ℓ_1, ℓ_2 to yield competitive parameters which means the right-hand side of (1) has low rank. A first attempt is therefore to find $\mathbf{F}$ and scalars $\mu_{s,i} \in \mathbb{F}_q$ so that $\mathbf{FB}_s - \sum_i \mu_{s,i} \mathbf{B}_i$ is of low rank for all $s = 1, \dots, km$. This resembles a MinRank instance, so one might try to adapt existing MinRank solvers. Solving the km problems independently, however, yields too many spurious solutions. Indeed, the low-rank matrices share structure and we need to exploit this fact.

Write the right-hand side of (1) as

$$\mathbf{P} \begin{bmatrix} \mathbf{0} & * \\ * & * \end{bmatrix} \mathbf{Q} = \mathbf{P} \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ * & * \end{bmatrix} \mathbf{Q} + \mathbf{P} \begin{bmatrix} \mathbf{0} & * \\ \mathbf{0} & * \end{bmatrix} \mathbf{Q} \in \mathbb{F}_q^{(m+\ell_1) \times (n+\ell_2)}$$

where the top-left $\mathbf{0}$ in each matrix is of size $m \times n$. We see the first summand has columns in the span of the last ℓ_1 columns of $\mathbf{P}$ and the second summand has rows in the span of the last ℓ_2 rows of $\mathbf{Q}$. Hence the s th equation can be written as

$$\mathbf{FB}_s - \sum_i \mu_{s,i} \mathbf{B}_i = \mathbf{L}_s^{(1)} + \mathbf{L}_s^{(2)}$$

where, over $s = 1, \dots, km$, the $\mathbf{L}_s^{(1)}$'s share the same ℓ_1 -dimensional column space and the $\mathbf{L}_s^{(2)}$'s share the same ℓ_2 -dimensional row space. Equivalently, the

$\mathbf{L}_s^{(1)}$'s (resp. the $\mathbf{L}_s^{(2)}$'s) share a large left (resp. right) null space. If we guess vectors $\mathbf{v}$ and $\mathbf{w}$ in these respective spaces, then

$$\mathbf{v}^\top \left(\mathbf{F}\mathbf{B}_s - \sum_i \mu_{s,i} \mathbf{B}_i \right) \mathbf{w} = 0 \quad \text{for all } s = 1, \dots, km.$$

Stacking b_1 such left guesses into $\mathbf{V} \in \mathbb{F}_q^{(m+\ell_1) \times b_1}$ and b_2 right guesses into $\mathbf{W} \in \mathbb{F}_q^{(n+\ell_2) \times b_2}$ yields

$$\mathbf{V}^\top \left(\mathbf{F}\mathbf{B}_s - \sum_i \mu_{s,i} \mathbf{B}_i \right) \mathbf{W} = \mathbf{0} \quad \text{for all } s = 1, \dots, km. \quad (2)$$

These are $b_1 b_2 km$ scalar equations and if we have sufficient equations then we can hope to solve (1) for $\mathbf{F}$ and $\mu_{s,i}$. The above idea is an adaptation of the kernel-search MinRank solver [9]. A caveat is that (2) turns out to have many spurious solutions for $\mathbf{F}$. But we will show that it is still possible to recover an equivalent $\mathcal{C}_{\text{mat}}$ with this information.

3.4 Recovering $\mathcal{C}_{\text{mat}}$ (Q2)

Proposition 2 (Section 5) states that if we know the span of the last ℓ_1 columns of $\mathbf{P}$, we can remove the ℓ_1 random rows from the basis matrices $\mathbf{B}_1, \dots, \mathbf{B}_{km}$ of $\mathcal{C}_{\text{pub}}$. That is, we get a new instance $\mathcal{C}'_{\text{pub}}$ with $\ell_1 = 0$. Similarly, if we know the span of the last ℓ_2 rows of $\mathbf{Q}$, we can get a new instance with $\ell_2 = 0$. (The idea of the proof is to project the basis matrices on the orthogonal complement of these spaces.) Once we have an instance with $\ell_1 = 0$ and $\ell_2 = 0$, the problem is solved.

First suppose we know a solution to the original system (1). In the simple $\ell_1 = 0$ case, the right-hand side of (1) looks like $\mathbf{P} [\mathbf{0} \ast] \mathbf{Q}$ whose rows reveal the row span of the last ℓ_2 rows of $\mathbf{Q}$, allowing us to remove the ℓ_2 random columns. Similarly, if $\ell_2 = 0$, we can remove the ℓ_1 random rows. When both $\ell_1 \neq 0$ and $\ell_2 \neq 0$, the right-hand side does not directly reveal either space. Appendix C shows it is possible to still recover the first space from $\mathbf{F}$, but, as mentioned earlier, our kernel-search method to solve (1) yields many spurious solutions for $\mathbf{F}$ and so we cannot use this. Recall the actual kernel-search system (2):

$$\mathbf{V}^\top \left(\mathbf{F}\mathbf{B}_s - \sum_i \mu_{s,i} \mathbf{B}_i \right) \mathbf{W} = \mathbf{0} \quad \text{for all } s = 1, \dots, km. \quad (2 \text{ revisited})$$

Experiments suggest that all solutions to (2) turn out to be, w.h.p., of the form $(\mathbf{F}' + \mathbf{K}, \{\mu'_{s,i}\}_{(s,i)=(1,1)}^{(km,km)})$ where the matrix $\mathbf{K}$ satisfies $\mathbf{V}^\top \mathbf{K} = \mathbf{0}$ and $(\mathbf{F}', \{\mu'_{s,i}\}_{(s,i)=(1,1)}^{(km,km)})$ is a solution to (1). Thus, the obtained $\mathbf{F}$ is corrupted by the addition of $\mathbf{K}$. We next show how to still recover an equivalent $\mathcal{C}_{\text{mat}}$.

Removing random columns If we substitute the solution into (2) but do not right-multiply by $\mathbf{W}$, we get

$$\begin{aligned}\mathbf{V}^\top \left((\mathbf{F}' + \mathbf{K})\mathbf{B}_s - \sum_i \mu'_{s,i} \mathbf{B}_i \right) &= \mathbf{V}^\top \left(\mathbf{F}'\mathbf{B}_s - \sum_i \mu'_{s,i} \mathbf{B}_i \right) = \mathbf{V}^\top \mathbf{P} \begin{bmatrix} \mathbf{0} & * \\ * & * \end{bmatrix} \mathbf{Q} \\ &= [\mathbf{0} \ *] \mathbf{Q}.\end{aligned}$$

The first equality follows from $\mathbf{V}^\top \mathbf{K} = \mathbf{0}$, the second from (1), and the third follows since, if $\mathbf{V}$ is guessed correctly, its columns are orthogonal to the last ℓ_1 columns of $\mathbf{P}$. This reveals the span of the last ℓ_2 rows of $\mathbf{Q}$, so that we can remove the random columns.

Removing random rows Let the resulting instance $\mathcal{C}'_{\text{pub}}$ (with $\ell_2 = 0$) have basis⁷ $\mathbf{B}'_s = \mathbf{P}' \begin{bmatrix} \mathbf{A}_s \\ * \end{bmatrix} \mathbf{Q}' \in \mathbb{F}_q^{(m+\ell_1) \times n}$ for $s = 1, \dots, km$. The scalars $\mu'_{s,i}$'s obtained from the previous solution defines a linear map on $\mathcal{C}_{\text{pub}}$ which maps the s th basis matrix $\mathbf{B}'_s$ to $\sum_{i=1}^{km} \mu'_{s,i} \mathbf{B}'_i$. In the underlying matrix code $\mathcal{C}_{\text{mat}}$, the basis matrix $\mathbf{A}_s$ is mapped to $\sum_{i=1}^{km} \mu'_{s,i} \mathbf{A}_i$. This actually represents a multiplication-by- t map for some $t \in \mathbb{F}_{q^m}$ in the “unexpanded” $\mathbb{F}_{q^m}$ -linear code $\mathcal{C}_{\text{vec}}$. Set up the linear system,

$$\mathbf{F}^{(1)} \mathbf{B}'_s = \mathbf{F}^{(2)} \sum_{i=1}^{km} \mu'_{s,i} \mathbf{B}'_i \quad \text{for all } s = 1, \dots, km,$$

in the unknowns $\mathbf{F}^{(1)}, \mathbf{F}^{(2)} \in \mathbb{F}_q^{m \times (m+\ell_1)}$. Experiments show it has solutions w.h.p. of the form $\mathbf{F}^{(1)} = [\mathbf{S} \ \mathbf{T} \ \mathbf{0}] \mathbf{P}'^{-1}$, and $\mathbf{F}^{(2)} = [\mathbf{S} \ \mathbf{0}] \mathbf{P}'^{-1}$ where $\mathbf{S}$ is invertible and $\mathbf{T}$ represents multiplication by $t \in \mathbb{F}_{q^m}$. Then $\mathbf{F}^{(1)}$ kills the random rows and applies the multiplication-by- t map, while $\mathbf{F}^{(2)}$ only kills the random rows. Thus, $\mathbf{F}^{(1)} \mathcal{C}'_{\text{pub}}$ (and also $\mathbf{F}^{(2)} \mathcal{C}'_{\text{pub}}$) equals $\mathcal{C}_{\text{mat}}$ up to left- and right-multiplication by invertible matrices.

4 Basis Recovery

The attacks in the next section use Algorithm 1 (**BasisRecover**) as a subroutine. Otherwise, this section is independent and can be read on its own.

Suppose we are given a matrix code $\mathcal{C}_{\text{mat}} \subseteq \mathbb{F}_q^{m \times n}$ where $\mathcal{C}_{\text{mat}} = \text{ext}_\gamma(\mathcal{C}_{\text{vec}})$ for an $[n, k]_{\mathbb{F}_{q^m}}$ code $\mathcal{C}_{\text{vec}} \subseteq \mathbb{F}_{q^m}^n$ and an $\mathbb{F}_q$ -basis γ of $\mathbb{F}_{q^m}$. Then **BasisRecover**, on input $\mathcal{C}_{\text{mat}}$, returns a basis $\gamma' \in \mathbb{F}_{q^m}^m$ such that $\gamma' = b\gamma^{[j]}$ for some $b \in \mathbb{F}_{q^m} \setminus \{0\}$ and $j \in \{0, \dots, m-1\}$. One can then show $\mathcal{C}_{\text{vec}}^{[j]} = \text{ext}_{\gamma'}^{-1}(\mathcal{C}_{\text{mat}})$, so the algorithm allows one to recover $\mathcal{C}_{\text{vec}}$ from $\mathcal{C}_{\text{mat}}$ up to the Frobenius automorphism.

The intuition for **BasisRecover** is as follows. The space $\text{Stab}(\mathcal{C}_{\text{mat}}) \subseteq \mathbb{F}_q^{m \times m}$ includes matrices that represent multiplication-by- b maps over $\mathbb{F}_q$ for $b \in \mathbb{F}_{q^m}$

⁷ When $\ell_2 = 0$, we can also omit $\mathbf{Q}' \in \mathbb{F}_q^{n \times n}$ without loss of generality.

with respect to γ . If $\dim \text{Stab}(\mathcal{C}_{\text{mat}}) = m$, then $\text{Stab}(\mathcal{C}_{\text{mat}})$ is exactly this space of matrices. Let $\alpha \in \mathbb{F}_{q^m}$ have minimal polynomial $f(x)$ of degree m . Then the multiplication-by- α matrix with respect to the basis $1, \alpha, \dots, \alpha^{m-1}$ is the companion matrix $\mathbf{C}_f$ of $f(x)$. With respect to a different basis γ , the matrix is $\mathbf{P}^{-1}\mathbf{C}_f\mathbf{P}$ where $\mathbf{P}$ is the change-of-basis matrix such that $(1, \alpha, \dots, \alpha^{m-1})\mathbf{P} = \gamma$. The space of multiplication maps is spanned by the powers of the multiplication-by- α map, so $\text{Stab}(\mathcal{C}_{\text{mat}}) = \mathbb{F}_q[\mathbf{P}^{-1}\mathbf{C}_f\mathbf{P}]$. Finding a matrix in $\text{Stab}(\mathcal{C}_{\text{mat}})$ with minimal polynomial $f(x)$ yields $\mathbf{P}^{-1}\mathbf{C}_f\mathbf{P}$ (up to a Frobenius power), from which we can recover (up to some appropriate equivalences) $\mathbf{P}$ and hence γ .

We now present the technical proof, starting with some lemmas. We note Couvreur, Debris-Alazard and Gaborit in [5, Lemma 26] prove similar results.

Lemma 1 (Linear Algebra). *Let $\mathbf{B} \in \mathbb{F}_q^{m \times m}$ and let $f(x) \in \mathbb{F}_q[x]$ be its minimal polynomial.*

- (a) *The following statements are equivalent.*
 - (i) *$f(x)$ has degree m (or equivalently, the characteristic polynomial of $\mathbf{B}$ is the same as its minimal polynomial).*
 - (ii) *$\mathbf{B}$ is similar to the companion matrix of $f(x)$.*
 - (iii) *A matrix commutes with $\mathbf{B}$ iff it is a polynomial in $\mathbf{B}$.*
 - (iv) *There exists a $\mathbf{v} \in \mathbb{F}_q^m$ such that $(\mathbf{v}, \mathbf{B}\mathbf{v}, \dots, \mathbf{B}^{m-1}\mathbf{v})$ is a basis of $\mathbb{F}_q^m$.*
- (b) *The following statements are equivalent.*
 - (i) *$f(x)$ has degree m and is irreducible (or equivalently, the characteristic polynomial of $\mathbf{B}$ is irreducible).*
 - (ii) *$\mathbb{F}_q[\mathbf{B}] \cong \mathbb{F}_{q^m}$.*
 - (iii) *For all non-zero $\mathbf{v} \in \mathbb{F}_q^m$, $(\mathbf{v}, \mathbf{B}\mathbf{v}, \dots, \mathbf{B}^{m-1}\mathbf{v})$ is a basis of $\mathbb{F}_q^m$.*

See Appendix D for the proof.

Lemma 2 (Matrix Space of Multiplication Maps). *Define $\mathcal{M}$ to be the set of matrices representing the multiplication map by some element in $\mathbb{F}_{q^m}$ with respect to the $\mathbb{F}_q$ -basis $\gamma = (\gamma_1, \dots, \gamma_m)^\top \in \mathbb{F}_{q^m}^m$. That is,*

$$\mathcal{M} = \{\mathbf{M} \in \mathbb{F}_q^{m \times m} \mid \text{for some } b \in \mathbb{F}_{q^m}, \text{ for all } x \in \mathbb{F}_{q^m}, \mathbf{M} \text{ext}_\gamma(x) = \text{ext}_\gamma(bx)\}.$$

Then,

- (a) $\mathcal{M} \cong \mathbb{F}_{q^m}$.
- (b) *For any $b \in \mathbb{F}_{q^m}$, the matrix representing the $\mathbb{F}_q$ -linear map $\mathbb{F}_{q^m} \rightarrow \mathbb{F}_{q^m}$ given by $x \mapsto bx$ with respect to the basis γ is $(\text{ext}_\gamma(b\gamma_1), \dots, \text{ext}_\gamma(b\gamma_m)) \in \mathbb{F}_q^{m \times m}$. Hence $\mathcal{M} = \{(\text{ext}_\gamma(b\gamma_1), \dots, \text{ext}_\gamma(b\gamma_m)) \in \mathbb{F}_q^{m \times m} \mid b \in \mathbb{F}_{q^m}\}$.*
- (c) *Let $\alpha \in \mathbb{F}_{q^m}$ and suppose its minimal polynomial $f(x) \in \mathbb{F}_q[x]$ has degree m . Then $\mathcal{M} = \mathbb{F}_q[\mathbf{P}^{-1}\mathbf{C}_f\mathbf{P}]$ where $\mathbf{C}_f$ is the companion matrix of $f(x)$ and $\mathbf{P} \in \mathbb{F}_q^{m \times m}$ is the change-of-basis matrix such that $(1, \alpha, \alpha^2, \dots, \alpha^{m-1})\mathbf{P} = (\gamma_1, \dots, \gamma_m)$. In particular, if $\gamma = (1, \alpha, \alpha^2, \dots, \alpha^{m-1})^\top$, then $\mathcal{M} = \mathbb{F}_q[\mathbf{C}_f]$.*
- (d) *There exists a $\mathbf{Q} \in \mathbb{F}_q^{m \times m}$ such that $\mathbf{Q}\mathbf{M}\mathbf{Q}^{-1} = \mathbf{M}^q$ for all $\mathbf{M} \in \mathcal{M}$ and $\mathbf{Q}^m = \mathbf{I}$ (where $\mathbf{M}^q = \mathbf{M} \cdots \mathbf{M}$ denotes the product of q copies of $\mathbf{M}$ under matrix multiplication).*

- Proof.* (a) If $\mathbf{M}$ represents the multiplication-by- b map and $\mathbf{M}'$ represents the multiplication-by- b' map for some $b, b' \in \mathbb{F}_{q^m}$, then it is clear that the matrices $\mathbf{M} + \mathbf{M}'$ and $\mathbf{M}\mathbf{M}'$ represent multiplication-by- $(b+b')$ and multiplication-by- (bb') respectively. Hence $\mathcal{M}$ with the operations of matrix addition and multiplication is isomorphic to $\mathbb{F}_{q^m}$.
- (b) Let $\mathbf{M} \in \mathcal{M}$ and $b \in \mathbb{F}_{q^m}$ be such that, for all $x \in \mathbb{F}_{q^m}$, $\mathbf{M} \text{ext}_\gamma(x) = \text{ext}_\gamma(bx)$. Write $x = \sum_i x_i \gamma_i$ where $x_i \in \mathbb{F}_q$. Then $\mathbf{M} \text{ext}_\gamma(x) = \text{ext}_\gamma(bx) = \text{ext}_\gamma(\sum_i b x_i \gamma_i) = \sum_i x_i \text{ext}_\gamma(b \gamma_i) = (\text{ext}_\gamma(b \gamma_1), \dots, \text{ext}_\gamma(b \gamma_m)) \text{ext}_\gamma(x)$.
- (c) Let $\mathbf{M} \in \mathcal{M}$ represent the multiplication-by- b map with respect to the basis γ for some $b \in \mathbb{F}_{q^m}$. Likewise, let $\mathbf{M}' \in \mathbb{F}_q^{m \times m}$ represent the same map but with respect to the basis $\alpha := (1, \alpha, \alpha^2, \dots, \alpha^{m-1})^\top$. It turns out $\mathbf{M} = \mathbf{P}^{-1} \mathbf{M}' \mathbf{P}$ where $\mathbf{P}$ is the change-of-basis matrix such that $\alpha^\top \mathbf{P} = \gamma^\top$. This follows because

$$\begin{aligned}
\gamma^\top \mathbf{P}^{-1} \mathbf{M}' \mathbf{P} &= \alpha^\top \mathbf{M}' \mathbf{P} \\
&= \alpha^\top (\text{ext}_\alpha(b), \text{ext}_\alpha(b\alpha), \dots, \text{ext}_\alpha(b\alpha^{m-1})) \mathbf{P} \quad (\text{Lemma 2(b)}) \\
&= (b, b\alpha, \dots, b\alpha^{m-1}) \mathbf{P} \\
&= b(\gamma_1, \gamma_2, \dots, \gamma_m) \\
&= \gamma^\top (\text{ext}_\gamma(b \gamma_1), \dots, \text{ext}_\gamma(b \gamma_m)) \\
&= \gamma^\top \mathbf{M}. \quad (\text{Lemma 2(b)})
\end{aligned}$$

Set $b = \alpha$, so that $\mathbf{M}$ and $\mathbf{M}'$ represent multiplication-by- α maps. Thus, $\mathcal{M} = \mathbb{F}_q[\mathbf{M}]$ since $\mathcal{M} \cong \mathbb{F}_{q^m}$ and $\mathbb{F}_{q^m} = \mathbb{F}_q[\alpha]$. Further, $\mathbf{M}' = \mathbf{C}_f$ by definition of the companion matrix and so $\mathbf{M} = \mathbf{P}^{-1} \mathbf{C}_f \mathbf{P}$.

- (d) The Frobenius automorphism $x \mapsto x^q$ on $\mathbb{F}_{q^m}$ is an $\mathbb{F}_q$ -linear map. Thus there is a matrix representation $\mathbf{Q}$ of this map with respect to the basis γ . Hence $\mathbf{Q}^m = \mathbf{I}$. Let $\mathbf{M} \in \mathcal{M}$ represent multiplication by some $b \in \mathbb{F}_{q^m}$ with respect to the basis γ . Then, for all $x \in \mathbb{F}_{q^m}$, we have $\mathbf{Q} \mathbf{M} \mathbf{Q}^{-1} \text{ext}_\gamma(x) = \mathbf{Q} \mathbf{M} \text{ext}_\gamma(x^{[-1]}) = \mathbf{Q} \text{ext}_\gamma(b x^{[-1]}) = \text{ext}_\gamma(b^q x) = \mathbf{M}^q \text{ext}_\gamma(x)$.

□

Remark 3 (Deterministic BasisRecover). Since steps 2 and 5 in Algorithm 1 are probabilistic, **BasisRecover** returns a basis w.h.p. but not always, i.e., it is a one-sided Monte Carlo algorithm. We chose to describe this version for simplicity, but it can be made deterministic while keeping it polynomial-time as follows.

Step 2 finds a matrix $\mathbf{B} \in \mathcal{M} \cong \mathbb{F}_{q^m}$ such that $\mathcal{M} = \mathbb{F}_q[\mathbf{B}]$. Lenstra [15, §2] describes the following method. Prime factorize m as $p_1^{t_1} p_2^{t_2} \dots p_r^{t_r}$. For each maximal prime power $p_i^{t_i}$, find $\mathbf{B}_i \in \mathcal{M}$ such that $\mathbb{F}_q[\mathbf{B}_i]$ is isomorphic to the subfield $\mathbb{F}_{q^{p_i^{t_i}}}$ where we let $p := p_i$ and $t := t_i$ to ease notation. To obtain such a $\mathbf{B}_i$, first compute a basis $\mathbf{C}_1, \dots, \mathbf{C}_{p^t} \in \mathcal{M}$ spanning $\mathbb{F}_{q^{p^t}}$. This amounts to solving $x^{q^{p^t}} - x = 0$ in $\mathcal{M}$, which is just linear algebra. At least one $\mathbf{C}_j$ lies outside the subfield $\mathbb{F}_{q^{p^{t-1}}}$, i.e., does not satisfy $x^{q^{p^{t-1}}} - x = 0$. Hence $\mathbb{F}_q[\mathbf{C}_j] \cong \mathbb{F}_{q^{p^t}}$ and so pick $\mathbf{B}_i := \mathbf{C}_j$. Finally, by [19, Lemma 3.3], $\mathbf{B} := \sum_i \mathbf{B}_i$ satisfies $\mathcal{M} = \mathbb{F}_q[\mathbf{B}]$.

Algorithm 1: BasisRecover($\mathbf{A}_1, \dots, \mathbf{A}_{km}$)

Input: A basis $\mathbf{A}_1, \dots, \mathbf{A}_{km}$ of a matrix code $\mathcal{C}_{\text{mat}} \subseteq \mathbb{F}_q^{m \times n}$.
Output: If $\dim \text{Stab}(\mathcal{C}_{\text{mat}}) = m$ and there exists an $[n, k]_{\mathbb{F}_{q^m}}$ code $\mathcal{C}_{\text{vec}} \subseteq \mathbb{F}_{q^m}^n$ and an $\mathbb{F}_q$ -basis γ of $\mathbb{F}_{q^m}$ such that $\mathcal{C}_{\text{mat}} = \text{ext}_{\gamma}(\mathcal{C}_{\text{vec}})$, return $b\gamma^{[j]}$ for $b \in \mathbb{F}_{q^m} \setminus \{0\}$ and $j \in \{0, \dots, m-1\}$. Else return $\perp$.
(See Remark 3 on how to make this algorithm deterministic.)

- 1 Set $\mathcal{M} \leftarrow \text{Stab}(\mathcal{C}_{\text{mat}}) \subseteq \mathbb{F}_q^{m \times m}$. Return $\perp$ if $\dim \mathcal{M} \neq m$.
- 2 Find $\mathbf{B} \in \mathcal{M}$ such that $\mathcal{M} = \mathbb{F}_q[\mathbf{B}]$: repeatedly draw $\mathbf{B} \xleftarrow{\$} \mathcal{M}$ until its minimal polynomial $f(x)$ over $\mathbb{F}_q$ has degree m . Return $\perp$ after at most $(1 - 2q^{-m/2} + 2q^{-m})^{-1} \leq 2$ tries.
- 3 Check $f(x)$ is irreducible, e.g., using Rabin's test [20]; else return $\perp$.
- 4 Sample $\mathbf{v} \xleftarrow{\$} \mathbb{F}_q^m \setminus \{\mathbf{0}\}$ and set $\mathbf{R} \leftarrow (\mathbf{v}, \mathbf{B}\mathbf{v}, \dots, \mathbf{B}^{m-1}\mathbf{v}) \in \mathbb{F}_q^{m \times m}$.
- 5 Find a root α of $f(x)$ in $\mathbb{F}_{q^m}$, e.g., using Cantor–Zassenhaus [4] (or Lenstra [15], see Remark 3). However, this is trivial if we construct $\mathbb{F}_{q^m}$ as $\frac{\mathbb{F}_q[x]}{\langle f(x) \rangle}$; see Remark 4.
- 6 Set $(\gamma'_1, \dots, \gamma'_m) \leftarrow (1, \alpha, \alpha^2, \dots, \alpha^{m-1})\mathbf{R}^{-1} \in \mathbb{F}_{q^m}^m$.
- 7 **return** $(\gamma'_1, \dots, \gamma'_m)$.

Step 5 finds a root of $f(x) \in \mathbb{F}_q[x]$ in $\mathbb{F}_{q^m}$ where $f(x)$ is irreducible over $\mathbb{F}_q$ and has degree m . This is an instance of the polynomial factorization problem which can be solved, e.g., using the probabilistic polynomial-time algorithm of Cantor–Zassenhaus [4]. However, our problem is a special case and is equivalent to finding an isomorphism between $\frac{\mathbb{F}_q[x]}{\langle f(x) \rangle}$ and $\mathbb{F}_{q^m}$, i.e., between two representations of the same finite field. This is in fact easier as Lenstra [15] gave a deterministic polynomial-time algorithm for this. More recent algorithms that are faster (but probabilistic) are described in [3].

Remark 4 (Trivial Root-Finding). For attacking the ACDGV scheme, the root-finding in Step 5 of **BasisRecover** is trivial since we can construct $\mathbb{F}_{q^m}$ as $\mathbb{F}_q[x]/\langle f(x) \rangle$, where the residue class of x is the desired root. We can do this since messages in the ACDGV scheme are in $\mathbb{F}_q^{km}$ and the particular representation of $\mathbb{F}_{q^m}$ used during decryption is irrelevant.

Remark 5 (The $\dim \text{Stab}(\mathcal{C}_{\text{mat}}) > m$ case). A complete characterization of stabilizer algebras that contain an isomorphic copy of $\mathbb{F}_{q^m}$ is given by Couvreur, Debris-Alazard and Gaborit in [5, Prop. 31]. Hence, we expect it possible to extend **BasisRecover** to the general case using their result.

Remark 6 (Sampled Stabilizer Algebra). The computation of the stabilizer algebra dominates the polynomial cost of all our algorithms, so we also consider a cheaper way to compute it. Let $\mathcal{C}_{\text{mat}} \subseteq \mathbb{F}_q^{m \times n}$ be a matrix code of dimension K . To compute $\text{Stab}(\mathcal{C}_{\text{mat}})$, we need to solve the following linear system in the

unknowns $\mathbf{T} \in \mathbb{F}_q^{m \times m}$ and $\mu_{s,i} \in \mathbb{F}_q$,

$$\mathbf{T}\mathbf{A}_s - \sum_{i=1}^K \mu_{s,i} \mathbf{A}_i = \mathbf{0} \quad \text{for } s = 1, \dots, \ell, \quad (3)$$

where $\ell := K$. This system has ℓmn scalar equations and $m^2 + K\ell$ unknowns, and typically, the former number is much higher. Thus, instead of using all K basis equations, we can choose the smallest ℓ such that the number of scalar equations is at least the number of unknowns. We make the heuristic assumption that w.h.p. this system will still have the same rank, and hence the same solution space $\text{Stab}(\mathcal{C}_{\text{mat}})$, thereby reducing our computational cost.

Note if we instead pick sufficiently many random linear combinations of the scalar equations, then the resulting system will provably have the same rank as the full system with at least constant (and thereby high after a few repetitions) probability. However, this also produces a dense system that is more expensive to solve. Thus, we select a subset of the original basis equations to preserve sparsity, though this step relies on a heuristic. To ease notation, we will call the solution space thus obtained $\text{Stab}_{\text{samp}}(\mathcal{C}_{\text{mat}})$, as defined below.

Definition 7 (Sampled Stabilizer Algebra $\text{Stab}_{\text{samp}}(\mathcal{C}_{\text{mat}})$ -distribution). Let $\mathcal{C}_{\text{mat}} \subseteq \mathbb{F}_q^{m \times n}$ be a matrix code of dimension K . Let ℓ be the smallest integer such that $\ell mn \geq m^2 + K\ell$, i.e., $\ell = \lceil m^2/(mn - K) \rceil$ and sample a random ℓ -dimensional $\mathcal{C}$ subcode of $\mathcal{C}_{\text{mat}}$. Then $\text{Stab}_{\text{samp}}(\mathcal{C}_{\text{mat}})$ returns

$$\{\mathbf{T} \in \mathbb{F}_q^{m \times m} \mid \mathbf{T}\mathcal{C} \subseteq \mathcal{C}_{\text{mat}}\}$$

Note we always have $\text{Stab}(\mathcal{C}_{\text{mat}}) \subseteq \mathcal{S}$ for $\mathcal{S} \xleftarrow{\$} \text{Stab}_{\text{samp}}(\mathcal{C}_{\text{mat}})$.

Theorem 1 (BasisRecover).

- (i) Algorithm 1 succeeds with at least constant probability and runs in polynomial time.
- (ii) Let $0 < k < m$ and $m = n$. Define the following heuristic.
 - (a) If $\mathcal{S} \xleftarrow{\$} \text{Stab}_{\text{samp}}(\mathcal{C}_{\text{mat}})$, then $\dim \mathcal{S} = \dim \text{Stab}(\mathcal{C}_{\text{mat}})$. Given Heuristic (a), the algorithm can be modified to run in at most $m^6 \mathbb{F}_q$ -operations (up to a constant factor). Without this heuristic, the complexity is at most $k(m - k)m^6 \mathbb{F}_q$ -operations. (See Corollary 3 in Appendix F for the cost in terms of ω .)

Proof. Suppose $\mathcal{C}_{\text{mat}} = \text{ext}_{\gamma}(\mathcal{C}_{\text{vec}})$ and $\dim \text{Stab}(\mathcal{C}_{\text{mat}}) = m$. For each $b \in \mathbb{F}_q^m$, multiplication by b is an $\mathbb{F}_q$ -linear map on $\mathbb{F}_q^m$. Let $\mathcal{M} \subseteq \mathbb{F}_q^{m \times m}$ be the space of all matrices representing such a map with respect to the (unknown) basis γ . For each $\mathbf{M} \in \mathcal{M}$, we have $\mathbf{M}\mathcal{C}_{\text{mat}} \subseteq \mathcal{C}_{\text{mat}}$. Thus, $\mathcal{M} \subseteq \text{Stab}(\mathcal{C}_{\text{mat}})$ but since both spaces have dimension m , these spaces are equal. Hence $\mathcal{M}$ can be found by computing $\text{Stab}(\mathcal{C}_{\text{mat}})$.

The next step finds a matrix $\mathbf{B}$ such that $\mathcal{M} = \mathbb{F}_q[\mathbf{B}]$. Suppose $\dim \text{Stab}(\mathcal{C}_{\text{mat}})$ has dimension m . Then $\mathcal{M}$ is isomorphic to $\mathbb{F}_q^m$ and so $\mathbf{B}$ must have an irreducible minimal polynomial of degree m . The number of degree- m irreducible

polynomials over $\mathbb{F}_q$ is at least $\frac{q^m}{m} - \frac{q}{m(q-1)}(q^{m/2} - 1)$ [16, p. 142, Exercise 3.27] and so the number of elements in $\mathbb{F}_{q^m}$ with such a minimal polynomial is at least m times as many. Thus, if we randomly sample from $\mathcal{M}$, such a $\mathbf{B}$ will be found with probability at least $1 - \frac{q}{q-1}(q^{-m/2} - q^{-m}) > 1 - 2q^{-m/2} + 2q^{-m} \geq 1/2$.

Let $f(x)$ be the irreducible minimal polynomial of $\mathbf{B}$. Since $\mathbf{B}$ is similar to its companion matrix $\mathbf{C}_f$ (Lemma 1(a)(ii)), there exists an $\mathbf{R} \in \mathbb{F}_q^{m \times m}$ such that $\mathbf{C}_f = \mathbf{R}^{-1}\mathbf{B}\mathbf{R}$. As shown in [5, Lemma 26(v)], such an $\mathbf{R}$ can be found as follows. Choose any nonzero $\mathbf{v} \in \mathbb{F}_q^m$. Then $\mathbf{v}, \mathbf{B}\mathbf{v}, \dots, \mathbf{B}^{m-1}\mathbf{v}$ form a basis of $\mathbb{F}_q^m$ (Lemma 1(b)(iii)). Define $\mathbf{R}$ as the change-of-basis matrix from the standard basis to this one, i.e., $\mathbf{R} = (\mathbf{v}, \mathbf{B}\mathbf{v}, \dots, \mathbf{B}^{m-1}\mathbf{v}) \in \mathbb{F}_q^{m \times m}$. Then $\mathbf{R}^{-1}\mathbf{B}\mathbf{R}$ is exactly the companion matrix $\mathbf{C}_f$ of $f(x)$.

Let $\alpha \in \mathbb{F}_{q^m}$ be a root of $f(x)$. By Lemma 2(c), $\mathcal{M} = \mathbb{F}_q[\mathbf{P}^{-1}\mathbf{C}_f\mathbf{P}]$ where $\mathbf{P}$ is the change-of-basis matrix such that $(1, \alpha, \alpha^2, \dots, \alpha^{m-1})\mathbf{P} = (\gamma_1, \dots, \gamma_m)$. Since $\mathbf{B}$ and $\mathbf{P}^{-1}\mathbf{C}_f\mathbf{P}$ are roots of the same minimal polynomial in a space isomorphic to $\mathbb{F}_{q^m}$, we have $\mathbf{B} = \mathbf{P}^{-1}\mathbf{C}_f^{[j]}\mathbf{P}$ for some unknown j , where $\mathbf{C}_f^{[j]}$ denotes the product of q^j copies of $\mathbf{C}_f$ under matrix multiplication. The latter can be written as $\mathbf{Q}^j\mathbf{P}^{-1}\mathbf{C}_f\mathbf{P}\mathbf{Q}^{-j}$ (Lemma 2(d)) where $\mathbf{Q}$ represents the Frobenius automorphism with respect to the basis γ . But also $\mathbf{B} = \mathbf{R}\mathbf{C}_f\mathbf{R}^{-1}$, so

$$\mathbf{R}\mathbf{C}_f\mathbf{R}^{-1} = \mathbf{Q}^j\mathbf{P}^{-1}\mathbf{C}_f\mathbf{P}\mathbf{Q}^{-j} \implies (\mathbf{P}\mathbf{Q}^{-j}\mathbf{R})\mathbf{C}_f = \mathbf{C}_f(\mathbf{P}\mathbf{Q}^{-j}\mathbf{R}).$$

Since $\mathbf{P}\mathbf{Q}^{-j}\mathbf{R}$ commutes with $\mathbf{C}_f$ we have $\mathbf{P}\mathbf{Q}^{-j}\mathbf{R} = g(\mathbf{C}_f)$ for some nonzero polynomial $g(x) \in \mathbb{F}_q[x]$ (Lemma 1(a)(iii)). Let $b \in \mathbb{F}_{q^m} \setminus \{0\}$ be such that $g(\mathbf{C}_f)$ represents the multiplication-by- b map with respect to the basis $\alpha := (1, \alpha, \alpha^2, \dots, \alpha^{m-1})^\top$. Then for $x \in \mathbb{F}_{q^m}$,

$$\begin{aligned} \mathbf{R} \text{ext}_\alpha(x) &= \mathbf{Q}^j\mathbf{P}^{-1}g(\mathbf{C}_f) \text{ext}_\alpha(x) = \mathbf{Q}^j\mathbf{P}^{-1} \text{ext}_\alpha(bx) = \mathbf{Q}^j \text{ext}_\gamma(bx) \\ &= \text{ext}_\gamma(b^{[j]}x^{[j]}) \\ &= \text{ext}_{b^{-1}\gamma^{[-j]}}(x). \end{aligned}$$

The third equality uses $\mathbf{P}^{-1} \text{ext}_\alpha(x) = \text{ext}_\gamma(x)$ which follows from

$$\gamma^\top \mathbf{P}^{-1} \text{ext}_\alpha(x) = \alpha^\top \text{ext}_\alpha(x) = x = \gamma^\top \text{ext}_\gamma(x).$$

The last equality follows because, if we let $(y_1, \dots, y_m)^\top := \text{ext}_\gamma(b^{[j]}x^{[j]}) \in \mathbb{F}_q^m$, then

$$b^{[j]}x^{[j]} = \sum_i y_i \gamma_i \implies x = \sum_i y_i b^{-1} \gamma_i^{[-j]}.$$

Thus $\mathbf{R}$ is the change-of-basis matrix such that

$$b^{-1}(\gamma_1, \dots, \gamma_m)^{[-j]} = (1, \alpha, \alpha^2, \dots, \alpha^{m-1})\mathbf{R}^{-1}.$$

Hence, the algorithm is correct when the input $\mathcal{C}_{\text{mat}}$ is the expansion of an $[n, k]_{\mathbb{F}_{q^m}}$ code with $\dim \text{Stab}(\mathcal{C}_{\text{mat}}) = m$. If the input $\mathcal{C}_{\text{mat}}$ does not satisfy this, then at least one of Steps 1–3 must fail because, otherwise, $\text{Stab}(\mathcal{C}_{\text{mat}}) \cong \mathbb{F}_{q^m}$ (Lemma 1(b)(ii)) which will contradict that $\mathcal{C}_{\text{mat}}$ is not such an expansion.

The most expensive step is the computation of $\text{Stab}(\mathcal{C}_{\text{mat}})$. Corollary 3 in Appendix F shows that this costs at most $k(m-k)m^6$ operations (up to a constant factor) when $\omega = 3$. Given Heuristic (a), we can instead do this computation using fewer basis equations as in Remark 6 in Section 4, and this costs m^6 by the same corollary. $\square$

5 ACDGV Key Recovery

We now describe algorithms to solve the HVCS problem, which can be used to perform key recovery on the ACDGV scheme. All the attacks in this paper rely on the following proposition.

Proposition 2. *Given an instance of the HVCS problem $\mathcal{C}_{\text{pub}}$ (Definition 3), let $\mathcal{C}_{\text{mat}}$ be the matrix code that was input to the hiding transform and $\mathbf{P} \in \mathbb{F}_q^{(m+\ell_1) \times (m+\ell_1)}$ and $\mathbf{Q} \in \mathbb{F}_q^{(n+\ell_2) \times (n+\ell_2)}$ be the (invertible) scrambling matrices.*

Let $\mathcal{L}_1 \subseteq \mathbb{F}_q^{m+\ell_1}$ and $\mathcal{L}_2 \subseteq \mathbb{F}_q^{n+\ell_2}$ be the span of the last ℓ_1 columns of $\mathbf{P}$ (with dimension ℓ_1) and the span of the last ℓ_2 rows of $\mathbf{Q}$ (with dimension ℓ_2) respectively. Draw uniformly random $\mathbf{V} \in \mathbb{F}_q^{(m+\ell_1) \times m}$ and $\mathbf{W} \in \mathbb{F}_q^{(n+\ell_2) \times n}$ such that $\text{colspan}(\mathbf{V}) = \mathcal{L}_1^\perp$ and $\text{colspan}(\mathbf{W}) = \mathcal{L}_2^\perp$.

Then the code $\mathbf{V}^\top \mathcal{C}_{\text{pub}} \in \mathbb{F}_q^{m \times (n+\ell_2)}$ (resp. $\mathcal{C}_{\text{pub}} \mathbf{W} \in \mathbb{F}_q^{(m+\ell_1) \times n}$) is an HVCS instance with $\ell_1 = 0$ (resp. $\ell_2 = 0$) and the same input $\mathcal{C}_{\text{mat}}$ to the hiding transform. In particular, $\mathbf{V}^\top \mathcal{C}_{\text{pub}} \mathbf{W} = \mathbf{S}' \mathcal{C}_{\text{mat}} \mathbf{S} \subseteq \mathbb{F}_q^{m \times n}$ for some invertible and random $\mathbf{S}' \in \mathbb{F}_q^{m \times m}$ and $\mathbf{S} \in \mathbb{F}_q^{n \times n}$.

Proof. Let $\mathbf{P} = [\mathbf{P}' \ \mathbf{P}''] \in \mathbb{F}_q^{(m+\ell_1) \times (m+\ell_1)}$ where $\mathbf{P}''$ denotes the last ℓ_1 columns of $\mathbf{P}$. Pick $\mathbf{V} \in \mathbb{F}_q^{(m+\ell_1) \times m}$ as above. Then for all $i = 1, \dots, km$,

$$\begin{aligned} \mathbf{V}^\top \mathbf{B}_i &= \mathbf{V}^\top \mathbf{P} \begin{bmatrix} \mathbf{A}_i & \mathbf{R}_i \\ \mathbf{R}'_i & \mathbf{R}''_i \end{bmatrix} \mathbf{Q} = \mathbf{V}^\top [\mathbf{P}' \ \mathbf{P}''] \begin{bmatrix} \mathbf{A}_i & \mathbf{R}_i \\ \mathbf{R}'_i & \mathbf{R}''_i \end{bmatrix} \mathbf{Q} = [\mathbf{V}^\top \mathbf{P}' \ 0] \begin{bmatrix} \mathbf{A}_i & \mathbf{R}_i \\ \mathbf{R}'_i & \mathbf{R}''_i \end{bmatrix} \mathbf{Q} \\ &= \mathbf{V}^\top \mathbf{P}' [\mathbf{A}_i \ \mathbf{R}_i] \mathbf{Q}. \end{aligned}$$

By Lemma 3 in Appendix E, $\mathbf{V}^\top \mathbf{P}' \in \mathbb{F}_q^{m \times m}$ is an invertible matrix drawn uniformly at random. Thus, we are reduced to an HVCS instance with $\ell_1 = 0$. The proof for $\mathbf{Q}$ is similar. $\square$

5.1 Naive Key Recovery

Proposition 2 suggests a naive key-recovery algorithm (Algorithm 2) where we guess these ℓ_1 - and ℓ_2 -dimensional subspaces.

Theorem 2 (Naive Key Recovery). *Let $\mathcal{C}_{\text{pub}} \stackrel{\$}{\leftarrow} \text{EMC}(\mathcal{D})$, where $\mathcal{C}_{\text{vec}}$ is the $\mathbb{F}_{q^m}$ -linear code such that $\mathcal{C}_{\text{mat}} := \text{ext}_\gamma(\mathcal{C}_{\text{vec}})$ is the input to the hiding transform (for a random $\mathbb{F}_q$ -basis $\gamma \in \mathbb{F}_{q^m}^m$). Assume $\dim \text{Stab}(\mathcal{C}_{\text{mat}}) = m$.*

- (i) *Algorithm 2 solves the HVCS problem for $\mathcal{D}$ (Definition 3) in time $q^{m\ell_1+n\ell_2}$, ignoring polynomial factors.*

Algorithm 2: NaiveKeyRecover($\mathbf{B}_1, \dots, \mathbf{B}_{km}$)

Input: A basis $\mathbf{B}_1, \dots, \mathbf{B}_{km} \in \mathbb{F}_q^{(m+\ell_1) \times (n+\ell_2)}$ of the matrix code $\mathcal{C}_{\text{pub}}$ sampled from the EMC($\mathcal{D}$)-distribution (Definition 2).
Output: $\mathbf{V} \in \mathbb{F}_q^{(m+\ell_1) \times m}$, $\mathbf{W} \in \mathbb{F}_q^{(n+\ell_2) \times n}$, $\gamma' \in \mathbb{F}_{q^m}^m$ such that γ' is an $\mathbb{F}_q$ -basis of $\mathbb{F}_{q^m}$ and $\mathcal{C}'_{\text{vec}} := \text{ext}_{\gamma'}^{-1}(\mathbf{V}^\top \mathcal{C}_{\text{pub}} \mathbf{W}) \subseteq \mathbb{F}_{q^m}^n$ is an $[n, k]$ code over $\mathbb{F}_{q^m}$.
1 repeat
2 Draw random full-rank matrices $\mathbf{V} \in \mathbb{F}_q^{(m+\ell_1) \times m}$ and $\mathbf{W} \in \mathbb{F}_q^{(n+\ell_2) \times n}$.
3 **for** $i \leftarrow 1$ **to** km **do** $\mathbf{A}'_i \leftarrow \mathbf{V}^\top \mathbf{B}_i \mathbf{W} \in \mathbb{F}_q^{m \times n}$.
4 $\gamma' \leftarrow$ run **BasisRecover** (Algorithm 1) on the matrix code $\langle \mathbf{A}'_1, \dots, \mathbf{A}'_{km} \rangle$.
5 until $\gamma' \neq \perp$.
6 return $(\mathbf{V}, \mathbf{W}, \gamma')$.

Define Heuristic (a) as follows. Let $\mathbf{V} \in \mathbb{F}_q^{(m+\ell_1) \times m}$ and $\mathbf{W} \in \mathbb{F}_q^{(n+\ell_2) \times n}$ be full-rank random matrices. Let $0 < k < m$; $m = n$; and $\ell_1, \ell_2 \in \mathcal{O}(m)$. (For the definition of $\text{Stab}_{\text{samp}}$, see Definition 7 in Section 4.)

- (a) If $\mathcal{S} \stackrel{\$}{\leftarrow} \text{Stab}_{\text{samp}}(\mathbf{V}^\top \mathcal{C}_{\text{pub}} \mathbf{W})$, then $\dim \mathcal{S} \neq m$ except with negligible probability in m .
(ii) Given Heuristic (a), the algorithm can be modified to require (up to a constant factor) at most $m^6 q^{m(\ell_1+\ell_2)}$ $\mathbb{F}_q$ -operations. Without this heuristic, the polynomial factor is $k(m-k)m^6$. (See Corollary 3 in Appendix F for the cost in terms of ω .)

Proof. Let $\mathcal{C}_{\text{mat}} := \text{ext}_{\gamma}(\mathcal{C}_{\text{vec}})$ be the input to the hiding transform, where $\gamma \in \mathbb{F}_{q^m}^m$ is a random $\mathbb{F}_q$ -basis. Let the scrambling matrices be $\mathbf{P} = [\mathbf{P}' \ \mathbf{P}''] \in \mathbb{F}_q^{(m+\ell_1) \times (m+\ell_1)}$ and $\mathbf{Q} = \begin{bmatrix} \mathbf{Q}' \\ \mathbf{Q}'' \end{bmatrix} \in \mathbb{F}_q^{(n+\ell_2) \times (n+\ell_2)}$ where $\mathbf{P}''$ denotes the last ℓ_1 columns of $\mathbf{P}$ and $\mathbf{Q}''$ denotes the last ℓ_2 rows of $\mathbf{Q}$.

Suppose in Step 1 we guess $\mathbf{V} \in \mathbb{F}_q^{(m+\ell_1) \times m}$ and $\mathbf{W} \in \mathbb{F}_q^{(n+\ell_2) \times n}$ such that $\text{colspan}(\mathbf{V}) = \text{colspan}(\mathbf{P}'')^\perp$ and $\text{colspan}(\mathbf{W}) = \text{rowspan}(\mathbf{Q}'')^\perp$. Then by Proposition 2, $\mathbf{V}^\top \mathcal{C}_{\text{pub}} \mathbf{W} = \mathbf{S}' \mathcal{C}_{\text{mat}} \mathbf{S}$ for some invertible $\mathbf{S}' \in \mathbb{F}_q^{m \times m}$ and $\mathbf{S} \in \mathbb{F}_q^{n \times n}$. Since $\mathcal{C}_{\text{mat}}$ is the expansion of $\mathcal{C}_{\text{vec}}$, $\mathbf{S}' \mathcal{C}_{\text{mat}} \mathbf{S}$ is the expansion of $\mathbf{S} \mathcal{C}_{\text{vec}}$. Further, $\text{Stab}(\mathbf{S}' \mathcal{C}_{\text{mat}} \mathbf{S}) = \mathbf{S}' \text{Stab}(\mathcal{C}_{\text{mat}}) \mathbf{S}'^{-1}$ so that $\dim \text{Stab}(\mathbf{S}' \mathcal{C}_{\text{mat}} \mathbf{S}) = m$. Hence **BasisRecover** returns a basis γ' such that $(\mathbf{S} \mathcal{C}_{\text{vec}})^{[j]} = \text{ext}_{\gamma'}^{-1}(\mathbf{S}' \mathcal{C}_{\text{mat}} \mathbf{S})$ for some $j \in \{0, \dots, m-1\}$. Since we have $\text{ext}_{\gamma'}^{-1}(\mathbf{V}^\top \mathcal{C}_{\text{pub}} \mathbf{W}) = \text{ext}_{\gamma'}^{-1}(\mathbf{S}' \mathcal{C}_{\text{mat}} \mathbf{S}) = (\mathbf{S} \mathcal{C}_{\text{vec}})^{[j]}$, the algorithm is correct.

The probability $\text{colspan}(\mathbf{V})$ equals the m -dimensional space $\text{colspan}(\mathbf{P}'')^\perp$ is $\binom{m+\ell_1}{m}_q^{-1}$. Likewise $\text{colspan}(\mathbf{W}) = \text{rowspan}(\mathbf{Q}'')^\perp$ with probability $\binom{n+\ell_2}{n}_q^{-1}$. Using $q^{(a-b)b} \leq \binom{a}{b}_q \leq 4q^{(a-b)b}$, the overall probability is at least $(16q^{m\ell_1+n\ell_2})^{-1}$ and the reciprocal of this is then the expected number of iterations.

The complexity of each iteration comes from the computation of the stabilizer algebra in the call to **BasisRecover**. In Corollary 3 in Appendix F, it is shown that this computation requires (up to a constant factor) at most $k(m-k)m^6$

$\mathbb{F}_q$ -operations when we pick $\omega = 3$. If we modify the stabilizer computation step to use fewer basis equations (Remark 6 in Section 4), then this cost becomes m^6 $\mathbb{F}_q$ -operations. But for this we also need to assume Heuristic (a). $\square$

5.2 ACDGV Distinguisher

A slight modification of the previous algorithm allows one to distinguish $\mathcal{C}_{\text{pub}}$ from a random matrix code with better complexity. Intuitively, the previous algorithm hopes to draw $\mathbf{V}$ and $\mathbf{W}$ that annihilate exactly the random rows and columns part of each codeword in $\mathcal{C}_{\text{pub}}$, so we can recover (a code equivalent to) $\mathcal{C}_{\text{mat}}$. However, if we only want to distinguish, then it does not matter if a few columns of $\mathcal{C}_{\text{mat}}$ are also annihilated. Thus, instead of letting $\mathbf{W}$ be an $(n + \ell_2) \times n$ matrix as in `NaiveKeyRecover` (Algorithm 2), we can let it be of size $(n + \ell_2) \times (k + 1)$.

Algorithm 3: ACDGV-KeyDistinguish($\mathbf{B}_1, \dots, \mathbf{B}_{km}$)

Input: A basis $\mathbf{B}_1, \dots, \mathbf{B}_{km} \in \mathbb{F}_q^{(m+\ell_1) \times (n+\ell_2)}$ of a km -dimensional code $\mathcal{C} \subseteq \mathbb{F}_q^{(m+\ell_1) \times (n+\ell_2)}$ where $\mathcal{C}$ is sampled from either the EMC($\mathcal{D}$)-distribution (Definition 2) or uniformly from the set of km -dimensional subspaces of $\mathbb{F}_q^{(m+\ell_1) \times (n+\ell_2)}$.

Output: If there exist $\mathbf{V} \in \mathbb{F}_q^{(m+\ell_1) \times m}$ and $\mathbf{W} \in \mathbb{F}_q^{(n+\ell_2) \times n}$ such that $\mathbf{V}^T \mathbf{C} \mathbf{W} = \text{ext}_{\gamma'}(\mathcal{C}'_{\text{vec}})$ for some $\mathbb{F}_q$ -basis γ' of $\mathbb{F}_{q^m}$ and some $[n, k]_{\mathbb{F}_{q^m}}$ code $\mathcal{C}'_{\text{vec}}$, return 1. Else return 0.

- 1 Draw random full-rank matrices $\mathbf{V} \in \mathbb{F}_q^{(m+\ell_1) \times m}$ and $\widetilde{\mathbf{W}} \in \mathbb{F}_q^{(n+\ell_2) \times (k+1)}$.
 - 2 **for** $i \leftarrow 1$ **to** km **do** $\mathbf{A}'_i \leftarrow \mathbf{V}^T \mathbf{B}_i \widetilde{\mathbf{W}} \in \mathbb{F}_q^{m \times (k+1)}$.
 - 3 Compute dimension d of $\text{Stab}(\langle \mathbf{A}'_1, \dots, \mathbf{A}'_{km} \rangle)$.
 - 4 **if** $d \geq m$ **then return** 1 **else return** 0.
-

Theorem 3 (ACDGV Distinguisher [1, p. 19]). *Let $\mathcal{C} \subseteq \mathbb{F}_q^{m \times (k+1)}$ be a random km -dimensional code.*

Define Heuristics (a) and (b) as follows.

- (a) *The probability that $\dim \text{Stab}(\mathcal{C}) \geq m$ is at most $cq^{-(m\ell_1 + (k+1)\ell_2)}$ for some constant $c < \frac{1}{16}$.*
- (b) *The probability that $\dim \text{Stab}(\mathcal{S}) \geq m$ is at most $cq^{-(m\ell_1 + (k+1)\ell_2)}$ for some constant $c < \frac{1}{16}$, where $\mathcal{S} \xleftarrow{\$} \text{Stab}_{\text{samp}}(\mathcal{C})$.*
- (i) *Given Heuristic (a), Algorithm 3 decides the HVCD problem (Definition 4) with advantage, up to a constant factor, at least $q^{-m\ell_1 - (k+1)\ell_2}$.*
- (ii) *Let $0 < k < m$; $m = n$; and $\ell_1, \ell_2 \in \mathcal{O}(m)$. Given Heuristics⁸ (a) and (b), the work factor⁹ is, up to a constant factor, at most $m^6 q^{m\ell_1 + (k+1)\ell_2}$ $\mathbb{F}_q$ -*

⁸ Heuristic (b) implies Heuristic (a), so we can also omit Heuristic (a) here.

⁹ The complexity of the algorithm divided by its advantage.

operations. Given only Heuristic (a), the polynomial factor becomes km^6 . (See Corollary 3 in Appendix F for the cost in terms of ω .)

Proof. The proof is similar to Theorem 2. Here we succeed if $\text{colspan}(\mathbf{V}) = \text{colspan}(\mathbf{P}'')^\perp$ and $\text{colspan}(\widetilde{\mathbf{W}}) \subseteq \text{rowspan}(\mathbf{Q}'')^\perp$ where subspaces $\text{colspan}(\mathbf{V})$ and $\text{colspan}(\widetilde{\mathbf{W}})$ are random and of dimension m and $k+1$ respectively. The probability of the former is $\begin{bmatrix} m+\ell_1 \\ m \end{bmatrix}_q^{-1}$, while for the latter it is $\begin{bmatrix} n \\ k+1 \end{bmatrix}_q \begin{bmatrix} n+\ell_2 \\ k+1 \end{bmatrix}_q^{-1}$. The product can be shown to be at least $\frac{1}{16}q^{-m\ell_1-(k+1)\ell_2}$.

Let p_1 and p_2 be the probability that the algorithm outputs 1 when the input is $\mathcal{C}_{\text{pub}}$ (from the $\text{EMC}(\mathcal{D})$ -distribution) and a random code respectively. The advantage is defined as $|p_1 - p_2|$, which by Heuristic (a), is at least $(1/16 - c)q^{-m\ell_1-(k+1)\ell_2}$ where $1/16 - c > 0$ is a constant. Corollary 3 in Appendix F shows that the dimension of the stabilizer algebra can be computed using km^6 $\mathbb{F}_q$ -operations (up to a constant factor). If we again modify the stabilizer algebra computation step to use fewer basis equations (as in Remark 6 in Section 4), then we only need m^6 operations. In this case, we need Heuristic (b) so that probability p_2 is small enough. Note using fewer basis equations can only increase the computed dimension d , so we do not need a heuristic for p_1 , i.e., when the input is $\mathcal{C}_{\text{pub}}$. $\square$

5.3 New Key Recovery

We need two technical assumptions for our key recovery. These involve claims about certain systems of linear equations that are easy to experimentally verify for any parameter set. In all of our experiments (see Appendix H), both these assumptions were always true.

Assumption 1. Let $\mathcal{C}_{\text{mat}}$ be as in Definition 2 ($\text{EMC}(\mathcal{D})$ -distribution). In other words, draw an $[n, k]_{\mathbb{F}_{q^m}}$ code $\mathcal{C}_{\text{vec}}$ from the distribution $\mathcal{D}$ and let $\mathcal{C}_{\text{mat}} := \text{ext}_\gamma(\mathcal{C}_{\text{vec}}) \subseteq \mathbb{F}_q^{m \times n}$ for a random $\mathbb{F}_q$ -basis γ of $\mathbb{F}_{q^m}$.

Draw a random basis $\mathbf{A}_1, \dots, \mathbf{A}_{km} \in \mathbb{F}_q^{m \times n}$ of $\mathcal{C}_{\text{mat}}$. Let $\mathbf{R}'_1, \dots, \mathbf{R}'_{km} \in \mathbb{F}_q^{\ell_1 \times n}$ be random matrices. Draw random full-rank matrices $\mathbf{U} \in \mathbb{F}_q^{b_1 \times m}$, $\mathbf{U}' \in \mathbb{F}_q^{n \times b_2}$ where $0 < b_1 \leq m$, $0 < b_2 \leq n$. Suppose

$$\mathbf{U} \left([\mathbf{J}_1 \ \mathbf{J}_2] \begin{bmatrix} \mathbf{A}_s \\ \mathbf{R}'_s \end{bmatrix} - \sum_{i=1}^{km} \mu_{s,i} \mathbf{A}_i \right) \mathbf{U}' = \mathbf{0} \quad \text{for } s = 1, \dots, km \quad (4)$$

for unknowns $\mathbf{J}_1 \in \mathbb{F}_q^{m \times m}$, $\mathbf{J}_2 \in \mathbb{F}_q^{m \times \ell_1}$ and $\mu_{s,i} \in \mathbb{F}_q$.

Suppose $b_1 \left(b_2 - \frac{(m+\ell_1)}{km} \right) \geq km - \frac{1}{k} + \frac{1}{k}m$. Then w.h.p. the only solutions of

$$\left(\mathbf{J}_1, \mathbf{J}_2, \{\mu_{s,i}\}_{(s,i)=(1,1)}^{(km,km)} \right) \in \mathbb{F}_q^{m \times m} \times \mathbb{F}_q^{m \times \ell_1} \times \mathbb{F}_q^{k^2 m^2}$$

belong to the sum of the following two subspaces:

$$\left\{ \left(\mathbf{J}_1, \mathbf{0}, \{\mu_{s,i}\}_{(s,i)=(1,1)}^{(km,km)} \right) \mid \mathbf{J}_1 \mathbf{A}_s - \sum_{i=1}^{km} \mu_{s,i} \mathbf{A}_i = \mathbf{0} \right\}, \quad (5)$$

$$\left\{ \left(\mathbf{K}_1, \mathbf{K}_2, \{0\}_{(s,i)=(1,1)}^{(km,km)} \right) \mid \mathbf{U} [\mathbf{K}_1 \ \mathbf{K}_2] = \mathbf{0} \right\}. \quad (6)$$

Remark 7 (on Assumption 1). Note that any tuple in the sum of the two subspaces (5) and (6) is clearly a solution of the system (4). The assumption states that these are the only solutions. The first subspace has dimension m since $\dim \text{Stab}(\mathcal{C}_{\text{mat}}) = m$ and the second $(m - b_1)(m + \ell_1)$, so the assumption implies that the solution space of the system has dimension $m + (m - b_1)(m + \ell_1)$. The condition $b_1 \left(b_2 - \frac{(m + \ell_1)}{km} \right) \geq km - \frac{1}{k} + \frac{1}{km}$ is equivalent to asserting that the number of equations is at least the number of unknowns minus the dimension of the solution space plus 1. That is, it is equivalent to $b_1 b_2 km \geq ((km)^2 + m(m + \ell_1)) - (m + (m - b_1)(m + \ell_1)) + 1$.

To understand the assumption better, consider the system of equations but without the $\mathbf{U}$ and $\mathbf{U}'$, i.e.,

$$[\mathbf{J}_1 \ \mathbf{J}_2] \begin{bmatrix} \mathbf{A}_s \\ \mathbf{R}'_s \end{bmatrix} - \sum_{i=1}^{km} \mu_{s,i} \mathbf{A}_i = \mathbf{0} \quad \text{for } s = 1, \dots, km.$$

Then a weak version of the assumption would claim that this system only has solutions satisfying $\mathbf{J}_1 \mathbf{A}_s - \sum_{i=1}^{km} \mu_{s,i} \mathbf{A}_i = \mathbf{0}$ and $\mathbf{J}_2 = \mathbf{0}$. A less weak version would claim that this conclusion does not change even if we right-multiply the system by a full-rank tall matrix $\mathbf{U}' \in \mathbb{F}_q^{n \times b_2}$. Finally, the actual claim, i.e. Assumption 1, says that if we also left-multiply by a full-rank wide matrix $\mathbf{U} \in \mathbb{F}_q^{b_1 \times m}$ then the only new solutions are obtained by adding vectors from the right null space of $\mathbf{U}$ to the matrix $[\mathbf{J}_1 \ \mathbf{J}_2]$ as columns.

Assumption 2. Let $\mathcal{C}_{\text{mat}}$ be as in Definition 2 (EMC($\mathcal{D}$)-distribution). Draw a random basis $\mathbf{A}_1, \dots, \mathbf{A}_{km} \in \mathbb{F}_q^{m \times n}$ of $\mathcal{C}_{\text{mat}}$. Let $\mathbf{R}'_1, \dots, \mathbf{R}'_{km} \in \mathbb{F}_q^{\ell_1 \times n}$ be random matrices. Let $\{\mu_{s,i}\}_{(s,i)=(1,1)}^{(km,km)}$ be such that $\mathbf{T} \mathbf{A}_s = \sum_{i=1}^{km} \mu_{s,i} \mathbf{A}_i$ for all $s = 1, \dots, km$ for some $\mathbf{T} \in \mathbb{F}_q^{m \times m}$ that is not a scalar matrix. Suppose

$$[\mathbf{J}_1 \ \mathbf{J}_2] \begin{bmatrix} \mathbf{A}_s \\ \mathbf{R}'_s \end{bmatrix} = [\mathbf{K}_1 \ \mathbf{K}_2] \sum_{i=1}^{km} \mu_{s,i} \begin{bmatrix} \mathbf{A}_i \\ \mathbf{R}'_i \end{bmatrix} \quad \text{for } s = 1, \dots, km$$

for unknowns $\mathbf{J}_1, \mathbf{K}_1 \in \mathbb{F}_q^{m \times m}$ and $\mathbf{J}_2, \mathbf{K}_2 \in \mathbb{F}_q^{m \times \ell_1}$. Then w.h.p. $\mathbf{J}_2 = \mathbf{K}_2 = \mathbf{0}$, so that $\mathbf{J}_1 = \mathbf{K}_1 \mathbf{T}$.

Remark 8 (on Assumption 2). Suppose we have an HVCS instance $\mathcal{C}_{\text{pub}} \subseteq \mathbb{F}_q^{(m + \ell_1) \times n}$ (Definition 3) with no random columns, i.e., $\ell_2 = 0$. Suppose we know two lists $\mathbf{C}_1, \dots, \mathbf{C}_{km} \in \mathcal{C}_{\text{pub}}$ and $\mathbf{C}'_1, \dots, \mathbf{C}'_{km} \in \mathcal{C}_{\text{pub}}$ that are related in the following sense. There exists a $\mathbf{T} \in \text{Stab}(\mathcal{C}_{\text{mat}}) \setminus \{\lambda \mathbf{I} \mid \lambda \in \mathbb{F}_q\} \subseteq \mathbb{F}_q^{m \times m}$ such that

all pairs $(\mathbf{C}_i, \mathbf{C}'_i) \in \mathbb{F}_q^{(m+\ell_1) \times n} \times \mathbb{F}_q^{(m+\ell_1) \times n}$ have the form $\left(\mathbf{P} \begin{bmatrix} \mathbf{A}_i \\ * \end{bmatrix}, \mathbf{P} \begin{bmatrix} \mathbf{T}\mathbf{A}_i \\ * \end{bmatrix} \right)$ for $\mathbf{A}_i \in \mathcal{C}_{\text{mat}} \subseteq \mathbb{F}_q^{m \times n}$ and invertible $\mathbf{P} \in \mathbb{F}_q^{(m+\ell_1) \times (m+\ell_1)}$. Then this assumption allows us to recover a matrix $\mathbf{F} \in \mathbb{F}_q^{m \times (m+\ell_1)}$ such that, for some invertible $\mathbf{S} \in \mathbb{F}_q^{m \times m}$, we have $\mathbf{F}\mathbf{C}_i = \mathbf{S}\mathbf{A}_i$ for all i . In other words, we obtain an $\mathbf{F}$ such that $\mathbf{F}\mathcal{C}_{\text{pub}} = \mathbf{S}\mathcal{C}_{\text{mat}}$. We need that $\mathbf{T}$ is not of the form $\lambda\mathbf{I}$ because otherwise we will just have $\mathbf{C}'_i = \lambda\mathbf{C}_i$ which is not useful.

Theorem 4 (New Key Recovery). *Let $\mathcal{C}_{\text{pub}} \stackrel{s}{\leftarrow} \text{EMC}(\mathcal{D})$, where $\mathcal{C}_{\text{vec}}$ is the $\mathbb{F}_{q^m}$ -linear code such that $\mathcal{C}_{\text{mat}} := \text{ext}_{\gamma}(\mathcal{C}_{\text{vec}})$ is the input to the hiding transform (for a random $\mathbb{F}_q$ -basis $\gamma \in \mathbb{F}_{q^m}^m$). Assume $\dim \text{Stab}(\mathcal{C}_{\text{mat}}) = m$ and suppose¹⁰ $\epsilon := \frac{m+\ell_1}{km} \leq 1$ and¹¹ $\ell_2 \leq km^2$. Define the following heuristic.*

(a) Assumptions 1 and 2 hold.

Given Heuristic (a), Algorithm 4 solves the HVCS problem for $\mathcal{D}$ (Definition 3) with at least constant probability, and its complexity is as follows.

- (i) (Exact) Set integers $\frac{\ell_2}{km} \leq b_1 \leq m$ and $\frac{m+\ell_1}{km} \leq b_2 \leq n$ such that $b_1\ell_1 + b_2\ell_2$ is minimum and $b_1(b_2 - \frac{m+\ell_1}{km}) \geq km - \frac{1}{k} + \frac{1}{km}$. The algorithm runs in time at most $q^{b_1\ell_1 + b_2\ell_2}$, ignoring polynomial factors. Let $0 < k < m$; $m = n$; and $\ell_1, \ell_2 \in \mathcal{O}(m)$. Then the algorithm requires at most $k(m-k)m^6 q^{b_1\ell_1 + b_2\ell_2}$ $\mathbb{F}_q$ -operations (up to a constant factor).
- (ii) (Closed-form) Let $R := k/m$. Suppose $\ell_2 \leq k^2m$. If $m = n$ and $\ell_1, \ell_2 \in \mathcal{O}(m)$, the algorithm requires at most

$$\begin{cases} P q^{2m\sqrt{R\ell_1\ell_2} + \ell_1 + (1+\epsilon)\ell_2} & \text{if } \ell_1\ell_2 \neq 0 \text{ and } R \leq \min \left\{ \frac{\ell_1}{\ell_2}, \left(1 - \frac{\epsilon}{m}\right)^2 \frac{\ell_2}{\ell_1} \right\}, \\ P \min \{ q^{m\ell_1 + (Rm+1)\ell_2}, q^{(Rm+1)\ell_1 + m\ell_2} \} & \text{otherwise,} \end{cases}$$

$\mathbb{F}_q$ -operations (up to a constant factor), where $P := k(m-k)m^6$. (See Corollary 3 in Appendix F for the cost in terms of ω .)

Proof. The constraints are required for Assumption 1 to hold and, as we will see, for Part (ii) (Lines 6–12) of the algorithm to succeed. We first prove it is possible to pick b_1, b_2 satisfying the constraints in Step 1, by showing the choice $(b_1, b_2) = (m, n)$ is feasible. The box constraints $\frac{\ell_2}{km} \leq b_1 \leq m$ and $\frac{m+\ell_1}{km} \leq b_2 \leq n$ are clearly satisfied by this choice since $\ell_2 \leq km^2$ and $\epsilon = \frac{m+\ell_1}{km} \leq 1$. The third constraint becomes $m(n - \epsilon) \geq km - \frac{1}{k}(1 - \frac{1}{m})$, which also holds since $\epsilon \leq 1$ and $k < n$.

Let $\mathbf{B}_i = \mathbf{P} \begin{bmatrix} \mathbf{A}_i & \mathbf{R}_i \\ \mathbf{R}'_i & \mathbf{R}''_i \end{bmatrix} \mathbf{Q}$ for $i = 1, \dots, km$ where the matrices $\mathbf{R}_i, \mathbf{R}'_i, \mathbf{R}''_i$ are random for each i , the basis $\mathbf{A}_1, \dots, \mathbf{A}_{km} \in \mathbb{F}_q^{m \times n}$ spans the code $\mathcal{C}_{\text{mat}}$ which was

¹⁰ This is a mild restriction as ℓ_1 is typically much smaller than even m .

¹¹ This is a trivial restriction. See Remark 12 in Appendix G.

Algorithm 4: NewKeyRecover($\mathbf{B}_1, \dots, \mathbf{B}_{km}$)

Input: A basis $\mathbf{B}_1, \dots, \mathbf{B}_{km} \in \mathbb{F}_q^{(m+\ell_1) \times (n+\ell_2)}$ of the matrix code $\mathcal{C}_{\text{pub}}$ sampled from the EMC($\mathcal{D}$)-distribution (Definition 2).

Output: $\mathbf{V} \in \mathbb{F}_q^{(m+\ell_1) \times m}$, $\mathbf{W} \in \mathbb{F}_q^{(n+\ell_2) \times n}$, $\gamma' \in \mathbb{F}_q^m$ such that γ' is an $\mathbb{F}_q$ -basis of $\mathbb{F}_q^m$ and $\mathcal{C}'_{\text{vec}} := \text{ext}_{\gamma'}^{-1}(\mathbf{V}^\top \mathcal{C}_{\text{pub}} \mathbf{W}) \subseteq \mathbb{F}_q^n$ is an $[n, k]$ code over $\mathbb{F}_q$.

- 1 Set integers b_1 and b_2 such that $b_1 \ell_1 + b_2 \ell_2$ is minimum subject to $\frac{\ell_2}{km} \leq b_1 \leq m$, $\frac{m+\ell_1}{km} \leq b_2 \leq n$ and $b_1 \left(b_2 - \frac{(m+\ell_1)}{km}\right) \geq km - \frac{1}{k} + \frac{1}{km}$.
- 2 **repeat**
 - 3 Draw random full-rank matrices $\mathbf{V} \in \mathbb{F}_q^{(m+\ell_1) \times b_1}$ and $\mathbf{W} \in \mathbb{F}_q^{(n+\ell_2) \times b_2}$.
 - 4 Find the dimension d of the solution space of the following system in the unknowns $\mathbf{F} \in \mathbb{F}_q^{(m+\ell_1) \times (m+\ell_1)}$ and $\mu_{s,i}$'s $\in \mathbb{F}_q$,
$$\mathbf{V}^\top \left(\mathbf{F} \mathbf{B}_s - \sum_{i=1}^{km} \mu_{s,i} \mathbf{B}_i \right) \mathbf{W} = \mathbf{0} \text{ for } s = 1, \dots, km. \quad (7)$$
 - 5 **until** $d \geq m + (m + \ell_1 - b_1)(m + \ell_1)$.

// remove the ℓ_2 random columns
- 6 $\mathcal{L} \leftarrow \{\mathbf{0}\} \subseteq \mathbb{F}_q^{n+\ell_2}$.
- 7 **repeat** $(1 - q^{-(m-1)})^{-1} \leq 2$ **times**
 - 8 Draw random $\left(\mathbf{F}, \{\mu_{s,i}\}_{(s,i)=(1,1)}^{(km,km)} \right)$ from the solution space of (7).
 - 9 **for** $s \leftarrow 1, \dots, km$ **do** $\mathcal{L} \leftarrow \mathcal{L} + \text{rowspan} \left(\mathbf{V}^\top \left(\mathbf{F} \mathbf{B}_s - \sum_{i=1}^{km} \mu_{s,i} \mathbf{B}_i \right) \right)$.
- 10 **if** $\dim(\mathcal{L}) \neq \ell_2$ **then** go to Step 2.
- 11 Set $\widetilde{\mathbf{W}} \in \mathbb{F}_q^{(n+\ell_2) \times n}$ such that $\text{colspan}(\widetilde{\mathbf{W}}) = \mathcal{L}^\perp$.
- 12 **for** $s \leftarrow 1, \dots, km$ **do** $\mathbf{B}'_s \leftarrow \mathbf{B}_s \widetilde{\mathbf{W}} \in \mathbb{F}_q^{(m+\ell_1) \times n}$.

// remove the ℓ_1 random rows
- 13 **repeat** $(0.288)^{-1} (1 - q^{-(m-1)})^{-1} \leq 7$ **times**
 - 14 Draw random $\{\mu_{s,i}\}_{(s,i)=(1,1)}^{(km,km)}$ from the solution space of (7).
 - 15 Draw random invertible $\mathbf{M} \in \mathbb{F}_q^{(m+\ell_1) \times (m+\ell_1)}$.
 - 16 Solve the following system for unknowns $\mathbf{F}^{(1)} \in \mathbb{F}_q^{m \times (m+\ell_1)}$ and $\widetilde{\mathbf{F}}^{(2)} \in \mathbb{F}_q^{m \times \ell_1}$.
$$\mathbf{F}^{(1)} \mathbf{B}'_s = \left[\mathbf{I}_m \quad \widetilde{\mathbf{F}}^{(2)} \right] \mathbf{M} \sum_{i=1}^{km} \mu_{s,i} \mathbf{B}'_i \text{ for } s = 1, \dots, km. \quad (8)$$
 - 17 **if** (8) has a unique solution **then break**.
- 18 **if** (8) does not have a unique solution **then** go to Step 2.
- 19 $\gamma' \leftarrow$ run BasisRecover (Algorithm 1) on the matrix code $\langle \mathbf{F}^{(1)} \mathbf{B}'_1, \dots, \mathbf{F}^{(1)} \mathbf{B}'_{km} \rangle$.
- 20 **if** $\gamma' = \perp$ **then** go to Step 2.
- 21 **return** $(\mathbf{F}^{(1)\top}, \widetilde{\mathbf{W}}, \gamma')$.

input to the hiding transform, and $\mathbf{P} \in \mathbb{F}_q^{(m+\ell_1) \times (m+\ell_1)}$ and $\mathbf{Q} \in \mathbb{F}_q^{(n+\ell_2) \times (n+\ell_2)}$ are random invertible matrices.

In the following, we consider the desired case where $\mathbf{V}$ and $\mathbf{W}$ are drawn in Line 3 such that $\text{colspan}(\mathbf{V}) \subseteq \text{colspan}(\mathbf{P}'')^\perp$ and $\text{colspan}(\mathbf{W}) \subseteq \text{rowspan}(\mathbf{Q}'')^\perp$.

Part (i) (Lines 2-5) We can always write $\mathbf{F}$ as $\mathbf{P} \begin{bmatrix} \mathbf{J}_1 & \mathbf{J}_2 \\ \mathbf{J}_3 & \mathbf{J}_4 \end{bmatrix} \mathbf{P}^{-1}$ for some matrices $\mathbf{J}_1 \in \mathbb{F}_q^{m \times m}$, $\mathbf{J}_2 \in \mathbb{F}_q^{m \times \ell_1}$, $\mathbf{J}_3 \in \mathbb{F}_q^{\ell_1 \times m}$ and $\mathbf{J}_4 \in \mathbb{F}_q^{\ell_1 \times \ell_1}$. Then one can show $\mathbf{F}\mathbf{B}_s - \sum_{i=1}^{km} \mu_{s,i} \mathbf{B}_i$ equals

$$\begin{aligned} & \mathbf{P} \begin{bmatrix} \mathbf{J}_1 \mathbf{A}_s + \mathbf{J}_2 \mathbf{R}'_s - \sum_{i=1}^{km} \mu_{s,i} \mathbf{A}_i & \mathbf{J}_1 \mathbf{R}_s + \mathbf{J}_2 \mathbf{R}''_s - \sum_{i=1}^{km} \mu_{s,i} \mathbf{R}_i \\ \mathbf{J}_3 \mathbf{A}_s + \mathbf{J}_4 \mathbf{R}'_s - \sum_{i=1}^{km} \mu_{s,i} \mathbf{R}'_i & \mathbf{J}_3 \mathbf{R}_s + \mathbf{J}_4 \mathbf{R}''_s - \sum_{i=1}^{km} \mu_{s,i} \mathbf{R}''_i \end{bmatrix} \mathbf{Q} \\ &= \mathbf{P} \begin{bmatrix} \mathbf{J}_1 \mathbf{A}_s + \mathbf{J}_2 \mathbf{R}'_s - \sum_{i=1}^{km} \mu_{s,i} \mathbf{A}_i & * \\ * & * \end{bmatrix} \mathbf{Q} \end{aligned}$$

where we have replaced the block matrices that are not relevant for now by $*$.

Let $\mathbf{P} = [\mathbf{P}' \ \mathbf{P}''] \in \mathbb{F}_q^{(m+\ell_1) \times (m+\ell_1)}$ and $\mathbf{Q} = \begin{bmatrix} \mathbf{Q}' \\ \mathbf{Q}'' \end{bmatrix} \in \mathbb{F}_q^{(n+\ell_2) \times (n+\ell_2)}$, such that $\mathbf{P}''$ denotes the last ℓ_1 columns of $\mathbf{P}$ and $\mathbf{Q}''$ the last ℓ_2 rows of $\mathbf{Q}$. Then $\mathbf{V}^\top \mathbf{P} = [\mathbf{V}^\top \mathbf{P}' \ \mathbf{0}_{b_1 \times \ell_1}]$ and $\mathbf{Q} \mathbf{W} = \begin{bmatrix} \mathbf{Q}' \mathbf{W} \\ \mathbf{0}_{\ell_2 \times b_2} \end{bmatrix}$. Consequently,

$$\begin{aligned} & \mathbf{V}^\top \left(\mathbf{F}\mathbf{B}_s - \sum_{i=1}^{km} \mu_{s,i} \mathbf{B}_i \right) \mathbf{W} \\ &= \mathbf{V}^\top \mathbf{P} \begin{bmatrix} \mathbf{J}_1 \mathbf{A}_s + \mathbf{J}_2 \mathbf{R}'_s - \sum_{i=1}^{km} \mu_{s,i} \mathbf{A}_i & * \\ * & * \end{bmatrix} \mathbf{Q} \mathbf{W} \\ &= \mathbf{V}^\top \mathbf{P}' \left(\mathbf{J}_1 \mathbf{A}_s + \mathbf{J}_2 \mathbf{R}'_s - \sum_{i=1}^{km} \mu_{s,i} \mathbf{A}_i \right) \mathbf{Q}' \mathbf{W}. \end{aligned}$$

By Lemma 3 in Appendix E, $\mathbf{V}^\top \mathbf{P}'$ and $\mathbf{Q}' \mathbf{W}$ are full-rank matrices drawn uniformly at random. Hence the previous expression is equivalent to the system

$$\mathbf{U} \left(\mathbf{J}_1 \mathbf{A}_s + \mathbf{J}_2 \mathbf{R}'_s - \sum_{i=1}^{km} \mu_{s,i} \mathbf{A}_i \right) \mathbf{U}' = \mathbf{0} \quad \text{for all } s = 1, \dots, km \quad (9)$$

where $\mathbf{U} \in \mathbb{F}_q^{b_1 \times m}$ and $\mathbf{U}' \in \mathbb{F}_q^{n \times b_2}$ are random full-rank matrices. One can check that the solutions

$$\left(\mathbf{J}_1, \mathbf{J}_2, \mathbf{J}_3, \mathbf{J}_4, \{\mu_{s,i}\}_{(s,i)=(1,1)}^{(km,km)} \right) \in \mathbb{F}_q^{m \times m} \times \mathbb{F}_q^{m \times \ell_1} \times \mathbb{F}_q^{\ell_1 \times m} \times \mathbb{F}_q^{\ell_1 \times \ell_1} \times \mathbb{F}_q^{k^2 m^2}$$

to this system include elements from the following three subspaces:

$$\begin{aligned} & \left\{ \left(\mathbf{0}, \mathbf{0}, \mathbf{J}_3, \mathbf{J}_4, \{0\}_{(s,i)=(1,1)}^{(km,km)} \right) \mid \mathbf{J}_3 \in \mathbb{F}_q^{\ell_1 \times m}, \mathbf{J}_4 \in \mathbb{F}_q^{\ell_1 \times \ell_1} \right\}, \\ & \left\{ \left(\mathbf{J}_1, \mathbf{0}, \mathbf{0}, \mathbf{0}, \{\mu_{s,i}\}_{(s,i)=(1,1)}^{(km,km)} \right) \mid \mathbf{J}_1 \mathbf{A}_s - \sum_{i=1}^{km} \mu_{s,i} \mathbf{A}_i = \mathbf{0} \right\} \quad \text{and} \\ & \left\{ \left(\mathbf{K}_1, \mathbf{K}_2, \mathbf{0}, \mathbf{0}, \{0\}_{(s,i)=(1,1)}^{(km,km)} \right) \mid \mathbf{U} [\mathbf{K}_1 \ \mathbf{K}_2] = \mathbf{0} \right\} \end{aligned}$$

These three subspaces have dimensions $\ell_1(m + \ell_1)$, m and $(m - b_1)(m + \ell_1)$ in that order. The second subspace has dimension m because it is equivalent to $\text{Stab}(\mathcal{C}_{\text{mat}})$. Further, the sum of these subspaces is a direct sum. In particular, the last two subspaces have the trivial intersection because $\mathbf{K}_1$ is of rank $(m - b_1) < m$ while any non-zero $\mathbf{J}_1 \in \text{Stab}(\mathcal{C}_{\text{mat}})$ must have full rank since $\mathbf{J}_1$ represents the multiplication-by- b map for some $b \in \mathbb{F}_{q^m} \setminus \{0\}$ (see first part of Theorem 1's proof). Thus, the “until” condition is fulfilled.

Part (ii) (Lines 6–12) Let $(\mathbf{F}, \{\mu_{s,i}\}_{(s,i)=(1,1)}^{(km,km)})$ be a random solution of (7). If we do not right-multiply by $\mathbf{W}$, then we get

$$\begin{aligned} & \mathbf{V}^\top \left(\mathbf{F} \mathbf{B}_s - \sum_{i=1}^{km} \mu_{s,i} \mathbf{B}_i \right) \\ &= \mathbf{V}^\top \mathbf{P}' \left[\mathbf{J}_1 \mathbf{A}_s + \mathbf{J}_2 \mathbf{R}'_s - \sum_{i=1}^{km} \mu_{s,i} \mathbf{A}_i \quad \mathbf{J}_1 \mathbf{R}_s - \sum_{i=1}^{km} \mu_{s,i} \mathbf{R}_i \right] \mathbf{Q} \\ &= \begin{bmatrix} \mathbf{0} & \mathbf{U} \left(\mathbf{J}_1 \mathbf{R}_s - \sum_{i=1}^{km} \mu_{s,i} \mathbf{R}_i \right) \end{bmatrix} \mathbf{Q} \quad \text{for all } s = 1, \dots, km. \end{aligned} \quad (10)$$

where the first block matrix equals $\mathbf{0}$ by Assumption 1. Write

$$\mathbf{J}_1 \mathbf{R}_s - \sum_{i=1}^{km} \mu_{s,i} \mathbf{R}_i = (\mathbf{J}_1 - \mu_{ss} \mathbf{I}) \mathbf{R}_s - \sum_{i=1, i \neq s}^{km} \mu_{s,i} \mathbf{R}_i \quad (11)$$

We will show that the expressions on the right-hand side are independent and uniformly random $m \times \ell_2$ matrices over $s = 1, \dots, km$. Collect all the “matrix coefficients” of each $\mathbf{R}_i$ into a block matrix such that the s th row corresponds to the s th expression.

$$\begin{bmatrix} \mathbf{J}_1 - \mu_{11} \mathbf{I} & -\mu_{1,2} \mathbf{I} & -\mu_{1,3} \mathbf{I} & \cdots & -\mu_{1,km} \mathbf{I} \\ -\mu_{2,1} \mathbf{I} & \mathbf{J}_1 - \mu_{22} \mathbf{I} & -\mu_{2,3} \mathbf{I} & \cdots & -\mu_{2,km} \mathbf{I} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ -\mu_{km,1} \mathbf{I} & -\mu_{km,2} \mathbf{I} & \cdots & \cdots & \mathbf{J}_1 - \mu_{km,km} \mathbf{I} \end{bmatrix}$$

With probability $1 - q^{-(m-1)}$, the drawn $\mathbf{F}$ will be such that $\mathbf{J}_1$ is not a scalar matrix (i.e., of the form $\lambda \mathbf{I}$ for some $\lambda \in \mathbb{F}_q$). Hence $\mathbf{J}_1$ represents an element in $\mathbb{F}_{q^m} \setminus \mathbb{F}_q$ and so $\mathbf{J}_1 - \mu_{ss} \mathbf{I}$ represents a nonzero $\mathbb{F}_{q^m}$ element. Further, notice all

these $m \times m$ “matrix elements” in the block matrix represent $\mathbb{F}_{q^m}$ elements. Thus, one can perform “Gaussian elimination” on it (where the “scalars” are $m \times m$ matrices that represent $\mathbb{F}_{q^m}$ elements), and since $\mathbf{J}_1 - \beta \mathbf{I}$ is always invertible for any $\beta \in \mathbb{F}_q$, one sees this block matrix is full rank. Hence we get the desired result about the right-hand side of (11).

Hence, the right block matrices $\mathbf{U}(\mathbf{J}_1 \mathbf{R}_s - \sum \dots)$ for $s = 1, \dots, km$ in (10) are independently and uniformly random of size $b_1 \times \ell_2$. Hence, the sum of the row spaces of these matrices fills the entire space $\mathbb{F}_q^{\ell_2}$ w.h.p. since $\ell_2 \leq b_1 km$ (by the box constraint on b_1). Thus, the expression in (10) reveals the row space of the last ℓ_2 rows of $\mathbf{Q}$ and by Proposition 2 we can effectively remove the ℓ_2 random columns from $\mathcal{C}_{\text{pub}}$.

Part (iii) (Lines 12–18) Let $\mathbf{B}'_i = \mathbf{P} \begin{bmatrix} \mathbf{A}_i \\ \mathbf{R}'_i \end{bmatrix}$ (up to right multiplication by an invertible matrix). Suppose $\{\mu_{s,i}\}_{(s,i)=(1,1)}^{(km,km)}$ are drawn randomly from the solution space of (7). Thus there exists a $\mathbf{T} \in \mathbb{F}_q^{m \times m}$ such that $\mathbf{T} \mathbf{A}_s = \sum_{i=1}^{km} \mu_{s,i} \mathbf{A}_i$ for all $s = 1, \dots, km$. With probability $1 - q^{-(m-1)}$, such a $\mathbf{T}$ will not be a scalar matrix. Equation (8) essentially solves

$$\mathbf{F}^{(1)} \mathbf{B}'_s = \tilde{\mathbf{F}}^{(2)} \sum_{i=1}^{km} \mu_{s,i} \mathbf{B}'_i \text{ for } s = 1, \dots, km. \quad (12)$$

for unknowns $\mathbf{F}^{(1)} \in \mathbb{F}_q^{m \times (m+\ell_1)}$ and $\tilde{\mathbf{F}}^{(2)} \in \mathbb{F}_q^{m \times (m+\ell_1)}$. In order to obtain a unique solution, $\tilde{\mathbf{F}}^{(2)}$ is specialized to $\begin{bmatrix} \mathbf{I}_m & \tilde{\mathbf{F}}'' \end{bmatrix} \mathbf{M}$ for unknown $\tilde{\mathbf{F}}'' \in \mathbb{F}_q^{m \times \ell_1}$ and a known invertible $\mathbf{M}$ which is guessed.

Write $\mathbf{F}^{(1)} = [\mathbf{J}_1 \ \mathbf{J}_2] \mathbf{P}^{-1}$ and $\tilde{\mathbf{F}}^{(2)} = [\mathbf{K}_1 \ \mathbf{K}_2] \mathbf{M}$, so that the system (12) becomes

$$\begin{aligned} [\mathbf{J}_1 \ \mathbf{J}_2] \begin{bmatrix} \mathbf{A}_s \\ \mathbf{R}'_s \end{bmatrix} &= [\mathbf{K}_1 \ \mathbf{K}_2] \mathbf{M} \mathbf{P} \sum_{i=1}^{km} \mu_{s,i} \begin{bmatrix} \mathbf{A}_i \\ \mathbf{R}'_i \end{bmatrix} \\ &= [\mathbf{K}_1 \ \mathbf{K}_2] \mathbf{M}' \sum_{i=1}^{km} \mu_{s,i} \begin{bmatrix} \mathbf{T} \mathbf{A}_i \\ * \end{bmatrix} \end{aligned}$$

for $s = 1, \dots, km$ and where we let $\mathbf{M}' = \mathbf{M} \mathbf{P} \in \mathbb{F}_q^{(m+\ell_1) \times (m+\ell_1)}$. If we partition $\mathbf{M}'$ as $\begin{bmatrix} \mathbf{M}'_1 & \mathbf{M}'_2 \\ \mathbf{M}'_3 & \mathbf{M}'_4 \end{bmatrix}$ where $\mathbf{M}'_1 \in \mathbb{F}_q^{m \times m}$, $\mathbf{M}'_2 \in \mathbb{F}_q^{m \times \ell_1}$, $\mathbf{M}'_3 \in \mathbb{F}_q^{\ell_1 \times m}$ and $\mathbf{M}'_4 \in \mathbb{F}_q^{\ell_1 \times \ell_1}$, then the equation becomes

$$[\mathbf{J}_1 \ \mathbf{J}_2] \begin{bmatrix} \mathbf{A}_s \\ \mathbf{R}'_s \end{bmatrix} = [\mathbf{K}_1 \mathbf{M}'_1 + \mathbf{K}_2 \mathbf{M}'_3 \quad \mathbf{K}_1 \mathbf{M}'_2 + \mathbf{K}_2 \mathbf{M}'_4] \sum_{i=1}^{km} \mu_{s,i} \begin{bmatrix} \mathbf{T} \mathbf{A}_i \\ * \end{bmatrix}.$$

Assumption 2 implies that the only solutions are $\mathbf{J}_2 = \mathbf{K}_1 \mathbf{M}'_2 + \mathbf{K}_2 \mathbf{M}'_4 = \mathbf{0}$ and $\mathbf{J}_1 = (\mathbf{K}_1 \mathbf{M}'_1 + \mathbf{K}_2 \mathbf{M}'_3) \mathbf{T}$. Since we specialize $\mathbf{K}_1 = \mathbf{I}$, we get

$$\begin{aligned} \mathbf{J}_2 &= \mathbf{M}'_2 + \mathbf{K}_2 \mathbf{M}'_4 = \mathbf{0}, \\ \mathbf{J}_1 &= (\mathbf{M}'_1 + \mathbf{K}_2 \mathbf{M}'_3) \mathbf{T}. \end{aligned}$$

If $\mathbf{M}'_4 \in \mathbb{F}_q^{\ell_1 \times \ell_1}$ is invertible, then $\mathbf{K}_2 = -\mathbf{M}'_2 \mathbf{M}'_4{}^{-1}$ and so

$$\mathbf{J}_2 = \mathbf{0},$$

$$\mathbf{J}_1 = (\mathbf{M}'_1 - \mathbf{M}'_2 \mathbf{M}'_4{}^{-1} \mathbf{M}'_3) \mathbf{T} = \mathbf{S} \mathbf{T},$$

where the Schur complement $\mathbf{S} = \mathbf{M}'_1 - \mathbf{M}'_2 \mathbf{M}'_4{}^{-1} \mathbf{M}'_3$ of $\mathbf{M}'$ is invertible since $\mathbf{M}'$ and $\mathbf{M}'_4$ both are. Thus, $\mathbf{F}^{(1)} = [\mathbf{S} \mathbf{T} \quad \mathbf{0}] \mathbf{P}^{-1}$ and so $\mathbf{F}^{(1)} \langle \mathbf{B}'_1, \dots, \mathbf{B}'_{km} \rangle = \mathbf{S} \mathbf{C}_{\text{mat}}$. The above argument requires that $\mathbf{M}'_4 \in \mathbb{F}_q^{\ell_1 \times \ell_1}$ is invertible, but this happens with high (i.e. with at least constant) probability¹², so $\mathbf{M}$ needs to be redrawn only a few times.

Complexity The success case is when both $\text{colspan}(\mathbf{V}) \subseteq \text{colspan}(\mathbf{P}'')^\perp$ and $\text{colspan}(\mathbf{W}) \subseteq \text{rowspan}(\mathbf{Q}'')^\perp$. This has probability

$$\frac{\begin{bmatrix} m \\ b_1 \end{bmatrix}}{\begin{bmatrix} m + \ell_1 \\ b_1 \end{bmatrix}} \frac{\begin{bmatrix} n \\ b_2 \end{bmatrix}}{\begin{bmatrix} n + \ell_2 \\ b_2 \end{bmatrix}} > \frac{1}{16} q^{-(b_1 \ell_1 + b_2 \ell_2)},$$

where we use $q^{(a-b)b} \leq \begin{bmatrix} a \\ b \end{bmatrix}_q \leq 4q^{(a-b)b}$, which shows the expected number of iterations is $q^{b_1 \ell_1 + b_2 \ell_2}$, ignoring polynomial factors. The closed-form complexity expressions are derived in Corollary 4 in Appendix G. In Corollary 3 in Appendix F, we also show that the polynomial cost of solving (7) is, up to a constant factor, $k(m-k)m^6$ (when $\omega = 3$). Moreover, we show that the portion of the algorithm after Line 5 (i.e., after Part (i)) also has complexity at most $k(m-k)m^6$. Thus, we do not need a heuristic that the “until” condition fails when $\mathbf{V}$ and $\mathbf{W}$ are chosen incorrectly to bound the polynomial cost. Note that if the algorithm reaches the last “return” line (Line 21), then by construction the return value is an HVCS solution. $\square$

6 Complexity and Comparison

6.1 Theoretical Complexity

Table 2. Complexity of various attacks in $\mathbb{F}_q$ -operations (up to a constant factor), assuming $m = n$.

Algorithm	Complexity
Naive Key Rec. (Thm 2)	$m^6 q^{m\ell_1 + m\ell_2}$
ACDGV Dist. (Thm 3)	$m^6 q^{m\ell_1 + (Rm+1)\ell_2}$
New Key Rec. (Thm 4)	$\begin{cases} k(m-k)m^6 q^{2m\sqrt{R\ell_1\ell_2} + \ell_1 + (1+\epsilon)\ell_2} & \text{if } R \lesssim \min\left(\frac{\ell_1}{\ell_2}, \frac{\ell_2}{\ell_1}\right) \\ k(m-k)m^6 \min\{q^{m\ell_1 + (mR+1)\ell_2}, q^{(Rm+1)\ell_1 + m\ell_2}\} & \text{else.} \end{cases}$

Consider Table 2. We will show that, ignoring polynomial factors, the complexities satisfy,

¹² A random matrix over any finite field is invertible with probability at least 0.2888.

New Recovery $\leq$ ACDGV Distinguisher $\leq$ Naive Recovery.

The second inequality is clear. For the first inequality, we will use the exact formula from Theorem 4 (the formula in Table 2 is just an upper bound) that says that the complexity of the new attack is $q^{b_1\ell_1+b_2\ell_2}$ where b_1, b_2 is chosen to minimize $b_1\ell_1 + b_2\ell_2$ subject to

$$\frac{\ell_2}{km} \leq b_1 \leq m, \quad \epsilon := \frac{m + \ell_1}{km} \leq b_2 \leq m, \quad b_1(b_2 - \epsilon) \geq km - \frac{1}{k} + \frac{1}{km}.$$

Since $\epsilon \leq 1$ for parameters of interest, the choice $b_1 = m$ and $b_2 = k + 1 = mR + 1$ is feasible and hence the complexity is at most $q^{m\ell_1 + (mR+1)\ell_2}$ which equals the ACDGV distinguisher's complexity, ignoring polynomial factors.

6.2 Concrete Complexity

Table 3 compares the work factor of various attacks on all proposed ACDGV parameter sets. Our attack generally performs best, except for a few cases where the ACDGV distinguisher is slightly better. However, unlike the distinguisher, our method recovers the full secret key. Compared to the naive algorithm, we are always at least as good, and often significantly better. Our attack breaks some of their proposed parameters.

7 Conclusion and Outlook

We presented a new key-recovery algorithm for the ACDGV scheme which exploits only the $\mathbb{F}_{q^m}$ -linear structure of the hidden vector code. It is better than the previously best-known distinguisher and significantly better than the naive approach.

It remains open whether the attack can be improved by assuming that the hidden code has Gabidulin structure. Indeed, if $m = n$, then a “right-side” variant of our attack can also be performed due to certain properties of Gabidulin codes (see [1, Lemma 1]). However, the complexity obtained is similar. It is unclear if the “left-” and “right-side” variants can be combined to obtain a better attack. Another approach is to study the dual code of the public code with respect to the Frobenius inner product (see [1, Prop. 1]). The corresponding search problem looks quite different and may allow for different attack techniques.

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ChatGPT 5.2 was valuable in finding mistakes in the tedious derivations of the polynomial cost, the convex optimization problems therein, and the integer programming problem of finding the closed-form expression of the new key recovery attack.

Table 3. Concrete work factors (in bits) for all proposed ACDGV parameter sets. An asterisk indicates lowest work factor; bold denotes values below security level. All parameters assume $m = n$. The naive, ACDGV distinguisher and our work factors are computed using Theorems 2, 3 and 4 (using the “Exact” formula) respectively. The costs include polynomial factors; these being, respectively, m^6 , m^6 and $k(m - k)m^6$. We assume an $\mathbb{F}_q$ -operation costs $\log_2(q)$ bits. The “pk size” and “ct size” columns refer to the public key and ciphertext size, respectively, where the Niederreiter-variant of the ACDGV scheme is assumed. The rate of the code k/m is denoted R .

NIST Security Level	q	R	m	ℓ_1	ℓ_2	pk size (kB)	ct size (B)	New	ACDGV Dist.	Naive
143 bits (AES-128)	16	0.30	23	0	5	39	242	196	*189	489
	16	0.57	23	1	1	41	138	*176	177	213
	2	0.46	37	4	0	70	111	* 112	179	179
	2	0.46	37	3	3	76	121	*193	196	253
	2	0.68	37	3	3	78	84	222	*220	253
	2	0.81	43	2	2	98	65	197	*191	205
	2	0.89	53	2	2	166	67	243	*236	246
207 bits (AES-192)	2	0.53	43	5	0	133	134	* 161	248	248
	2	0.70	47	5	0	173	112	*212	268	268
	2	0.77	53	4	0	230	106	*211	246	246
	2	0.86	59	2	2	268	89	264	*257	271
272 bits (AES-256)	16	0.31	29	2	1	87	334	* 223	303	379
	16	0.59	29	2	1	108	218	*295	335	379
	2	0.49	47	3	3	192	177	* 240	246	315
	2	0.70	53	3	2	274	140	* 262	269	299
	2	0.90	79	2	2	667	119	347	*340	354

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A ACDGV Scheme and Key Recovery (Section 2)

Definition 8 (ACDGV Public-Key Encryption Scheme [1, Sec. 5]).

- KeyGen* Draw a random $\mathbb{F}_{q^m}$ -linear $[n, k]$ Gabidulin code $\mathcal{C}_{vec}$ with $n = m$ and a random $\mathbb{F}_q$ -basis $\gamma \in \mathbb{F}_{q^m}^m$ of $\mathbb{F}_{q^m}$. Apply the hiding transform in Definition 1 to $\mathcal{C}_{mat} := \text{ext}_\gamma(\mathcal{C}_{vec})$ to obtain $\mathcal{C}_{pub} \subseteq \mathbb{F}_q^{(m+\ell_1) \times (n+\ell_2)}$. The public key is a random basis $(\mathbf{B}_1, \dots, \mathbf{B}_{km})$ of $\mathcal{C}_{pub}$. The secret key is $(\mathcal{C}_{vec}, \gamma, \mathbf{P}, \mathbf{Q})$ where $\mathbf{P} \in \mathbb{F}_q^{(m+\ell_1) \times (m+\ell_1)}$ and $\mathbf{Q} \in \mathbb{F}_q^{(n+\ell_2) \times (n+\ell_2)}$ are the scrambling matrices used in the hiding transform.
- Encrypt* To encrypt $\mathbf{m} \in \mathbb{F}_q^{km}$, draw a random matrix $\mathbf{E} \in \mathbb{F}_q^{(m+\ell_1) \times (n+\ell_2)}$ of rank $w \leq \frac{n-k}{2}$. The ciphertext is $\mathbf{C} = \sum_{i=1}^{km} m_i \mathbf{B}_i + \mathbf{E}$.
- Decrypt* Compute $\mathbf{P}^{-1} \mathbf{C} \mathbf{Q}^{-1}$ and then delete the last ℓ_1 rows and ℓ_2 columns. The result is $\sum_{i=1}^{km} m_i \mathbf{A}_i + \tilde{\mathbf{E}} \in \mathbb{F}_q^{m \times n}$ where $(\mathbf{A}_1, \dots, \mathbf{A}_{km})$ is a basis of $\mathcal{C}_{mat}$ and $\tilde{\mathbf{E}}$ has rank at most w . By applying $\text{ext}_\gamma^{-1}(\cdot)$ on this matrix, one obtains a corrupted codeword in the Gabidulin code $\mathcal{C}_{vec}$ with rank at most w . Hence it can be corrected to obtain $\mathbf{m}$.

We note ACDGV [1] also present a “Niederreiter”-variant of their scheme. However, the security is equivalent to the “McEliece”-variant described above.

Proposition 3 (Stabilizer Algebra of Matrix Gabidulin Code when $m \leq n$). Let $\mathcal{C}_{vec} \subseteq \mathbb{F}_{q^m}^n$ be an $[n, k]_{\mathbb{F}_{q^m}}$ Gabidulin code such that $m \leq n$. Then $\dim \text{Stab}(\text{ext}_\gamma(\mathcal{C}_{vec})) = m$ for any $\mathbb{F}_q$ -basis γ of $\mathbb{F}_{q^m}$.

Proof. Let $\mathcal{C}_{vec}$ be a Gabidulin code defined using the evaluation points $\mathbf{g} = (g_0, \dots, g_{n-1})^T \in \mathbb{F}_{q^m}^n$ where $\mathbf{g}$ spans $\mathbb{F}_{q^m}$. Thus, $\mathcal{C}_{vec}$ is the set of vectors obtained by evaluating all q -polynomials of q -degree less than k at $\mathbf{g}$, i.e., $\mathcal{C}_{vec} =$

$\{u(\mathbf{g}) \mid \deg_q u < k\}$. Any matrix $\mathbf{T}$ in $\mathbb{F}_q^{m \times m}$ can be uniquely represented as a q -polynomial f of q -degree at most $m-1$ where matrix multiplication corresponds to function composition. Thus, $\mathbf{T}\mathcal{C}_{\text{mat}} \subseteq \mathcal{C}_{\text{mat}}$ if and only if

$$\begin{aligned} \{f \circ u(\mathbf{g}) \bmod (x^{q^m} - x) \mid \deg_q u < k\} &\subseteq \{u(\mathbf{g}) \mid \deg_q u < k\} \\ \text{or } \{f \circ u \bmod (x^{q^m} - x) \mid \deg_q u < k\} &\subseteq \{u \mid \deg_q u < k\} \end{aligned}$$

as evaluation at $\mathbf{g} \in \mathbb{F}_{q^m}^n$ completely determines the q -polynomial since $m \leq n$. If $k \leq \deg_q f < m$, then $\deg_q f \circ u \geq k$ for $u = x$ and if $0 < \deg_q f < k$ then $\deg_q f \circ u = k$ for any u with $\deg_q u = k - \deg_q f$. Thus the above condition is satisfied if and only if $\deg_q f = 0$ which means that $f(x) = \alpha x$ for some $\alpha \in \mathbb{F}_{q^m}$. Thus $\mathbf{T}$ is isomorphic to $\mathbb{F}_{q^m}$ and the space of all such $\mathbf{T}$'s has dimension m . $\square$

Proposition 4 (ACDGV key recovery reduces to HVCS). *Let $\mathcal{D}$ be the uniform distribution over all $[m, k]_{\mathbb{F}_{q^m}}$ Gabidulin codes. Suppose a solver for the HVCS problem (Definition 3) returns a random solution in time $\frac{T}{N}$ where N is the number of solutions to the HVCS instance up to semilinear $\mathbb{F}_q$ -equivalence (Definition 5). Then such a solver can be used to obtain an equivalent secret key for the ACDGV scheme in time T .*

Proof. Run the solver on the ACDGV public key $\mathcal{C}_{\text{pub}} \subseteq \mathbb{F}_q^{(m+\ell_1) \times (n+\ell_2)}$ to obtain

$$(\mathbf{V}, \mathbf{W}, \gamma) \in \mathbb{F}_q^{(m+\ell_1) \times m} \times \mathbb{F}_q^{(n+\ell_1) \times n} \times \mathbb{F}_{q^m}^m$$

and let $\mathcal{C}_{\text{vec}} := \text{ext}_{\gamma}^{-1}(\mathbf{V}^T \mathcal{C}_{\text{pub}} \mathbf{W})$ be the corresponding $[n, k]_{\mathbb{F}_{q^m}}$ code. Since there can be multiple solutions, check if $\mathcal{C}_{\text{vec}}$ is a Gabidulin code (which can be done efficiently). If it is not, repeat until we do. A code that is semilinear $\mathbb{F}_q$ -equivalent to a Gabidulin code is still a Gabidulin code, so we will succeed when we hit any code equivalent to the secret Gabidulin code that was used to generate the ACDGV public key. For this we need about N repetitions and hence time $N \frac{T}{N} = T$.

Let the ciphertext be $\mathbf{C} = \sum_{i=1}^{km} m_i \mathbf{B}_i + \mathbf{E}$. Then,

$$\mathbf{V}^T \mathbf{C} \mathbf{W} = \sum_{i=1}^{km} m_i \mathbf{A}_i + \mathbf{V}^T \mathbf{E} \mathbf{W}$$

where $\mathbf{A}_i$ is a basis of $\text{ext}_{\gamma}(\mathcal{C}_{\text{vec}})$. Thus $\text{ext}_{\gamma}^{-1}(\mathbf{V}^T \mathbf{C} \mathbf{W})$ is a corrupted codeword of $\mathcal{C}_{\text{vec}}$ with an error of rank at most $\text{rank}(\mathbf{E})$. Given an arbitrary basis of a Gabidulin code, it is possible to efficiently decode [6] and hence correct this error. $\square$

B Proof of Proposition 1 (Multiple Solutions) in Section 2

Proof. For any invertible $\mathbf{S} \in \mathbb{F}_q^{n \times n}$ and integer j , we have

$$\begin{aligned} \mathbf{S}\mathcal{C}_{\text{vec}}^{[j]} &= \mathbf{S} \left(\text{ext}_{\gamma}^{-1}(\mathbf{V}^T \mathcal{C}_{\text{pub}} \mathbf{W}) \right)^{[j]} \\ &= \mathbf{S} \text{ext}_{\gamma^{[j]}}^{-1}(\mathbf{V}^T \mathcal{C}_{\text{pub}} \mathbf{W}) \\ &= \text{ext}_{\gamma^{[j]}}^{-1}(\mathbf{V}^T \mathcal{C}_{\text{pub}} \mathbf{W} \mathbf{S}^T). \end{aligned}$$

The second equality follows because, for all $\mathbf{x} \in \mathbb{F}_q^m$,

$$\left(\text{ext}_{\gamma}^{-1}(\mathbf{x}) \right)^{[j]} = \left(\sum_i x \gamma_i \right)^{[j]} = \sum_i x \gamma_i^{[j]} = \text{ext}_{\gamma^{[j]}}^{-1}(\mathbf{x}).$$

The third equality follows because, if $\mathcal{C}_{\text{vec}} = \text{ext}_{\gamma}^{-1}(\mathcal{C}_{\text{mat}})$ with $\mathcal{C}_{\text{mat}} \subseteq \mathbb{F}_q^{m \times n}$, then $\mathbf{S}\mathcal{C}_{\text{vec}} = \text{ext}_{\gamma}^{-1}(\mathcal{C}_{\text{mat}} \mathbf{S}^T)$, since $\mathcal{C}_{\text{vec}}$ consists of column vectors in $\mathbb{F}_q^n$ while the expanded codewords in $\mathcal{C}_{\text{mat}}$ are $\mathbb{F}_q^{m \times n}$ matrices.

Thus $(\mathbf{V}, \mathbf{W} \mathbf{S}^T, \gamma^{[j]})$ is also a solution with corresponding vector code $\mathbf{S}\mathcal{C}_{\text{vec}}^{[j]}$. $\square$

C Note on using $\mathbf{F}$ to remove random columns (Section 3)

If we obtain an $\mathbf{F}$ of the form $\mathbf{P} \begin{bmatrix} \mathbf{T} & \mathbf{0} \\ * & * \end{bmatrix} \mathbf{P}^{-1}$ where $\mathbf{T} \in \text{Stab}(\mathcal{C}_{\text{mat}})$, then one can recover the span of the last ℓ_1 columns of $\mathbf{P}$. We also assume $\dim \text{Stab}(\mathcal{C}_{\text{mat}}) = m$, so that $\text{Stab}(\mathcal{C}_{\text{mat}}) \cong \mathbb{F}_{q^m}$ and hence $\mathbf{T}^{q^m} = \mathbf{T}$. Then,

$$\mathbf{F}^{q^m} - \mathbf{F} = \mathbf{P} \left(\begin{bmatrix} \mathbf{T} & \mathbf{0} \\ * & * \end{bmatrix}^{q^m} - \begin{bmatrix} \mathbf{T} & \mathbf{0} \\ * & * \end{bmatrix} \right) \mathbf{P}^{-1} = \mathbf{P} \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ * & * \end{bmatrix} \mathbf{P}^{-1},$$

which indeed reveals the desired span. Thus, we can get rid of the random columns by Proposition 2.

D Proof of Lemma 1 (Linear Algebra) in Section 4

Lemma 1 (restated). *Let $\mathbf{B} \in \mathbb{F}_q^{m \times m}$ and let $f(x) \in \mathbb{F}_q[x]$ be its minimal polynomial.*

- (a) *The following statements are equivalent.*
 - (i) *$f(x)$ has degree m (or equivalently, the characteristic polynomial of $\mathbf{B}$ is the same as its minimal polynomial).*
 - (ii) *$\mathbf{B}$ is similar to the companion matrix of $f(x)$.*
 - (iii) *A matrix commutes with $\mathbf{B}$ iff it is a polynomial in $\mathbf{B}$.*

- (iv) There exists a $\mathbf{v} \in \mathbb{F}_q^m$ such that $(\mathbf{v}, \mathbf{B}\mathbf{v}, \dots, \mathbf{B}^{m-1}\mathbf{v})$ is a basis of $\mathbb{F}_q^m$.
- (b) The following statements are equivalent.
- (i) $f(x)$ has degree m and is irreducible (or equivalently, the characteristic polynomial of $\mathbf{B}$ is irreducible).
- (ii) $\mathbb{F}_q[\mathbf{B}] \cong \mathbb{F}_{q^m}$.
- (iii) For all non-zero $\mathbf{v} \in \mathbb{F}_q^m$, $(\mathbf{v}, \mathbf{B}\mathbf{v}, \dots, \mathbf{B}^{m-1}\mathbf{v})$ is a basis of $\mathbb{F}_q^m$.

Proof. (a)

((i) $\leftrightarrow$ (ii)) We have (i) $\rightarrow$ (ii) since the rational canonical form of $\mathbf{B}$ has a single block that equals the companion matrix $\mathbf{C}_f$ of $f(x)$. The reverse direction is clear.

((ii) $\leftrightarrow$ (iv)) The equivalence follows since the companion matrix of $\mathbf{B}$ is just $\mathbf{B}$ expressed with respect to the basis $\mathbf{v}, \mathbf{B}\mathbf{v}, \dots, \mathbf{B}^{m-1}\mathbf{v}$.

((iii) $\leftrightarrow$ (iv)) Suppose (iv). Certainly any polynomial in $\mathbf{B}$ commutes with $\mathbf{B}$. Now let $\mathbf{A} \in \mathbb{F}_q^m$ commute with $\mathbf{B}$. Since $\mathbf{v}, \mathbf{B}\mathbf{v}, \dots, \mathbf{B}^{m-1}\mathbf{v}$ is a basis, we can write $\mathbf{A}\mathbf{v} = \sum_{i=1}^m a_i \mathbf{B}^{i-1}\mathbf{v} = g(\mathbf{B})\mathbf{v}$ where we let $g(x) := a_0 + a_1x + \dots + a_{m-1}x^{m-1}$. Let $\mathbf{x} = \sum_i \lambda_i \mathbf{B}^{i-1}\mathbf{v} \in \mathbb{F}_q^m$ be arbitrary. Then

$$\begin{aligned} \mathbf{A}\mathbf{x} &= \mathbf{A} \sum_i \lambda_i \mathbf{B}^{i-1}\mathbf{v} = \sum_i \lambda_i \mathbf{B}^{i-1}\mathbf{A}\mathbf{v} = \sum_i \lambda_i \mathbf{B}^{i-1}g(\mathbf{B})\mathbf{v} \\ &= g(\mathbf{B}) \sum_i \lambda_i \mathbf{B}^{i-1}\mathbf{v} = g(\mathbf{B})\mathbf{x}. \end{aligned}$$

The reverse direction is more involved and appears in Exercise 19 in Hoffman and Kunze [10, Sec. 7.2]. It can be proved using the so-called cyclic decomposition theorem or, more generally, the structure theorem for finitely generated modules over a principal ideal domain.

(b)

((i) $\leftrightarrow$ (ii)) Both directions follow from $\mathbb{F}_q[\mathbf{B}] \cong \mathbb{F}_q[x]/\langle f(x) \rangle \cong \mathbb{F}_{q^m}$.

((i) $\leftrightarrow$ (iii)) Suppose (i). Let $\mathbf{x} \in \mathbb{F}_q^m$ be nonzero. Then $\{g(x) \in \mathbb{F}_q[x] \mid g(\mathbf{B})\mathbf{x} = \mathbf{0}\}$ is a principal ideal that contains the minimal polynomial $f(x)$ of $\mathbf{B}$. Since $f(x)$ is irreducible, the ideal is generated by $f(x)$ and so there are no polynomials $g(x)$ of degree smaller than m such that $g(\mathbf{B})\mathbf{x} = \mathbf{0}$. Hence, $\mathbf{x}, \mathbf{B}\mathbf{x}, \dots, \mathbf{B}^{m-1}\mathbf{x}$ are always linearly independent. For the reverse direction, we prove its contrapositive. That is, if $\deg f(x) < m$ or $f(x)$ is reducible, then there are nonzero vectors that are non-cyclic. If $\deg f(x) < m$ then no vector would be cyclic, so now suppose $\deg f(x) = m$ and that it $f(x)$ factors non-trivially as $f(x) = p(x)q(x)$. So $p(T)q(T) = 0$ and the kernel of $q(T)$ must be non-trivial since otherwise $p(T) = 0$. Thus, any vector in the kernel of $q(T)$ is non-cyclic.

□

E Lemma for Proposition 3 in Section 5

Lemma 3. Let $\mathbf{P} \in \mathbb{F}_q^{(m+\ell_1) \times (m+\ell_1)}$ and $\mathbf{Q} \in \mathbb{F}_q^{(n+\ell_2) \times (n+\ell_2)}$ be invertible matrices. Write $\mathbf{P} = [\mathbf{P}' \ \mathbf{P}'']$ and $\mathbf{Q} = \begin{bmatrix} \mathbf{Q}' \\ \mathbf{Q}'' \end{bmatrix}$ such that $\mathbf{P}''$ denotes the last ℓ_1 columns of $\mathbf{P}$ and $\mathbf{Q}''$ the last ℓ_2 rows of $\mathbf{Q}$. Suppose $\mathbf{V} \in \mathbb{F}_q^{b_1 \times m}$ and $\mathbf{W} \in \mathbb{F}_q^{b_2 \times n}$ are uniformly random matrices with full rank such that $\text{colspan}(\mathbf{V}) \subseteq \text{colspan}(\mathbf{P}'')^\perp$ and $\text{colspan}(\mathbf{W}) \subseteq \text{rowspan}(\mathbf{Q}'')^\perp$. Then $\mathbf{V}^\top \mathbf{P}' \in \mathbb{F}_q^{b_1 \times m}$ and $\mathbf{Q}' \mathbf{W} \in \mathbb{F}_q^{b_2 \times n}$ are uniformly random matrices with full rank where the probability is taken over the randomness of $\mathbf{V}$ and $\mathbf{W}$.

Proof. Since $\mathbf{P}$ is invertible, $\text{colspan}(\mathbf{P}')^\perp \cap \text{colspan}(\mathbf{P}'')^\perp = \{\mathbf{0}\}$ and, due to dimensionality reasons, $\text{colspan}(\mathbf{P}')^\perp \oplus \text{colspan}(\mathbf{P}'')^\perp = \mathbb{F}_q^{m+\ell_1}$. Note $\text{colspan}(\mathbf{P}')^\perp$ is just the left null space of $\mathbf{P}'$ and hence $\mathbf{x} \rightarrow \mathbf{x}^\top \mathbf{P}'$ is an isomorphism from $\text{colspan}(\mathbf{P}')^\perp$ to $\text{rowspan}(\mathbf{P}') = \mathbb{F}_q^m$. Thus, $\mathbf{V}^\top \mathbf{P}' \in \mathbb{F}_q^{b_1 \times m}$ is a random full-rank matrix. Similar arguments apply to $\mathbf{Q}' \mathbf{W} \in \mathbb{F}_q^{b_2 \times n}$. $\square$

F Polynomial Costs for Sections 4 and 5

Remark 9 (*$\mathcal{O}$ -notation for multivariate functions and linear dominance [21]*). Let $f, g: D \rightarrow \mathbb{R}_{\geq 0}$ be functions with common domain D . We write

$$f \preceq g$$

if there exists a constant $c > 0$ such that

$$f(x) \leq c g(x) \quad \text{for all } x \in D.$$

We also write

$$f \sim g$$

if both $f \preceq g$ and $g \preceq f$ hold.

Throughout the paper, the phrase “up to a constant factor” is used in precisely this sense. In Rutanen’s thesis [21], this relation is called “linear dominance.” It provides a reasonable notion of $\mathcal{O}$ -comparison for multivariate functions, especially for functions measuring algorithmic cost. As has been noted previously (e.g., [12] and [21, Sec. 1.6, “Problem 3”]), there is no universally agreed-upon definition of multivariate $\mathcal{O}$ -notation, and usage in the literature can be often inconsistent.

Remark 10 (*Cost of matrix multiplication and PLE (row echelon form) decomposition*). Let $\text{MM}(m, n, p)$ be the cost of multiplying an $m \times n$ and an $n \times p$ matrix. We have

$$\text{MM}(m, n, p) \preceq mnp \min\{m, n, p\}^{\omega-3}$$

field operations [14, Sec. 2.1], where $\omega \in [2, 3]$ is the matrix multiplication exponent. Jeannerod et al. [13, Sec. 5] define the *PLE decomposition* of a matrix $\mathbf{A} \in \mathbb{F}^{m \times n}$, that allows one to write

$$\bar{\mathbf{L}} \bar{\mathbf{P}} \mathbf{A} = \mathbf{E},$$

where $\bar{\mathbf{P}} \in \mathbb{F}^{m \times m}$ is a permutation matrix, $\bar{\mathbf{L}} \in \mathbb{F}^{m \times m}$ is lower triangular with the $(m-r) \times (m-r)$ identity $\mathbf{I}_{m-r}$ at the bottom right, and $\mathbf{E} \in \mathbb{F}^{m \times n}$ is in row echelon form. (The original definition in [13] writes $\mathbf{A} = \mathbf{P}\mathbf{L}\mathbf{E}$, where $\mathbf{P} := \bar{\mathbf{P}}^{-1}$ and $\mathbf{L} := \bar{\mathbf{L}}^{-1}$ have the same form as $\mathbf{P}$ and $\mathbf{L}$ respectively.) They show that such a decomposition can be computed using¹³

$$\text{PLE-Cost}(m, n, r) \preceq mn r^{\omega-2} \quad (\text{if } r > 0)$$

field operations.

Remark 11 (On the lemmas and corollaries that follow). The following treatment is somewhat lengthy, even though its purpose is only to determine the polynomial factors in the cost expressions. Moreover, a looser analysis might conceivably lead to the same polynomial factors. We nevertheless present the argument in this level of detail for the following reasons: First, we exploit the sparse structure as fully as possible in order to figure out what a tight cost bound looks like. Second, to simplify the cost expressions, we generally assume $n = m$ (or $n = k + 1$ for ACDGV-KeyDistinguish) and $\ell_1, \ell_2 \in \mathcal{O}(m)$. However, we do not assume any relation between k and m beyond $0 < k < m$. We think this is a reasonable choice, since that allows us to see how the cost depends on the rate of the code. Third, Lemmas 4 and 5 are intentionally stated in a general form, so that they can be specialized to other parameter regimes if desired.

Lemma 4 (General Cost of Computing Stabilizer Algebra Dimension).

Let $\mathbf{A}_1, \dots, \mathbf{A}_K \in \mathbb{F}_q^{m \times n}$ be a list of K matrices that span a space over $\mathbb{F}_q$ of dimension r where $1 \leq r \leq K < mn$. Let $\mathbf{A}'_1, \dots, \mathbf{A}'_\ell \in \mathbb{F}_q^{(m+c) \times n}$ be another list of ℓ matrices. Then the cost to determine the dimension of the solution space of the following system,

$$\mathbf{T}\mathbf{A}'_s - \sum_{i=1}^K \mu_{s,i} \mathbf{A}_i = \mathbf{0} \text{ for } s = 1, \dots, \ell,$$

in the unknowns $\mathbf{T} \in \mathbb{F}_q^{m \times (m+c)}$ and $\mu_{s,i} \in \mathbb{F}_q$ is at most

$$mnK r^{\omega-2} + \ell(mn-r)(m+c) \left(1 + m^{3-\omega} r^{\omega-2} + \min\{mn-r, m+c\}^{\omega-3} r \right. \\ \left. + m \min\{\ell(mn-r), m(m+c)\}^{\omega-2} \right).$$

$\mathbb{F}_q$ -operations (up to some constant factor). Here $\omega \in [2, 3]$ is the matrix multiplication exponent.

Proof. (See also Remarks 9 and 10 at the beginning of this section. In particular, $\text{MM}(\cdot)$ and $\text{PLE-Cost}(\cdot)$ are defined in the latter remark.) For a matrix $\mathbf{B} \in \mathbb{F}_q^{a \times b}$,

¹³ They essentially state it as $\text{PLE-Cost}(m, n, r) = \mathcal{O}(mn r^{\omega-2})$ but it seems this is the interpretation they meant. Indeed, their proof [13, Prop. 6] implies this.

let $\text{vec}(\mathbf{B}) \in \mathbb{F}_q^{ab}$ denote the “vectorization” of $\mathbf{B}$ obtained by stacking the rows of $\mathbf{B}$ into a single column vector. One can show the vectorized system is

$$(\mathbf{I}_m \otimes \mathbf{A}_s'^T) \text{vec}(\mathbf{T}) - \sum_{i=1}^K \mu_{s,i} \text{vec}(\mathbf{A}_i) = \mathbf{0} \text{ for } s = 1, \dots, \ell,$$

where $\mathbf{I}_m$ is the identity of size $m \times m$ and $\otimes$ denotes Kronecker product. We can write this as

$$\begin{bmatrix} \mathbf{U} & & \mathbf{S}_1 \\ & \mathbf{U} & \mathbf{S}_2 \\ & & \ddots \\ & & & \mathbf{U} \mathbf{S}_\ell \end{bmatrix} \begin{bmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \\ \vdots \\ \boldsymbol{\mu}_\ell \\ \mathbf{t} \end{bmatrix} = \mathbf{0} \quad (13)$$

where $\mathbf{U} := -[\text{vec}(\mathbf{A}_1) \dots \text{vec}(\mathbf{A}_K)] \in \mathbb{F}_q^{mn \times K}$, $\mathbf{S}_s := \mathbf{I}_m \otimes \mathbf{A}_s'^T \in \mathbb{F}_q^{mn \times m(m+c)}$ are known, and the unknowns are $\boldsymbol{\mu}_s := [\mu_{s,1}, \dots, \mu_{s,K}]^T$ and $\mathbf{t} := \text{vec}(\mathbf{T}) \in \mathbb{F}_q^{m(m+c)}$. Note that $\mathbf{U}$ has rank r .

To find the solution space dimension, we fill the rank of the matrix on the left-hand side. The high-level algorithm is as follows. First, we bring $\mathbf{U} \in \mathbb{F}_q^{mn \times K}$ to a row echelon form $\mathbf{E}$ (which has $mn - r$ zero rows at the bottom) by left-multiplying by an appropriate invertible matrix. Second, we apply the same invertible transformation to $\mathbf{S}_s \in \mathbb{F}_q^{mn \times m(m+c)}$ for each $s = 1, \dots, \ell$ to obtain $\tilde{\mathbf{S}}_s$. Thus the full system has rank

$$\ell r + \text{rank} \left(\begin{bmatrix} \text{last } mn - r \text{ rows of } \tilde{\mathbf{S}}_1 \\ \text{last } mn - r \text{ rows of } \tilde{\mathbf{S}}_2 \\ \vdots \\ \text{last } mn - r \text{ rows of } \tilde{\mathbf{S}}_\ell \end{bmatrix} \right). \quad (14)$$

Thus, as the third and final step, we need to compute the rank of the matrix on the right.

Now we look at each step in more detail. First, we do a PLE decomposition [13] of $\mathbf{U} \in \mathbb{F}_q^{mn \times K}$, so that $\bar{\mathbf{L}} \mathbf{P} \mathbf{U} = \mathbf{E}$, where $\mathbf{P} \in \mathbb{F}_q^{mn \times mn}$ is a permutation matrix, $\mathbf{E} \in \mathbb{F}_q^{mn \times K}$ is a row echelon form of $\mathbf{U}$, and $\bar{\mathbf{L}} \in \mathbb{F}_q^{mn \times mn}$ is lower triangular of the form

$$\bar{\mathbf{L}} = \begin{pmatrix} \mathbf{L}_1 & \mathbf{0} \\ \mathbf{L}_2 & \mathbf{I}_{mn-r} \end{pmatrix},$$

where $\mathbf{L}_1 \in \mathbb{F}_q^{r \times r}$ is also lower triangular and $\mathbf{L}_2 \in \mathbb{F}_q^{(mn-r) \times r}$. This costs $\text{PLE-Cost}(mn, K, r)$.

Second, we now replace all occurrences of $\mathbf{U}$ in the system above by $\mathbf{E}$. This corresponds to left-multiplying each block row by $\bar{\mathbf{L}} \mathbf{P}$ and hence, for each $s = 1, \dots, \ell$, we also need to left-multiply $\mathbf{S}_s \in \mathbb{F}_q^{mn \times m(m+c)}$ by $\bar{\mathbf{L}} \mathbf{P}$ to obtain

$\bar{\mathbf{L}} \bar{\mathbf{P}} \mathbf{S}_s \in \mathbb{F}_q^{mn \times m(m+c)}$. Let

$$\bar{\mathbf{P}} = \begin{pmatrix} \mathbf{P}' \\ \mathbf{P}'' \end{pmatrix} \quad \text{where } \mathbf{P}' \in \mathbb{F}_q^{r \times mn} \text{ and } \mathbf{P}'' \in \mathbb{F}_q^{(mn-r) \times mn}.$$

Then

$$\bar{\mathbf{P}} \mathbf{S}_s = \begin{pmatrix} \mathbf{P}' \mathbf{S}_s \\ \mathbf{P}'' \mathbf{S}_s \end{pmatrix} \quad \text{so that } \mathbf{P}' \mathbf{S}_s \in \mathbb{F}_q^{r \times m(m+c)} \text{ and } \mathbf{P}'' \mathbf{S}_s \in \mathbb{F}_q^{(mn-r) \times m(m+c)}.$$

Thus

$$\bar{\mathbf{L}} \bar{\mathbf{P}} \mathbf{S}_s = \begin{pmatrix} \mathbf{L}_1 & \mathbf{0} \\ \mathbf{L}_2 & \mathbf{I}_{mn-r} \end{pmatrix} \begin{pmatrix} \mathbf{P}' \mathbf{S}_s \\ \mathbf{P}'' \mathbf{S}_s \end{pmatrix} = \begin{pmatrix} \mathbf{L}_1 \mathbf{P}' \mathbf{S}_s \\ \mathbf{L}_2 \mathbf{P}' \mathbf{S}_s + \mathbf{P}'' \mathbf{S}_s \end{pmatrix}.$$

Now we only need to determine the rank of the last $mn - r$ rows. Hence we only need to compute $\mathbf{L}_2 \mathbf{P}' \mathbf{S}_s + \mathbf{P}'' \mathbf{S}_s$ for each $s = 1, \dots, \ell$. Now recall that $\mathbf{S}_s$ has a sparse block diagonal structure that we can exploit to further reduce this computation. We have

$$\mathbf{S}_s := \mathbf{I}_m \otimes \mathbf{A}_s'^T \in \mathbb{F}_q^{mn \times m(m+c)} = \begin{pmatrix} \mathbf{A}_s'^T & & \mathbf{0} \\ & \mathbf{A}_s'^T & \\ & & \ddots \\ \mathbf{0} & & & \mathbf{A}_s'^T \end{pmatrix}$$

where we have m diagonal blocks $\mathbf{A}_s'^T \in \mathbb{F}_q^{n \times (m+c)}$. The first term of $\mathbf{L}_2 \mathbf{P}' \mathbf{S}_s + \mathbf{P}'' \mathbf{S}_s$ is the multiplication of $\mathbf{L}_2 \in \mathbb{F}_q^{(mn-r) \times r}$ and $\mathbf{P}' \mathbf{S}_s \in \mathbb{F}_q^{r \times m(m+c)}$. Here $\mathbf{P}' \mathbf{S}_s \in \mathbb{F}_q^{r \times m(m+c)}$ equals some r rows of $\mathbf{S}_s$ in some permutation. If we permute the columns of $\mathbf{L}_2$ appropriately, then we can instead assume these r rows are in their original order. Thus, we can write

$$\mathbf{L}_2 \mathbf{P}' \mathbf{S}_s = \tilde{\mathbf{L}}_2 \begin{pmatrix} \mathbf{D}_1 & & \mathbf{0} \\ & \mathbf{D}_2 & \\ & & \ddots \\ \mathbf{0} & & & \mathbf{D}_m \end{pmatrix},$$

where $\tilde{\mathbf{L}}_2 \in \mathbb{F}_q^{(mn-r) \times r}$ is a column permutation of $\mathbf{L}_2$, and $\mathbf{D}_i \in \mathbb{F}_q^{r_i \times (m+c)}$ for $i = 1, \dots, m$, with $\sum_{i=1}^m r_i = r$. The block diagonal matrix is obtained by selecting r_i rows from the i -th diagonal block $\mathbf{A}_s'^T$ of $\mathbf{S}_s$. Thus, the cost of this multiplication is $\sum_{i=1}^m \text{MM}(mn - r, r_i, m + c)$. Finally, the cost of adding $\mathbf{P}'' \mathbf{S}_s$ to $\mathbf{L}_2 \mathbf{P}' \mathbf{S}_s$ is $(mn - r)(m + c)$ $\mathbb{F}_q$ -operations which is the number of entries that can be non-zero in the sparse $\mathbf{P}'' \mathbf{S}_s$. Thus the total cost, over all $s = 1, \dots, \ell$ to compute $\mathbf{L}_2 \mathbf{P}' \mathbf{S}_s + \mathbf{P}'' \mathbf{S}_s$ is $\ell((mn - r)(m + c) + \sum_{i=1}^m \text{MM}(mn - r, r_i, m + c))$.

Finally, we need to compute the rank of the stacked $\ell(mn - r) \times m(m + c)$ matrix on the right in (14). if r' is the rank of this matrix, then this has cost at most (note $mn - r > 0$),

$$\begin{aligned} & \text{PLE-Cost}(\ell(mn - r), m(m + c), r') \\ & \leq \text{PLE-Cost}(\ell(mn - r), m(m + c), \min\{\ell(mn - r), m(m + c)\}). \end{aligned}$$

The final cost is the sum of these three costs:

$$\begin{aligned} & \text{PLE-Cost}(mn, K, r) + \ell(mn - r)(m + c) + \sum_{i=1}^m \text{MM}(mn - r, r_i, m + c) \\ & + \text{PLE-Cost}(\ell(mn - r), m(m + c), \min\{\ell(mn - r), m(m + c)\}) \end{aligned}$$

Recall $\text{MM}(m, n, p) = mnp \min\{m, n, p\}^{\omega-3}$. Then we can upper bound the expression $\sum_{i=1}^m \text{MM}(mn - r, r_i, m + c)$ by

$$(mn - r)(m + c) \left(m^{3-\omega} r^{\omega-2} + \min\{mn - r, m + c\}^{\omega-3} r \right).$$

This can be seen as follows. Let $M := mn - r$, $P := m + c$, and $d := \min\{M, P\}$. Then

$$\begin{aligned} \sum_{i=1}^m \text{MM}(M, r_i, P) &= MP \sum_{i=1}^m r_i \min\{M, r_i, P\}^{\omega-3} \\ &\leq MP \left(\sum_{r_i \leq d} r_i^{\omega-2} + d^{\omega-3} \sum_{r_i > d} r_i \right). \end{aligned}$$

Let $\alpha := \omega - 2$. Since x^α is concave,

$$\sum_{r_i \leq d} r_i^\alpha \leq k \left(\frac{1}{k} \sum_{r_i \leq d} r_i \right)^\alpha \leq m^{1-\alpha} \left(\sum_i r_i \right)^\alpha,$$

where $k := |\{i \mid r_i \leq d\}| \leq m$. Thus, $\sum_{r_i \leq d} r_i^{\omega-2} \leq m^{3-\omega} r^{\omega-2}$, and also trivially, $\sum_{r_i > d} r_i \leq r$, which gives the stated upper bound. (If $\omega = 2$ so that $\alpha = 0$, we restrict to only positive r_i 's in the summation above.)

Hence, together with the substitution $\text{PLE-Cost}(m, n, r) = mnr^{\omega-2}$, we get

$$\begin{aligned} & mnK r^{\omega-2} + \ell(mn - r)(m + c) \left(1 + m^{3-\omega} r^{\omega-2} + \min\{mn - r, m + c\}^{\omega-3} r \right) \\ & + \ell(mn - r)(m + c) m \min\{\ell(mn - r), m(m + c)\}^{\omega-2} \\ & = mnK r^{\omega-2} + \ell(mn - r)(m + c) \left(1 + m^{3-\omega} r^{\omega-2} + \min\{mn - r, m + c\}^{\omega-3} r \right. \\ & \quad \left. + m \min\{\ell(mn - r), m(m + c)\}^{\omega-2} \right). \end{aligned}$$

□

Lemma 5 (General Cost of Computing Stabilizer Algebra Basis). *Consider the system in Lemma 4. If we want to also compute the kernel of the system, then we need to additionally do*

$$\ell r(m + c) \left(m^{3-\omega} r^{\omega-2} + r \min\{r, m + c\}^{\omega-3} \right)$$

field operations up to a constant factor.

Proof. Note that in the proof of Lemma 4, we can easily bring (13) into row echelon form (REF) after we also compute $\mathbf{L}_1 \mathbf{P}' \mathbf{S}_s$ (which was ignored in the second step). After this, we only need to bring it to reduced row echelon form (RREF), and then we can directly read off a basis of the kernel from this RREF. We can ignore the cost of turning the REF form to RREF since this will be easily dominated by the complexity of the obtaining the REF form. Thus we only consider the cost of computing $\mathbf{L}_1 \mathbf{P}' \mathbf{S}_s$.

In the second step, we only accounted for the cost of forming the last $mn - r$ rows $\mathbf{L}_2 \mathbf{P}' \mathbf{S}_s + \mathbf{P}'' \mathbf{S}_s$. We now bound the cost of also forming the first r rows $\mathbf{L}_1 \mathbf{P}' \mathbf{S}_s$ (ignoring any triangular structure in $\mathbf{L}_1$ and treating it as dense). As above, $\mathbf{P}' \mathbf{S}_s \in \mathbb{F}_q^{r \times m(m+c)}$ consists of r selected rows of $\mathbf{S}_s$, and after a suitable column permutation of $\mathbf{L}_1$ we may assume these rows are in their original order. Hence

$$\mathbf{P}' \mathbf{S}_s = \text{diag}(\mathbf{D}_1, \dots, \mathbf{D}_m), \quad \mathbf{D}_i \in \mathbb{F}_q^{r_i \times (m+c)}, \quad \sum_{i=1}^m r_i = r.$$

Partition $\mathbf{L}_1 \in \mathbb{F}_q^{r \times r}$ conformally as $\mathbf{L}_1 = [\mathbf{L}_1^{(1)} \dots \mathbf{L}_1^{(m)}]$ with $\mathbf{L}_1^{(i)} \in \mathbb{F}_q^{r \times r_i}$. Then

$$\mathbf{L}_1 \mathbf{P}' \mathbf{S}_s = [\mathbf{L}_1^{(1)} \mathbf{D}_1 \dots \mathbf{L}_1^{(m)} \mathbf{D}_m],$$

so the cost for a fixed s is $\sum_{i=1}^m \text{MM}(r, r_i, m+c)$. Over all $s = 1, \dots, \ell$ this contributes $\ell \sum_{i=1}^m \text{MM}(r, r_i, m+c)$ operations. (If $r_i = 0$, then define $\text{MM}(\cdot, r_i, \cdot) = 0$.)

Using $\text{MM}(a, b, p) = abp \min\{a, b, p\}^{\omega-3}$ and the same concavity argument as before (with $P := m+c$ and $d := \min\{r, P\}$), we obtain

$$\sum_{i=1}^m \text{MM}(r, r_i, m+c) \leq r(m+c) \left(m^{3-\omega} r^{\omega-2} + \min\{r, m+c\}^{\omega-3} r \right).$$

Thus computing all matrices $\mathbf{L}_1 \mathbf{P}' \mathbf{S}_s$ for $s = 1, \dots, \ell$ costs at most

$$\ell r(m+c) \left(m^{3-\omega} r^{\omega-2} + \min\{r, m+c\}^{\omega-3} r \right)$$

field operations up to a constant factor. $\square$

Corollary 1 (Specialized Costs of Computing Stabilizer Algebra). *Let $\mathbf{A}_1, \dots, \mathbf{A}_K \in \mathbb{F}_q^{m \times n}$ be matrices with $K := km$ spanning a km -dimensional $\mathbb{F}_q$ -subspace, where $0 < k < m$ and $0 < k < n$. Consider the system*

$$\mathbf{T} \mathbf{A}_s - \sum_{i=1}^K \mu_{s,i} \mathbf{A}_i = \mathbf{0} \quad \text{for } s = 1, \dots, \ell,$$

in the unknowns $\mathbf{T} \in \mathbb{F}_q^{m \times m}$ and $\mu_{s,i} \in \mathbb{F}_q$. Let $C_{\dim}$ denote the cost of computing the dimension of the solution space and $C_{\ker}$ the cost of computing a basis of the solution space. Let $\ell_{\min}$ be the smallest positive integer satisfying $\ell mn \geq K\ell + m^2$. (In particular, $\ell_{\min} = \lceil m/(m-k) \rceil$ when $n = m$, and $\ell_{\min} = m$ when $n = k+1$.)

Then the costs $C_{\dim}$ and $C_{\ker}$ in $\mathbb{F}_q$ -operations, for various choices of (n, ℓ) , are upper bounded up to a constant factor as follows.

#	(n, ℓ)	$C_{\dim}$	$C_{\ker}$
1	(m, km)	$k(m-k)m^{2\omega}$	$\max\{k(m-k)m^{2\omega}, k^\omega m^4\}$
2	$(m, \ell_{\min})$	$m^{2\omega}$	$\max\left\{m^{2\omega}, \frac{k^{\omega-1}m^4}{m-k}\right\}$
3	$(k+1, km)$	$k m^{2\omega}$	$\max\{k m^{2\omega}, k^\omega m^4\}$
4	$(k+1, \ell_{\min})$	$m^{2\omega}$	$\max\{m^{2\omega}, k^{\omega-1}m^4\}$

(The values of $C_{\ker}$ in (3) and (4) are not needed later and just stated for completeness.)

Proof. (See Remark 9 at the beginning of this section for the definitions of $\preceq$ and $\sim$.) Throughout, apply Lemma 4 with $c = 0$ and $K = r = km$. This gives

$$\begin{aligned}
C_{\dim} &\preceq k^{\omega-1}m^\omega n + \ell m^2(n-k) \left(1 + k^{\omega-2}m + km^{\omega-2} \right. \\
&\quad \left. + m^{\omega-1} \min\{\ell(n-k), m\}^{\omega-2} \right) \\
&\sim k^{\omega-1}m^\omega n + \ell m^2(n-k) \left(k^{\omega-2}m + m^{\omega-1} \min\{\ell(n-k), m\}^{\omega-2} \right) \\
&\quad (\text{since } k^{\omega-2}m \text{ dominates } 1 \text{ and } km^{\omega-2}).
\end{aligned}$$

Now for the cost to compute the kernel, we have by Lemma 5,

$$\begin{aligned}
C_{\ker} &\preceq C_{\dim} + \ell r m \left(m^{3-\omega} r^{\omega-2} + r m^{\omega-3} \right) \\
&= C_{\dim} + \ell(km)m \left(k^{\omega-2}m + k m^{\omega-2} \right) \\
&\sim C_{\dim} + \ell k^{\omega-1}m^3.
\end{aligned}$$

(1) Set $n = m$, $\ell = km$. Therefore,

$$\begin{aligned}
C_{\dim} &\preceq k^{\omega-1}m^{\omega+1} + km^3(m-k) \left(k^{\omega-2}m + m^{2\omega-3} \right) \\
&\sim k(m-k)m^{2\omega} \quad (\text{the term } m^{2\omega-3} \text{ dominates in the parentheses}).
\end{aligned}$$

For the kernel computation,

$$C_{\ker} \preceq C_{\dim} + k^\omega m^4 \preceq \max\{k(m-k)m^{2\omega}, k^\omega m^4\}.$$

(2) Set $n = m$, $\ell = \ell_{\min} = \lceil m/(m-k) \rceil$. Therefore,

$$\begin{aligned}
C_{\dim} &\preceq k^{\omega-1}m^{\omega+1} + \ell m^2(m-k) \left(k^{\omega-2}m + m^{2\omega-3} \right) \\
&\sim k^{\omega-1}m^{\omega+1} + k^{\omega-2}m^4 + m^{2\omega} \\
&\sim m^{2\omega}.
\end{aligned}$$

For the kernel computation,

$$C_{\text{ker}} \preceq C_{\text{dim}} + \ell k^{\omega-1} m^3 \preceq \max \left\{ m^{2\omega}, \frac{k^{\omega-1} m^4}{m-k} \right\}.$$

(3) Set $n = k + 1$, $\ell = km$. Hence,

$$\begin{aligned} C_{\text{dim}} &\preceq k^{\omega-1} m^{\omega} (k+1) + km^3 (k^{\omega-2} m + m^{2\omega-3}) \\ &\sim k^{\omega} m^{\omega} + k^{\omega-1} m^4 + km^{2\omega} \\ &\sim km^{2\omega}. \end{aligned}$$

For the kernel computation,

$$C_{\text{ker}} \preceq C_{\text{dim}} + k^{\omega} m^4 \preceq \max \{ km^{2\omega}, k^{\omega} m^4 \}.$$

(4) Set $n = k + 1$, $\ell = \ell_{\min} = m$. Thus,

$$\begin{aligned} C_{\text{dim}} &\preceq k^{\omega-1} m^{\omega} (k+1) + m^3 (k^{\omega-2} m + m^{2\omega-3}) \\ &\sim k^{\omega} m^{\omega} + k^{\omega-2} m^4 + m^{2\omega} \\ &\sim m^{2\omega}. \end{aligned}$$

For the kernel computation,

$$C_{\text{ker}} \preceq C_{\text{dim}} + k^{\omega-1} m^4 \preceq \max \{ m^{2\omega}, k^{\omega-1} m^4 \}.$$

□

Corollary 2 (Cost of Solving New Key Recovery System). *Suppose the matrices $\mathbf{B}_1, \dots, \mathbf{B}_{km} \in \mathbb{F}_q^{(m+\ell_1) \times (m+\ell_2)}$ span a km -dimensional space where $0 < k < m$. Let $\mathbf{V} \in \mathbb{F}_q^{(m+\ell_1) \times b_1}$ and $\mathbf{W} \in \mathbb{F}_q^{(m+\ell_2) \times b_2}$ be full-rank matrices. Assume $\mathbf{V}^T \mathbf{B}_i \mathbf{W}$ for $i = 1, \dots, km$ span a km -dimensional space (note in particular $b_1 b_2 > km$). Consider the following system in the unknowns $\mathbf{F} \in \mathbb{F}_q^{(m+\ell_1) \times (m+\ell_1)}$ and $\mu_{s,i} \in \mathbb{F}_q$,*

$$\mathbf{V}^T \left(\mathbf{F} \mathbf{B}_s - \sum_{i=1}^{km} \mu_{s,i} \mathbf{B}_i \right) \mathbf{W} = \mathbf{0} \text{ for } s = 1, \dots, km.$$

If $b_1, b_2 \leq m$ and $\ell_1, \ell_2 \in \mathcal{O}(m)$, then the cost C_{dim} of computing the dimension of the solution space and the cost C_{ker} of computing a basis of the solution of the above system is, in $\mathbb{F}_q$ -operations,

$$\begin{aligned} C_{\text{dim}} &\preceq \max \{ k(m-k)m^{2\omega}, k^2(m-k)m^4 \} \\ C_{\text{ker}} &\preceq \max \{ k(m-k)m^{2\omega}, k^2(m-k)m^4, k^{\omega} m^4 \}. \end{aligned}$$

Proof. (See also Remarks 9 and 10 at the beginning of this section for the definitions of $\preceq$, $\sim$ and $\text{MM}(\cdot)$.) Note $\ell = km$. Rewrite the equations as

$$(\mathbf{V}^\top \mathbf{F})(\mathbf{B}_s \mathbf{W}) - \sum_{i=1}^{km} \mu_{s,i} (\mathbf{V}^\top \mathbf{B}_i \mathbf{W}) = \mathbf{0} \quad (s = 1, \dots, km).$$

Set

$$\mathbf{T} := \mathbf{V}^\top \mathbf{F} \in \mathbb{F}_q^{b_1 \times (m + \ell_1)}, \quad \mathbf{A}'_s := \mathbf{B}_s \mathbf{W} \in \mathbb{F}_q^{(m + \ell_1) \times b_2}, \quad \mathbf{A}_i := \mathbf{V}^\top \mathbf{B}_i \mathbf{W} \in \mathbb{F}_q^{b_1 \times b_2}.$$

Then the system is of the form from Lemma 4:

$$\mathbf{T} \mathbf{A}'_s - \sum_{i=1}^{km} \mu_{s,i} \mathbf{A}_i = \mathbf{0} \quad (s = 1, \dots, km).$$

These matrix multiplications in total cost no more than $km \cdot \text{MM}(m, m, m) \leq (km)m^\omega$ which can be neglected since the cost of solving the system will turn out to be much higher, as we will see. Hence in Lemma 4 we substitute

$$m \leftarrow b_1, \quad n \leftarrow b_2, \quad c \leftarrow (m + \ell_1) - b_1, \quad K \leftarrow km, \quad r \leftarrow km, \quad \ell \leftarrow km,$$

which gives

$$\begin{aligned} C_{\dim} &\preceq b_1 b_2 (km) (km)^{\omega-2} \\ &\quad + km (b_1 b_2 - km) (m + \ell_1) \\ &\quad \left(1 + b_1^{3-\omega} (km)^{\omega-2} + (km) \min\{b_1 b_2 - km, m + \ell_1\}^{\omega-3} \right. \\ &\quad \left. + b_1 \min\{km(b_1 b_2 - km), b_1(m + \ell_1)\}^{\omega-2} \right). \end{aligned}$$

If we substitute $b_1, b_2, \ell_1 \leftarrow m$, except for in the $\min\{b_1 b_2 - km, m + \ell_1\}^{\omega-3}$ term which we replace by 1 since it has a non-positive exponent, we get the upper bound,

$$\begin{aligned} C_{\dim} &\preceq m^2 (km) (km)^{\omega-2} \\ &\quad + km (m^2 - km) m \\ &\quad \left(1 + k^{\omega-2} m + km + m \min\{km(m^2 - km), m^2\}^{\omega-2} \right) \\ &= k^{\omega-1} m^{\omega+1} + k m^3 (m - k) \left(1 + k^{\omega-2} m + km + m^{2\omega-3} \right) \\ &\sim k^{\omega-1} m^{\omega+1} + k m^3 (m - k) (km + m^{2\omega-3}) \\ &\sim k m^3 (m - k) (km + m^{2\omega-3}) \\ &\sim \max\{k(m - k)m^{2\omega}, k^2(m - k)m^4\}. \end{aligned}$$

To additionally compute a basis of the kernel, we add the cost from Lemma 5. With the same substitutions

$$m \leftarrow b_1, \quad n \leftarrow b_2, \quad c \leftarrow (m + \ell_1) - b_1, \quad K \leftarrow km, \quad r \leftarrow km, \quad \ell \leftarrow km,$$

we get

$$\begin{aligned} C_{\ker} &\preceq C_{\dim} + \ell r (m + c) \left(m^{3-\omega} r^{\omega-2} + \min\{r, m + c\}^{\omega-3} r \right) \\ &= C_{\dim} + (km)(km)(m + \ell_1) \left(b_1^{3-\omega} (km)^{\omega-2} + \min\{km, m + \ell_1\}^{\omega-3} (km) \right). \end{aligned}$$

since $b_1 \leq m$; $m + \ell_1 \sim m$; and $\min\{km, m + \ell_1\} \sim m$,

$$\begin{aligned} &\preceq C_{\dim} + (km)(km)m \left(m^{3-\omega} (km)^{\omega-2} + m^{\omega-3} (km) \right) \\ &= C_{\dim} + k^\omega m^4 + k^3 m^{\omega+1} \\ &\sim C_{\dim} + k^\omega m^4 \\ &\sim \max\{k(m-k)m^{2\omega}, k^2(m-k)m^4, k^\omega m^4\}. \end{aligned}$$

□

Corollary 3 (Polynomial Factors of all Algorithms). *The polynomial factors in the cost expression, i.e., the cost of “one iteration” when applicable, of BasisRecover, NaiveRecover, ACDGV-KeyDistinguish, and NewKeyRecover (Algorithms 1, 2, 3, 4 respectively) is at most the following (see Theorems 1, 2, 3, 4). These costs are in terms of $\mathbb{F}_q$ -operations and are stated up to a constant factor. The matrix multiplication exponent is $\omega \in [2, 3]$. We assume $m = n$ (except for ACDGV-KeyDistinguish where $n = k + 1$), $0 < k < m = n$, and $\ell_1, \ell_2 \in \mathcal{O}(m)$.*

Algorithm	With heuristic	Without heuristic
BasisRecover		
if $\dim \text{Stab}(C_{\text{mat}}) \neq m$	$m^{2\omega}$	$k(m-k)m^{2\omega}$
if $\dim \text{Stab}(C_{\text{mat}}) = m$	$\max\left\{m^{2\omega}, \frac{k^{\omega-1}m^4}{m-k}\right\}$	$\max\{k(m-k)m^{2\omega}, k^\omega m^4\}$
NaiveRecover	$m^{2\omega}$	$\max\{k(m-k)m^{2\omega}, k^\omega m^4\}$
ACDGV-KeyDistinguish	$m^{2\omega}$	$k m^{2\omega}$
NewKeyRecover	$\max\left\{k(m-k)m^{2\omega}, k^2(m-k)m^4\right\}$	$\max\left\{k(m-k)m^{2\omega}, k^2(m-k)m^4, k^\omega m^4\right\}$

If we let $\omega = 3$,

Algorithm	With heuristic	Without heuristic
BasisRecover		
if $\dim \text{Stab}(\mathcal{C}_{\text{mat}}) \neq m$	m^6	$k(m-k)m^6$
if $\dim \text{Stab}(\mathcal{C}_{\text{mat}}) = m$	m^6	$k(m-k)m^6$
NaiveRecover	m^6	$k(m-k)m^6$
ACDGV-KeyDistinguish	m^6	$k m^6$
NewKeyRecover	$k(m-k)m^6$	$k(m-k)m^6$

(Here $\mathcal{C}_{\text{mat}}$ represents the matrix code whose basis is input to **BasisRecover**.)

The “with/without heuristic” columns refer to the following heuristics:

1. **BasisRecover**: If $\mathcal{S} \xleftarrow{\$} \text{Stab}_{\text{samp}}(\mathcal{C}_{\text{mat}})$, then $\dim \mathcal{S} = \dim \text{Stab}(\mathcal{C}_{\text{mat}})$.
2. **NaiveKeyRecover**: If $\mathcal{S} \xleftarrow{\$} \text{Stab}_{\text{samp}}(\mathbf{V}^T \mathcal{C}_{\text{pub}} \mathbf{W})$, then $\dim \mathcal{S} \neq m$ except with negligible probability in m (over randomness of choice of $\mathbf{V}$, $\mathbf{W}$ and $\mathcal{S}$).
3. **ACDGV-KeyDistinguish**: The probability that $\dim \text{Stab}(\mathcal{S}) \geq m$ is at most $cq^{-(m\ell_1 + (k+1)\ell_2)}$ for some constant $c < 1/16$, where $\mathcal{S} \xleftarrow{\$} \text{Stab}_{\text{samp}}(\mathcal{C})$ for a random km -dimensional code $\mathcal{C}$.
4. **NewKeyRecover**: The probability that (7) in Algorithm 4 has a solution space of dimension greater than $m + (m + \ell_1 - b_1)(m + \ell_1)$ is negligible in m (over randomness of choice of $\mathbf{V}$ and $\mathbf{W}$ in the algorithm).

Proof. The proofs largely follow from Corollaries 1 and 2.

1. **BasisRecover** computes the dimension of the stabilizer algebra and aborts if it is not equal to m . If it equals m , then it continues to compute a basis of the stabilizer algebra. These two cases cost C_{dim} and C_{ker} respectively with $(n, \ell) = (m, km)$ in Corollary 1. The heuristic allows us to do the computation using fewer basis equations as in Remark 6 in Section 4. Thus, now the costs are C_{dim} and C_{ker} with $\ell = \ell_{\min}$ in the corollary. Note these costs easily dominate the cost of the entire algorithm.
2. **NaiveRecover** is similar: it computes the dimension of the stabilizer algebra and aborts (i.e., starts a new iteration) if it is not equal to m . If it equals m , then it continues to compute a basis of the stabilizer algebra. These respectively cost C_{dim} and C_{ker} , corresponding to $(n, \ell) = (m, km)$ in the corollary. Without any heuristic, the cost is the maximum of the two which is C_{ker} . With the heuristic, we can use fewer basis equations *and* we are guaranteed to abort in all iterations except for a negligible number of them. Thus, the cost reduces to C_{dim} with $\ell = \ell_{\min}$.
3. **ACDGV-KeyDistinguish** only needs to compute the dimension of the stabilizer algebra and hence costs C_{dim} with $(n, \ell) = (k+1, km)$. With the heuristic, we can safely use $\ell = \ell_{\min}$ basis equations.
4. **NewKeyRecover** computes a basis of the solution space of (7) with complexity C_{ker} as in Corollary 2. This is also the polynomial cost of the algorithm, since

as we will show, finding the dimension of the solution space of (7) dominates the complexity of the entire algorithm. The heuristic says that (7) does not meet the dimension condition (i.e., fails the “until” condition) except a negligible number of times, so we abort early after computing the dimension, which costs C_{dim} , almost all the time.

We now show that, in **NewKeyRecover**, all the following steps after Line 5 cost no more than the cost of computing the dimension of (7). Consider Part (ii) (Lines 6–12). The most expensive step is Line 9:

$$\mathcal{L} \leftarrow \mathcal{L} + \text{rowspan} \left(\mathbf{V}^\top \left(\mathbf{F}\mathbf{B}_s - \sum_{i=1}^{km} \mu_{s,i} \mathbf{B}_i \right) \right)$$

for $s = 1, \dots, km$. Multiplying out the $\mathbf{V}^\top$ costs nothing as we can directly draw $\mathbf{V}^\top \mathbf{F}$ from the solution space of (7). This is because we can solve (7) by treating $\mathbf{V}^\top \mathbf{F}$ as an unknown (in fact, this is how we do it in Corollary 2). Further, $\mathbf{V}^\top \mathbf{B}$ can be computed and stored while solving (7). To compute $\mathcal{L} + \text{rowspan}(\dots)$, we can compute the row echelon form of the matrix obtained by vertically concatenating the matrix in the argument of $\text{rowspan}(\cdot)$ with a basis of $\mathcal{L}$ as rows (which cannot be more than ℓ_2 since otherwise we can already abort the loop). The number of rows and columns in this matrix is $\mathcal{O}(m)$ (recall we assume $\ell_1, \ell_2 \in \mathcal{O}(\ell)$ and $m = n$), so up to a constant factor this computation costs m^ω . Doing this km times, the cost becomes $km^{\omega+1}$ which is dominated by the cost of solving (7).

Now consider Part (iii) (Lines 12–18). The most expensive step is solving (8) in Line 16 which we write as,

$$\mathbf{F}^{(1)} \mathbf{B}'_s = \left[\mathbf{I}_m \quad \widetilde{\mathbf{F}}^{(2)} \right] \mathbf{Y}_s \quad \text{where} \quad \mathbf{Y}_s := \mathbf{M} \sum_{i=1}^{km} \mu_{s,i} \mathbf{B}'_i \in \mathbb{F}_q^{(m+\ell_1) \times n} \quad (15)$$

for $s = 1, \dots, km$. Here the unknowns are $\mathbf{F}^{(1)} \in \mathbb{F}_q^{m \times (m+\ell_1)}$ and $\widetilde{\mathbf{F}}^{(2)} \in \mathbb{F}_q^{m \times \ell_1}$. If we partition $\mathbf{Y}_s$ as

$$\mathbf{Y}_s = \begin{bmatrix} \mathbf{Y}'_s \\ \mathbf{Y}''_s \end{bmatrix}, \quad \mathbf{Y}'_s \in \mathbb{F}_q^{m \times n}, \quad \mathbf{Y}''_s \in \mathbb{F}_q^{\ell_1 \times n},$$

then it can be shown that (15) over $s = 1, \dots, km$ can be rewritten as

$$\begin{bmatrix} \mathbf{F}^{(1)} & \mathbf{F}^{(2)} \end{bmatrix} \begin{bmatrix} \mathbf{B}'_1 & \cdots & \mathbf{B}'_{km} \\ -\mathbf{Y}''_1 & \cdots & -\mathbf{Y}''_{km} \end{bmatrix} = \begin{bmatrix} \mathbf{Y}'_1 & \cdots & \mathbf{Y}'_{km} \end{bmatrix}.$$

This can be solved by computing the row echelon form of

$$\begin{bmatrix} \mathbf{B}'_1 & \cdots & \mathbf{B}'_{km} \\ \mathbf{Y}''_1 & \cdots & \mathbf{Y}''_{km} \end{bmatrix}$$

which costs up to a constant factor, $km^2 \cdot m \cdot m^{\omega-2} = km^{\omega+1}$. This again does not affect the overall cost.

The same is true for the final call to **BasisRecover** (see the costs in the table of the corollary statement).

As a side remark, note that Corollaries 1 and 2 assume the input list of matrices are linearly independent over $\mathbb{F}_q$, i.e., they span a km -dimensional code. However, in the actual algorithms we do not have such a guarantee, e.g., in **NaiveRecover**, if $\mathbf{A}'_i \leftarrow \mathbf{V}^\top \mathbf{B}_i \mathbf{W}$, then $\mathbf{A}'_1, \dots, \mathbf{A}'_{km}$ may not be linearly independent. But this does not matter because (1) the computation of the dimension of the stabilizer algebra implicitly includes finding the dimension of this code (this happens when we find the rank of $\mathbf{U}$ in (13)); and (2) so, if we find the list of matrices do not form a basis, we can abort early since this cannot correspond to a success case in any of the algorithms. $\square$

G Closed-form Exponential Cost of New Key Recovery (Section 5)

Lemma 6. *Let m, n, ℓ_1, ℓ_2 be integers where $m > 0$, $n > 0$, and $\ell_1, \ell_2 \geq 0$, and let $z, \epsilon_1, \epsilon_2$ be real numbers such that $0 \leq \epsilon_1 \leq m$, $0 \leq \epsilon_2 \leq n$, and $0 < z$. Let b_1, b_2 range over the positive integers subject to the constraints*

$$\begin{aligned} \epsilon_1 &\leq b_1 \leq m, \\ \epsilon_2 &\leq b_2 \leq n, \quad \text{and} \\ z &\leq b_1(b_2 - \epsilon_2). \end{aligned}$$

Then assuming $m(n - \epsilon_2) \geq z$ (so that there is at least one feasible solution), the minimum value of $b_1\ell_1 + b_2\ell_2$ is upper bounded by

$$\begin{cases} 2\sqrt{z\ell_1\ell_2} + \ell_1 + (1 + \epsilon_2)\ell_2 & \text{if } \ell_1\ell_2 \neq 0 \\ \text{and } \epsilon_1^2 \frac{\ell_1}{\ell_2} \leq z \leq \min \left\{ m^2 \frac{\ell_1}{\ell_2}, (n - \epsilon_2)^2 \frac{\ell_2}{\ell_1} \right\}, \\ \min \left\{ m\ell_1 + \left\lceil \frac{z}{m} + \epsilon_2 \right\rceil \ell_2, \max \left\{ \lceil \epsilon_1 \rceil, \left\lceil \frac{z}{n - \epsilon_2} \right\rceil \right\} \ell_1 + n\ell_2 \right\} & \text{otherwise.} \end{cases}$$

Moreover, in the first case (i.e., “if $\ell_1\ell_2 \neq 0$ and ...”), the following choice of (b_1, b_2) is feasible and, when substituted into $b_1\ell_1 + b_2\ell_2$, satisfies the stated upper bound (not necessarily with equality):

$$(b_1, b_2) = \left(\left\lceil \sqrt{z \frac{\ell_2}{\ell_1}} \right\rceil, \left\lceil \sqrt{z \frac{\ell_1}{\ell_2}} + \epsilon_2 \right\rceil \right).$$

In the second (“otherwise”) case, the same holds for one of the points

$$(b_1, b_2) = \left(m, \left\lceil \frac{z}{m} + \epsilon_2 \right\rceil \right) \quad \text{or} \quad (b_1, b_2) = \left(\max \left\{ \lceil \epsilon_1 \rceil, \left\lceil \frac{z}{n - \epsilon_2} \right\rceil \right\}, n \right).$$

Proof. Suppose $\ell_1\ell_2 \neq 0$. Consider the continuous relaxation of the problem, i.e., where b_1 and b_2 are allowed to take on any real values. Since $b_1\ell_1 + b_2\ell_2$ is increasing in both b_1 and b_2 , it is clear the optimal solution will have $b_1(b_2 - \epsilon_2) =$

z , provided they also satisfy the box constraints $\epsilon_1 \leq b_1 \leq m$ and $\epsilon_2 \leq b_2 \leq n$. We have

$$b_1 \ell_1 + b_2 \ell_2 = b_1 \ell_1 + (b_2 - \epsilon_2) \ell_2 + \epsilon_2 \ell_2 \stackrel{(i)}{\geq} 2\sqrt{b_1(b_2 - \epsilon_2)\ell_1 \ell_2} + \epsilon_2 \ell_2 = 2\sqrt{z\ell_1 \ell_2} + \epsilon_2 \ell_2$$

where (i) follows from the AM-GM inequality and we have equality when $b_1 \ell_1 = (b_2 - \epsilon_2) \ell_2$. Solving for (b_1, b_2) along with $b_1(b_2 - \epsilon_2) = z$ gives

$$(b'_1, b'_2) := \left(\sqrt{z \frac{\ell_2}{\ell_1}}, \sqrt{z \frac{\ell_1}{\ell_2}} + \epsilon_2 \right).$$

If $\epsilon_1 \leq b'_1 \leq m$ and $b'_2 \leq n$ (note $\epsilon_2 \leq b'_2$ is trivially true), or equivalently,

$$\epsilon_1^2 \frac{\ell_1}{\ell_2} \leq z \leq \min \left\{ m^2 \frac{\ell_1}{\ell_2}, (n - \epsilon_2)^2 \frac{\ell_2}{\ell_1} \right\},$$

then (b'_1, b'_2) is the optimal solution of the continuous problem. Now $(\lceil b'_1 \rceil, \lceil b'_2 \rceil)$ still satisfies all the constraints so the integer-constrained problem has minimum no more than

$$\begin{aligned} & \lceil b'_1 \rceil \ell_1 + \lceil b'_2 \rceil \ell_2 \\ & < (b'_1 + 1) \ell_1 + (b'_2 + 1) \ell_2 \\ & = \left(\sqrt{z \frac{\ell_2}{\ell_1}} + 1 \right) \ell_1 + \left(\sqrt{z \frac{\ell_1}{\ell_2}} + \epsilon_2 + 1 \right) \ell_2 \\ & = 2\sqrt{z\ell_1 \ell_2} + \ell_1 + (1 + \epsilon_2) \ell_2. \end{aligned}$$

This establishes the upper bound in the first case.

For the “otherwise” case (which also includes the case $\ell_1 \ell_2 = 0$), we simply exhibit feasible integer points. First, consider

$$(\hat{b}_1, \hat{b}_2) := \left(m, \left\lceil \frac{z}{m} + \epsilon_2 \right\rceil \right).$$

Since $m(n - \epsilon_2) \geq z$, we have $\frac{z}{m} + \epsilon_2 \leq n$, hence $\hat{b}_2 \leq n$. Also $\hat{b}_2 \geq \frac{z}{m} + \epsilon_2 \geq \epsilon_2$, so the box constraints hold. Finally,

$$\hat{b}_1(\hat{b}_2 - \epsilon_2) \geq m \left(\frac{z}{m} + \epsilon_2 - \epsilon_2 \right) = z,$$

so $(\hat{b}_1, \hat{b}_2)$ is feasible, and therefore the integer optimum is at most

$$\hat{b}_1 \ell_1 + \hat{b}_2 \ell_2 = m \ell_1 + \left\lceil \frac{z}{m} + \epsilon_2 \right\rceil \ell_2.$$

Second, consider

$$(\tilde{b}_1, \tilde{b}_2) := \left(\max \left\{ \lceil \epsilon_1 \rceil, \left\lceil \frac{z}{n - \epsilon_2} \right\rceil \right\}, n \right).$$

Since $\epsilon_1 \leq m$ and $\frac{z}{n-\epsilon_2} \leq m$ (because $m(n-\epsilon_2) \geq z$), we have $\tilde{b}_1 \leq m$. Further,

$$\tilde{b}_1(n-\epsilon_2) \geq \left\lceil \frac{z}{n-\epsilon_2} \right\rceil (n-\epsilon_2) \geq z,$$

so $(\tilde{b}_1, \tilde{b}_2)$ is feasible, and the integer optimum is at most

$$\tilde{b}_1 \ell_1 + \tilde{b}_2 \ell_2 = \max \left\{ \lceil \epsilon_1 \rceil, \left\lceil \frac{z}{n-\epsilon_2} \right\rceil \right\} \ell_1 + n \ell_2.$$

□

Corollary 4. *Let k, m, ℓ_1, ℓ_2 be integers such that $0 < k < m$ and $\ell_1, \ell_2 \geq 0$. Let $\epsilon_1 = \frac{\ell_2}{km}$ and $\epsilon_2 = \frac{m+\ell_1}{km}$, and assume $\epsilon_1 \leq k$ and $\epsilon_2 \leq 1$. Suppose b_1, b_2 range over the integers subject to the constraints*

$$\begin{aligned} \epsilon_1 &\leq b_1 \leq m, \\ \epsilon_2 &\leq b_2 \leq m, \quad \text{and} \\ km - \frac{1}{k} + \frac{1}{km} &\leq b_1(b_2 - \epsilon_2). \end{aligned}$$

If $R := k/m$, then the minimum of $b_1 \ell_1 + b_2 \ell_2$ is upper bounded by

$$\begin{cases} 2m\sqrt{R\ell_1\ell_2} + \ell_1 + (1+\epsilon_2)\ell_2 & \text{if } \ell_1\ell_2 \neq 0 \text{ and } R \leq \min \left\{ \frac{\ell_1}{\ell_2}, (1 - \frac{\epsilon_2}{m})^2 \frac{\ell_2}{\ell_1} \right\} \\ m \min \left\{ \ell_1 + (R + \frac{1}{m})\ell_2, (R + \frac{1}{m})\ell_1 + \ell_2 \right\} & \text{otherwise.} \end{cases}$$

Moreover, in the first case (i.e., “if $\ell_1\ell_2 \neq 0$ and $R \leq \dots$ ”), the following choice of (b_1, b_2) is feasible and, when substituted into $b_1 \ell_1 + b_2 \ell_2$, satisfies the stated upper bound (not necessarily with equality):

$$(b_1, b_2) = \left(\left\lceil \sqrt{km \frac{\ell_2}{\ell_1}} \right\rceil, \left\lceil \sqrt{km \frac{\ell_1}{\ell_2}} + \epsilon_2 \right\rceil \right).$$

In the second (“otherwise”) case, the same holds for one of the points

$$(b_1, b_2) = (m, k+1) \quad \text{or} \quad (b_1, b_2) = (k+1, m).$$

Proof. This follows from Lemma 6 if we let $m = n$ and $z := km > km - \frac{1}{k} + \frac{1}{km}$. The condition $\epsilon_2 \leq 1$ implies $m(m-\epsilon_2) \geq m(m-1) \geq z = km$ so there is at least one feasible solution. For the first case, note that

$$\epsilon_1^2 \frac{\ell_1}{\ell_2} \leq z \leq \min \left\{ m^2 \frac{\ell_1}{\ell_2}, (m-\epsilon_2)^2 \frac{\ell_2}{\ell_1} \right\},$$

is equivalent to

$$\frac{\ell_2^2}{k^2 m^4} \frac{\ell_1}{\ell_2} \leq R \leq \min \left\{ \frac{\ell_1}{\ell_2}, (1 - \frac{\epsilon_2}{m})^2 \frac{\ell_2}{\ell_1} \right\}$$

and the first inequality is always true since the left-hand side is

$$\frac{\ell_2^2}{k^2 m^4} \frac{\ell_1}{\ell_2} = \frac{\ell_1 \ell_2}{k^2 m^4} \leq \frac{(k-1)m \cdot k^2 m}{k^2 m^4} = \frac{k-1}{m^2} < \frac{k}{m} = R,$$

where we used $\epsilon_1 = \frac{\ell_2}{km} \leq k$ and $\epsilon_2 = \frac{m+\ell_1}{km} \leq 1$.

For the “otherwise” case, first note that

$$\left\lceil \frac{z}{m} + \epsilon_2 \right\rceil = \lceil k + \epsilon_2 \rceil = k + 1$$

and $\left\lfloor \frac{z}{m - \epsilon_2} \right\rfloor = \left\lfloor \frac{km}{m - \epsilon_2} \right\rfloor = k + 1,$

where the last equality follows since the condition $k < \frac{km}{m - \epsilon_2} \leq k + 1$ is equivalent to $0 < \epsilon_2 \leq \frac{m}{k+1}$, which in turn follows from $0 < \epsilon_2 \leq 1$. Thus,

$$m\ell_1 + \left\lceil \frac{z}{m} + \epsilon_2 \right\rceil \ell_2 = m\ell_1 + (k+1)\ell_2 = m(\ell_1 + (R + \frac{1}{m})\ell_2)$$

and $\left\lfloor \frac{z}{m - \epsilon_2} \right\rfloor \ell_1 + m\ell_2 = (k+1)\ell_1 + m\ell_2 = m((R + \frac{1}{m})\ell_1 + \ell_2).$

Hence, by the lemma, the points that achieve this are one of

$$(b_1, b_2) = \left(m, \left\lceil \frac{z}{m} + \epsilon_2 \right\rceil \right) = (m, k+1), \quad \text{or}$$

$$(b_1, b_2) = \left(\max \left\{ \lceil \epsilon_1 \rceil, \left\lfloor \frac{z}{m - \epsilon_2} \right\rfloor \right\}, m \right) = (\max\{\lceil \epsilon_1 \rceil, k+1\}, m) = (k+1, m).$$

□

Remark 12 (Trivial ACDGV Parameters (Section 5)). In the ACDGV scheme, the parameter choice $\ell_2 \geq km^2$ yields a trivial public key with at least constant probability. To see this, stack the basis $\mathbf{B}_1, \dots, \mathbf{B}_{km} \in \mathbb{F}_q^{(m+\ell_1) \times (n+\ell_2)}$ of $\mathcal{C}_{\text{pub}}$ vertically to obtain a $km(m+\ell_1) \times (n+\ell_2)$ matrix. Further, without loss of generality, we can assume that the scrambling $\mathbf{P}$ and $\mathbf{Q}$ are the identity. If we remove the ℓ_1 bottom rows of each basis matrix, then this becomes the $km^2 \times (n+\ell_2)$ matrix

$$\begin{bmatrix} \mathbf{A}_1 & \mathbf{R}_1 \\ \mathbf{A}_2 & \mathbf{R}_2 \\ \vdots & \vdots \\ \mathbf{A}_{km} & \mathbf{R}_{km} \end{bmatrix}.$$

Now the last ℓ_2 columns of this matrix form a wide matrix and with at least constant probability have full rank. Hence, by column scrambling, the first n columns can be chosen arbitrarily. Thus, the distribution of the public code is independent of the $\mathbf{A}_i$'s. (A similar argument can be made to show that w.h.p. $\ell_1 \geq kmn$ is trivial.)

H Python Code

We used Python with the `galois` library [11] to implement and test the following.

1. A proof-of-concept of our attack, which given only the public code $\mathcal{C}_{\text{pub}} \subseteq \mathbb{F}_q^{m \times n}$ and correctly guessed matrices $\mathbf{V} \in \mathbb{F}_q^{(m+\ell_1) \times b_1}$ and $\mathbf{W} \in \mathbb{F}_q^{(n+\ell_2) \times b_2}$, derives a matrix code equivalent to the secret $\mathcal{C}_{\text{mat}} \subseteq \mathbb{F}_q^{m \times n}$ up to left- and right-scrambling. The code is implemented without any optimizations in order to make verifiability easier. Thus, it is only suitable for small parameters.
2. Tests verifying Assumptions 1 and 2. We ran the code on all ACDGV parameter sets (except those with $q = 16$) for 50 iterations each, with 100% success in both tests. Parameter sets with $q = 16$ could not be tested as our implementation does not support prime power subfields. The tests were run with `seed=5` and `iterations=50`.
3. A script to reproduce the parameter costs in Table 3 (Section 6).

The code is available at <https://github.com/mnap/acdgv-key-attack> and will also be submitted to the call of artifacts of Eurocrypt 2026 for archival.

I Errata

1. The published version contains a minor error in the formulation of the heuristics for **BasisRecover** and **NaiveRecover** (Theorems 1 and 2, respectively). The heuristics stated there are not strong enough to support the claimed reduced polynomial complexity. The full version (this paper) gives correct and simpler heuristics.